

GENERAL APTITUDE

Year-wise & Session-wise **Solutions**

GATE — ALL BRANCHES

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in General Aptitude (all GATE branches), gathered from the original papers and arranged year by year and session by session. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by year and by session, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q24.5.3

24	GATE exam year — 2024 (two digits; 98 = 1998)
5	Session number of that year's exam
3	Question number within that session

So **Q24.5.3** is the 3rd aptitude question of Session 5, GATE 2024. Numbers are therefore not sequential through the book — they tell you the year and session at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a session to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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YEAR

2014

9 Sessions · Solutions

Sessions Summary

Session	Participating Branches
Session 1	CE, CH, XE, AR, AE, MA, MT
Session 2	XL, BT, CE, CH, PI, TF, IN
Session 3	EC, ME
Session 4	EC, ME
Session 5	EC, ME
Session 6	EC, ME
Session 7	CS, EE
Session 8	CS, EE
Session 9	CS

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Year Snapshot

Total Questions: 90 **MCQ:** 71 **MSQ:** 0 **NAT:** 19

Session : 1 | Forenoon | 02/02/14

Q14.1.1 MCQ 2014 1M Ans: A ● ○ ○

The sentence states that a student must demonstrate a high level of comprehension of a subject. In this context, comprehension refers to the capacity to grasp, process, and mentally perceive the nature or meaning of the material being studied. Among the given options:

- **Understanding:** This is a direct synonym for comprehension, representing the ability to perceive meaning or explain concepts.
- **Meaning:** This refers to the definition or significance of a word or concept rather than the act of grasping it.
- **Concentration:** This refers to the level of focus or attention applied, which is a prerequisite for comprehension but not the comprehension itself.
- **Stability:** This refers to being firm or unchanging, which is unrelated to the cognitive process of learning.

A. Understanding

Key Points

- **Vocabulary in Context:** Selecting the correct synonym requires analyzing how the word functions within the specific sentence.
- **Comprehension:** Defined as the power of understanding or the mental grasping of ideas.

Q14.1.2 MCQ 2014 1M Ans: B ● ○ ○

The sentence describes "**his ability to forgive**," which is a positive moral trait. The phrase "One of his biggest..." functions as a superlative structure that must be followed by a plural noun.

- **virtues:** This is the plural form of "virtue," referring to behavior showing high moral standards. Forgiveness is fundamentally a moral virtue.
- **vice:** This refers to a moral fault or bad habit, which is the antonym of the trait described.
- **choices:** While forgiveness involves a choice, "virtues" is the standard categorical term for moral excellences.
- **strength:** Although forgiveness is a strength, this word is singular and fails the grammatical requirement for a plural noun.

B: virtues

Key Points

- **Grammatical Agreement:** The construction "One of the [Superlative]..." always requires a plural countable noun.
- **Contextual Connotation:** The blank must be filled with a word that matches the positive ethical value of "forgiveness."

Q14.1.3 MCQ 2014 1M Ans: D ● ○ ○

The passage establishes that Rajan was unhappy specifically because Sajan decided to work alone. Sajan responded to this unhappiness by explaining his preference for independent work.

- **Option A**: The text does not support that Rajan *only* works in groups, just that he was unhappy in this specific instance.

- **Option B**: There is no mention of an external force or assignment that placed them in a group against their will.
- **Option C**: Sajan "explained" his preference; the text implies he maintained his decision rather than giving in.
- **Option D**: If Rajan was unhappy about Sajan working alone, it logically follows that Rajan had expected or assumed they would be working as a pair.

D: Rajan had believed that Sajan and he would be working together

Key Points

- **Logical Inference:** Inferred statements must be direct consequences of the information provided without introducing external assumptions.

Q14.1.4 MCQ 2014 1M Ans: C ●○○

To determine the nature of the tangent to the curve $y = 5x^2 + 3$, we calculate the derivative, which represents the slope m.

$$\frac{dy}{dx} = \frac{d}{dx}(5x^2 + 3) = 10x$$

Evaluating the slope at the given point $(0, 3)$:

$$m = \left.\frac{dy}{dx}\right|_{x=0} = 10(0) = 0$$

A line with a slope of 0 is a horizontal line.

- The equation of the tangent line is $y - 3 = 0(x - 0)$, which simplifies to $y = 3$.
- Since $y = 3$ is a horizontal line, it is parallel to the x-axis.
- It does not pass through the origin $(0, 0)$, nor does it have a slope of ± 1.

C: is parallel to the x axis

Key Points

- **Slope of Tangent:** The value of the first derivative $f'(x)$ at a point x_0 gives the slope of the tangent at that point.
- **Zero Slope:** A slope of $m = 0$ indicates a line parallel to the x-axis (horizontal).

Q14.1.5 NAT 2014 1M Ans: 1300 ●○○

The total cost of production C is the sum of fixed costs and variable costs.

Given:

Fixed daily cost = $50,000$

Variable cost = $800Q$, where Q is production in tonnes.

For a daily production of $Q = 100$ tonnes:

$$\text{Total Cost} = 50,000 + (800 \times 100) = 50,000 + 80,000 = 1,30,000$$

The cost of production per tonne is:

$$\text{Cost per tonne} = \frac{\text{Total Cost}}{Q} = \frac{1,30,000}{100} = 1300$$

1300

Key Points

- **Total Cost (C):** $C = F + V(Q)$, where F is fixed cost and $V(Q)$ is variable cost.
- **Average Cost:** The cost per unit is calculated by dividing the total cost by the quantity produced.

Q14.1.6 MCQ 2014 2M Ans: D ● ○ ○

To find the odd one out, we check the numerical positions of the letters in the English alphabet:

- **ALRVX:** A(1), L(12), R(18), V(22), X(24). Intervals: +11, +6, +4, +2.
- **EPVZB:** E(5), P(16), V(22), Z(26), B(2). Intervals: +11, +6, +4, +2.
- **ITZDF:** I(9), T(20), Z(26), D(4), F(6). Intervals: +11, +6, +4, +2.
- **OYEIK:** O(15), Y(25), E(5), I(9), K(11). Intervals: +10, +6, +4, +2.

The first interval in OYEIK is +10, whereas all other groups have a first interval of +11.

D: OYEIK

Key Points

- **Pattern Recognition:** Logical series questions often rely on constant or progressive arithmetic intervals between letter positions.

Q14.1.7 MCQ 2014 2M Ans: B ● ● ○

Let the floors be 1, 2, 3, 4, 5, 6.

	Anuj	Bhola	Chandan	Dilip	Eswar	Faisal
(A)	6	2	5	1	3	4
(B)	2	6	5	1	3	4
(C)	4	2	6	3	1	5
(D)	2	4	6	1	3	5

1. Anuj (A) and Bhola (B) are on even floors: $\{2, 4, 6\}$.
2. Faisal (F) is 3 floors above Dilip (D): (D, F) can be $(1, 4)$, $(2, 5)$, or $(3, 6)$.
3. Dilip is not on floor 2 $\implies$ (D, F) is $(1, 4)$ or $(3, 6)$.
4. Chandan (C) is not below Faisal $\implies$ $C > F$.
5. If $(D, F) = (3, 6)$, then C has no floor > 6, so (D, F) must be $(1, 4)$.
6. Since $C > F$ and $F = 4$, C must be floor 5 or 6.
7. Even floors remaining for A and B are $\{2, 6\}$.
8. If $B = 6$, then $A = 2$ and $C = 5$. Eswar (E) takes the last floor 3.
9. Considering Eswar's condition: Eswar (3) is not immediately above/below Bhola (6). This is satisfied.

Arrangement: $D = 1, A = 2, E = 3, F = 4, C = 5, B = 6$. Matching with Option (B).

B

Key Points

- **Constraint Satisfaction:** Logical puzzles are solved by systematically eliminating impossible positions based on given negative and relative constraints.

Q14.1.8 NAT | 2014 | 2M | Ans: 180 ● ● ○

To find the required sum, we process the geometric properties of the quadrilateral and the triangle in sequential steps.

Step 1: Quadrilateral Angles The internal angles of the quadrilateral are in the ratio 3 : 4 : 5 : 6. The sum of internal angles for any quadrilateral is 360°.

- **Total parts**: $3 + 4 + 5 + 6 = 18$
- **Smallest angle (Q_{min})**: $\frac{3}{18} \times 360^\circ = 60^\circ$
- **Largest angle (Q_{max})**: $\frac{6}{18} \times 360^\circ = 120^\circ$

Step 2: Triangle Angles The triangle's angles are derived from the smallest angle of the quadrilateral (Q_{min}). The sum of angles in a triangle is 180°.

- **Smallest angle (T_{min})**: $\frac{2}{3} \times Q_{\text{min}} = \frac{2}{3} \times 60^\circ = 40^\circ$
- **Largest angle (T_{max})**: $2 \times T_{\text{min}} = 2 \times 40^\circ = 80^\circ$
- **Second largest angle (T_2)**: $180^\circ - (40^\circ + 80^\circ) = 60^\circ$

Step 3: Required Sum The problem asks for the sum of the second largest angle of the triangle (T_2) and the largest angle of the quadrilateral (Q_{max}).

$$\text{Sum} = T_2 + Q_{\text{max}} = 60^\circ + 120^\circ = 180^\circ$$

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180

Key Points

- **Angle Sum Property**: The sum of internal angles is 360° for a quadrilateral and 180° for a triangle.
- **Proportional Division**: Specific angles are calculated by dividing the total angular sum by the sum of ratio parts and multiplying by the specific part.

Q14.1.9 MCQ | 2014 | 2M | Ans: D ● ● ○

Let the population of Country Y be N.

Then Country X population is $3N$.

Total population = $4N$.

People > 6 ft in X = 1% of $3N$ = $0.03N$.

People > 6 ft in Y = 2% of N = $0.02N$.

Total people > 6 ft = $0.05N$.

Percentage = $\frac{0.05N}{4N} \times 100 = 1.25\%$.

D: 1.25

Key Points

- **Weighted Average:** The overall percentage is the total count of the specific trait divided by the total combined population.

Q14.1.10 MCQ 2014 2M Ans: B ● ● ○

Analysis of statements based on the chart:

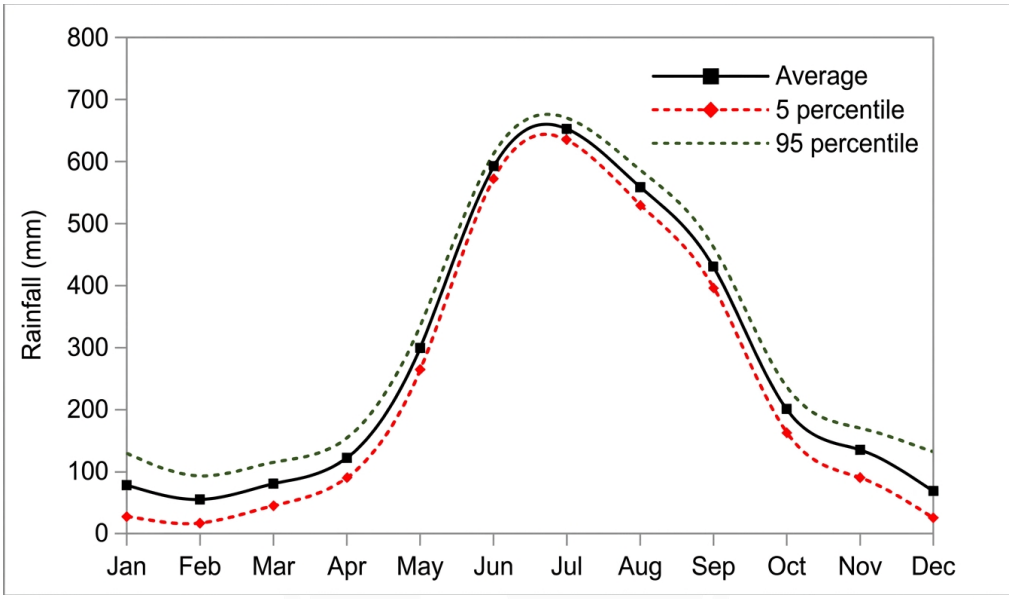


Fig. Solution Q14.1.10

- **(i):** Average rainfall in July (~ 650 mm) is significantly higher than December (~ 50 mm). **True**.
- **(ii):** The 5th percentile for August is around 500 mm i.e in 2 or 3 years August receives less than 500 mm, and the 95th percentile for January is around 125 mm. While it is highly probable, "every year" is a definitive claim that cannot be strictly proven for all 50 years just by looking at these three statistical lines.**Fale**.
- **(iii):** The variation in July between 5 and 95 percentile is small than in February, meaning July rainfall is less variable and thus easier to estimate with confidence. **True**.
- **(iv):** The 5th percentile for August is below 500 mm, meaning in some years, rainfall is less than 500 mm. **False**.

Only (i) and (iii) are logically supported.

B: (i) and (iii)

Key Points

- **Data Interpretation:** Percentiles describe the distribution of historical data; the spread between low and high percentiles indicates predictability or variability.

Session : 2 | Afternoon | 02/02/14

Q14.2.1 MCQ 2014 1M Ans: D ● ○ ○

The sentence describes a general characteristic or a factual state associated with a medical condition (Alzheimer's disease). For general truths, habits, or scientific facts, **the simple present tense** is used.

- **experiences:** This is the simple present tense third-person singular form, which correctly matches the subject "A person".
- **experienced:** Simple past tense; implies the condition happened only in the past.
- **has experienced:** Present perfect tense; suggests an action that happened at an unspecified time before now.
- **is experiencing:** Present continuous tense; suggests the action is happening only at this exact moment rather than being a characteristic of the disease.

D: experiences

Key Points

- **Simple Present Tense:** Used to express general truths, permanent situations, or scientific facts.

Q14.2.2 MCQ 2014 1M Ans: A ● ○ ○

The second part of the sentence states, "they are satisfied with what they have". This context directly defines the missing word.

- **Contentment:** The state of being satisfied and happy with one's current situation or possessions.
- **Ambition:** A strong desire to achieve something, often implying a lack of satisfaction with the current status.
- **Perseverance:** Steadfastness in doing something despite difficulty or delay in achieving success.
- **Hunger:** A strong desire or need for something (often food or power), indicating a lack rather than satisfaction.

A: Contentment

Key Points

- **Contextual Vocabulary:** The correct choice must align with the explanatory clause "satisfied with what they have".

Q14.2.3 MCQ 2014 1M Ans: C ● ○ ○

The quote "As a woman, I have no country" suggests that the identity and struggles of women transcend national borders and patriotic affiliations.

- **Option C:** Suggests that the shared experiences or unity (solidarity) among women are not limited by national boundaries, which aligns with the global perspective of the quote.
- **Option A & B:** Take the quote too literally as a statement of statelessness or lack of citizenship.
- **Option D:** Discusses legal rights, which is a specific subset of political status not directly captured by the philosophical nature of the quote.

C:Women's solidarity knows no national boundaries.

Key Points

- **Critical Thinking:** Interpreting philosophical statements requires looking beyond literal meanings to find underlying themes of universal identity or solidarity.

Q14.2.4 NAT | 2014 | 1M | Ans: 25 ● ○ ○

The probability of an earthquake occurring in any given year is $P = 0.04$. The average time (return period) between successive occurrences is the reciprocal of the annual probability.

$$T = \frac{1}{P} = \frac{1}{0.04}$$
$$T = \frac{100}{4} = 25$$

25

Key Points

- **Return Period (T):** Calculated as $T = 1/P$, where P is the annual probability of occurrence.

Q14.2.5 MCQ | 2014 | 1M | Ans: A ● ○ ○

The population grows according to the compound growth formula: $P = P_0(1 + r)^n$. To double the population, we set $P = 2P_0$:

$$2 = (1 + 0.20)^n \implies 2 = (1.2)^n$$

Taking natural logs on both sides:

$$\ln(2) = n \ln(1.2)$$
$$n = \frac{\ln(2)}{\ln(1.2)} \approx \frac{0.693}{0.182} \approx 3.8 \text{ years}$$

Since 3.8 falls between 3 and 4, the correct interval is 3 – 4 years.

A: 3 - 4 years

Key Points

- **Compound Growth:** Doubling time at a constant percentage growth rate is found using the formula $n = \ln(2)/\ln(1+r)$.

Q14.2.6 MCQ | 2014 | 2M | Ans: A ● ○ ○

Given Information:

1. Som < Riaz
2. Shiv > Ansu
3. Ansu is the youngest child.

From (2) and (3), we know Ansu is at the bottom. We need to find if Riaz or Shiv is at the top. - We have: Riaz > Som > Ansu and Shiv > Ansu.

- Statement 1: "Shiv is younger to Riaz" (Riaz > Shiv). Combined with Riaz > Som, this confirms Riaz is the eldest.

- Statement 2: "Shiv is elder to Som" (Shiv > Som). This tells us Shiv > Som > Ansu, but we still don't know the relative position of Riaz and Shiv.

Therefore, Statement 1 alone is sufficient.

A: Statement 1 by itself determines the eldest child

Key Points

- **Data Sufficiency:** Identifying the eldest child requires enough relative comparisons to place one individual above all others in the hierarchy.

Q14.2.7 MCQ 2014 2M Ans: C ● ○ ○

The paragraph argues that while the "analog era" required extreme exactitude due to sparse data, the modern "big data" world requires a change in thinking. It suggests that obsessing over exactness in a digital, connected world is "missing a crucial point".

Therefore, the main point is that absolute exactitude is no longer the primary requirement or the most critical factor when dealing with massive datasets.

C:Exactitude is not critical in dealing with big data

Key Points

- **Reading Comprehension:** Identifying the core argument involves recognizing the shift in requirements from low-data environments to high-data environments.

Q14.2.8 MCQ 2014 2M Ans: D ● ● ○

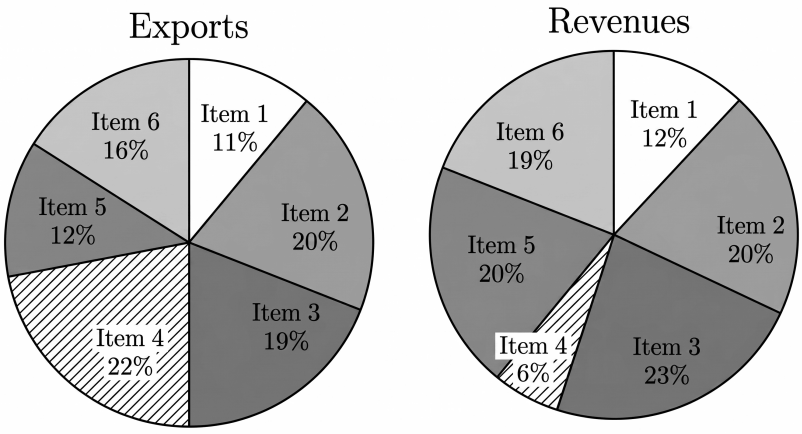


Fig. Solution Q14.2.8

Total Exports = 5 lakh tonnes = $5,00,000$ kg.
Total Revenue = 250 crore rupees = $2,50,00,00,000$ rupees.

For Item 1:

Export quantity = 11% of $5,00,000 = 55,000$ kg.
Revenue = 12% of 250 = 30 crore.
Revenue per kg (R_1) = $\frac{30 \text{ crore}}{55,000}$.

For Item 4:

Export quantity = 22% of $5,00,000 = 1,10,000$ kg.
Revenue = 6% of 250 = 15 crore.
Revenue per kg (R_4) = $\frac{15 \text{ crore}}{1,10,000}$.

Ratio ($R_1 : R_4$):

$$\text{Ratio} = \frac{30/55,000}{15/1,10,000} = \frac{30}{55} \times \frac{110}{15} = 2 \times 2 = 4$$

D: 4:1

Key Points

- **Revenue per Unit:** Calculated as (Total Revenue × Revenue %) divided by (Total Weight × Export %).

Q14.2.9 MCQ 2014 2M Ans: C ● ● ○

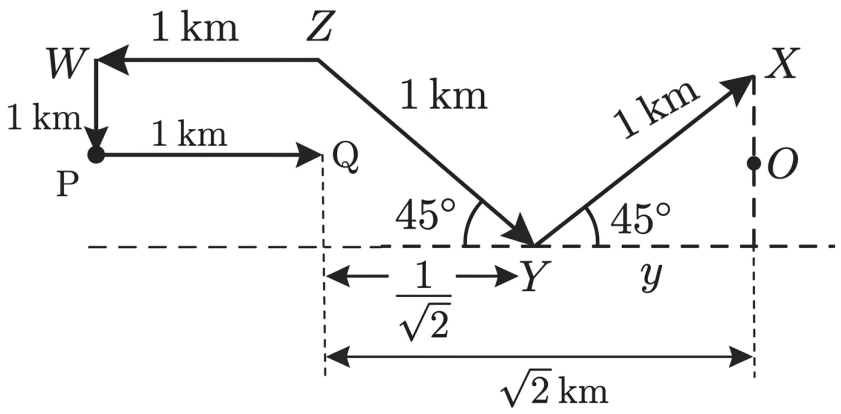


Fig. Solution Q14.2.9

OQ=$\sqrt{2}$ km, OX=1 km
Using Pythagoras theorem

$$\text{Distance(XQ)} = \sqrt{(\sqrt{2})^2 + 1^2} = \sqrt{2+1} = \sqrt{3}$$

C: $\sqrt{3}$

Key Points

- **Coordinate Geometry:** Complex navigation problems can be simplified by mapping directions onto a 2D Cartesian plane.

Q14.2.10 NAT 2014 2M Ans: 0.49 ● ● ○

To find the probability that the individual is actually positive given a positive test result, we apply Bayes' Theorem.
Step 1: Identify given probabilities**

- $P(H^+) = 0.10$: Prior probability of being HIV positive.
- $P(H^-) = 1 - 0.10 = 0.90$: Prior probability of being HIV negative[cite: 1, 3].
- $P(+|H^+) = 0.95$: Probability of testing positive given the person is H^+ (Sensitivity).
- $P(-|H^-) = 0.89$: Probability of testing negative given the person is H^- (Specificity).
- $P(+|H^-) = 1 - 0.89 = 0.11$: Probability of testing positive given the person is H^- (False Positive Rate).

Given the test is positive 2 situation can occur
(i) The person is positive and kit gives confirmed result

(ii) The person is negative and kit give Wrong result

Step 2: Apply Bayes' Theorem The probability $P(H^+|+)$ is calculated as:

$$P(H^+|+) = \frac{P(i)}{P(i) + P(ii)}$$

$$P(H^+|+) = \frac{P(+|H^+)P(H^+)}{P(+|H^+)P(H^+) + P(+|H^-)P(H^-)}$$

$$P(H^+|+) = \frac{0.95 \times 0.10}{(0.95 \times 0.10) + (0.11 \times 0.90)}$$

$$P(H^+|+) = \frac{0.095}{0.095 + 0.099} = \frac{0.095}{0.194} \approx 0.4896$$

Rounding to two decimal places as per standard NAT conventions for this paper:

$$\boxed{0.49}$$

Key Points

- **Bayes' Theorem:** A fundamental formula in probability used to update the likelihood of a hypothesis (H^+) given new evidence (+).
- **False Positive Impact:** Even with high sensitivity (95%), if the false positive rate is significant (11%) and the base rate is low, the probability of actually having the condition after a positive test can be surprisingly low.

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Session : 3 | Forenoon | 15/02/14

Q14.3.1 MCQ 2014 1M Ans: C ● ○ ○

The sentence describes a sequence of events in the past. The second part of the sentence uses the past tense ("was filed"), indicating that the entire action occurred in the past. Therefore, the main clause must also use a past tense form to maintain tense consistency.

- **is allowed to / has been allowed to:** These indicate present or ongoing relevance, which conflicts with "was filed."
- **will be allowed to:** This indicates a future action.
- **was allowed to:** This is the simple past passive form, which correctly aligns with the timeline of the flight plan being filed.

C: was allowed to

Key Points

- **Tense Consistency:** In complex sentences, the tense of the main clause should align with the subordinate clause to establish a logical timeline.

Q14.3.2 MCQ 2014 1M Ans: D ● ○ ○

Let W be the set of women, E be the set of entrepreneurs, and D be the set of doctors.

1. Statement (i) means $W \subseteq E$ (All women are entrepreneurs).
2. Statement (ii) means $W \cap D \neq \emptyset$ (Some women are doctors).

Since some women are doctors and all women are entrepreneurs, there must be a subset of doctors who are also entrepreneurs (specifically, the women who are doctors). Therefore, "Some entrepreneurs are doctors" is the only logical conclusion.

D: Some entrepreneurs are doctors

Key Points

- **Syllogism:** If $A \subset B$ and some A are C, then it is logically necessary that some B are C.

Q14.3.3 MCQ 2014 1M Ans: B ● ○ ○

The sentence contrasts ancient beliefs (supernatural causes) with the impact of modern science.

- **impel / propel:** These mean to drive forward or push.
- **repel:** To drive back or resist.
- **dispel:** To make a doubt, feeling, or belief disappear. Modern science helps eliminate or disprove ancient notions of supernatural causes.

B: dispel

Key Points

- **Contextual Vocabulary:** The word "dispel" is specifically used for removing doubts, myths, or erroneous beliefs.

Q14.3.4 MCQ 2014 1M Ans: A ● ○ ○

Consistency in statistics is measured by the Standard Deviation (SD). A lower standard deviation indicates that the data points (runs scored) are closer to the average, representing higher consistency.

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

- Batsman K: $SD = 5.21$
- Batsman L: $SD = 6.35$
- Batsman M: $SD = 6.22$
- Batsman N: $SD = 5.90$

Batsman K has the lowest standard deviation and is therefore the most consistent.

A: K

Key Points

- **Standard Deviation:** A measure of the amount of variation or dispersion of a set of values. Lower dispersion implies higher consistency.

Q14.3.5 NAT 2014 1M Ans: 725 ● ○ ○

The series is 12, 35, 81, 173, 357 Let the series be $a_1, a_2, a_3, a_4, a_5, a_6$.

$$a_2 = (a_1 \times 2) + 11 = (12 \times 2) + 11 = 35$$
$$a_3 = (a_2 \times 2) + 11 = (35 \times 2) + 11 = 81$$
$$a_4 = (a_3 \times 2) + 11 = (81 \times 2) + 11 = 173$$
$$a_5 = (a_4 \times 2) + 11 = (173 \times 2) + 11 = 357$$
$$a_6 = (a_5 \times 2) + 11 = (357 \times 2) + 11 = 725$$

725

Key Points

- **Arithmetic Series Pattern:** Identifying the recurrence relation $a_n = 2a_{n-1} + 11$ allows for term-by-term calculation.

Q14.3.6 MCQ 2014 2M Ans: D ● ● ○

Examine the alphabetical intervals between the letters in each group:

- **W,E,K,O:** W(23) $\xrightarrow{+8}$ E(5+26=31) $\xrightarrow{+6}$ K(11) $\xrightarrow{+4}$ O(15). Intervals: +8, +6, +4.
- **I,Q,W,A:** I(9) $\xrightarrow{+8}$ Q(17) $\xrightarrow{+6}$ W(23) $\xrightarrow{+4}$ A(1+26=27). Intervals: +8, +6, +4.
- **F,N,T,X:** F(6) $\xrightarrow{+8}$ N(14) $\xrightarrow{+6}$ T(20) $\xrightarrow{+4}$ X(24). Intervals: +8, +6, +4.

- **N,V,B,D:** N(14) $\xrightarrow{+8}$ V(22) $\xrightarrow{+6}$ B(2+26=28) $\xrightarrow{+2}$ D(4). Intervals: +8, +6, +2.

The group N,V,B,D is the odd one because its last interval is +2 instead of +4.

D: N,V,B,D

Key Points

- **Letter Position Arithmetic:** Converting letters to their numerical positions (A=1, B=2...) helps identify underlying mathematical patterns in strings.

Q14.3.7 MCQ 2014 2M Ans: B ● ● ○

Let's analyze the requirements:

- Resident Male ($< 10L$): Form P
- Resident Female ($< 8L$): Form Q
- Income $> 10L$ (All): Form R (Unless non-resident $> 15L$)
- Non-resident ($> 15L$): Form S
- Form T: For "All others" not covered by the specific rules above.

Check the options for Form T:

- A: Resident male ($9L$) falls under Form P.
- B: Resident female ($9L$) is not covered by Form Q ($< 8L$) or Form R ($> 10L$). Therefore, she must fill Form T.
- C: Non-resident male ($16L$) falls under Form S ($> 15L$).
- D: Non-resident female ($16L$) falls under Form S ($> 15L$).

B: a resident female with annual income Rs 9 lakh

Key Points

- **Conditional Logic:** Solving eligibility problems requires checking each case against a set of mutually exclusive or hierarchical conditions.

Q14.3.8 NAT 2014 2M Ans: 560 ● ● ○

Let L_p be the length of the platform and v be the speed of the train. Length of the train $L_t = 280\,\text{m}$. 1. Crossing the platform:

$$v = \frac{L_t + L_p}{60} = \frac{280 + L_p}{60}$$

2. Passing the man:

$$v = \frac{L_t}{20} = \frac{280}{20} = 14\,\text{m/s}$$

3. Equating speed:

$$14 = \frac{280 + L_p}{60}$$
$$840 = 280 + L_p$$
$$L_p = 560\,\text{m}$$

560

Key Points

- **Relative Motion:** When a train crosses a platform, the total distance covered is the sum of the lengths of the train and the platform.

Q14.3.9 MCQ 2014 2M Ans: D ●●○

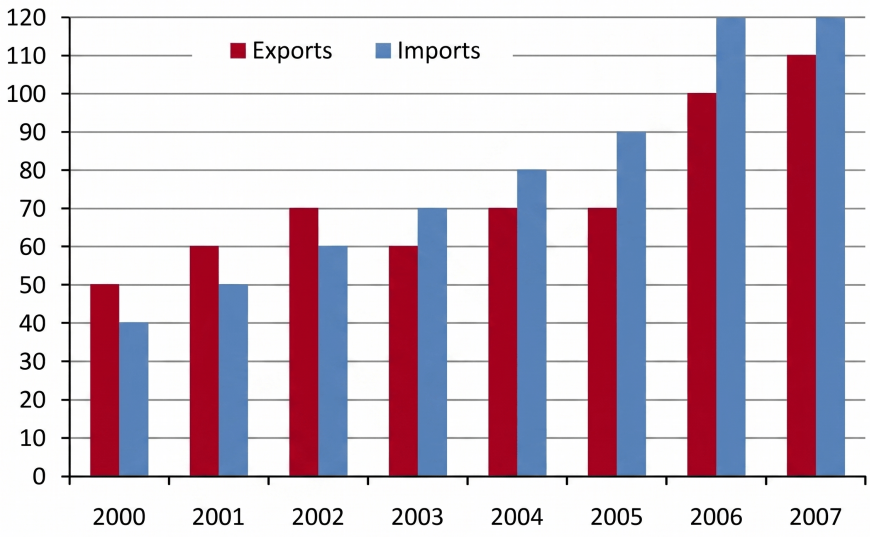


Fig. Solution Q14.3.9

Trade Deficit = Imports − Exports. We need to find the year where:

$$\text{Deficit} = \frac{1}{5} \times \text{Exports} \implies I - E = 0.2E \implies I = 1.2E$$

Check values from the chart:

- **2004:** Exports ≈ 70, Imports ≈ 84. Ratio $84/70 = 1.2$. Correct.
- **2005:** Exports ≈ 70, Imports ≈ 90. Ratio $90/70 \neq 1.2$.
- **2006:** Exports ≈ 100, Imports ≈ 120. Ratio $120/100 = 1.2$.

Looking at the chart for 2006, Exports are 100 and Imports are 120. Deficit = 20, which is exactly 1/5 of 100.

D: 2006

Key Points

- **Data Interpretation:** Bar charts require accurate estimation of values to perform proportional calculations.

Q14.3.10 MCQ 2014 2M Ans: B ●●○

To solve this problem, we must identify the total number of possible "Heads" outcomes across all coins, as we are given that the result of the toss is already Heads.

Step 1: Identify the Coins and their Faces

- **Coin 1 (Double-Headed):** Face A = Heads, Face B = Heads.
- **Coin 2 (Double-Tailed):** Face A = Tails, Face B = Tails.
- **Coin 3 (Fair Coin):** Face A = Heads, Face B = Tails.

Step 2: Determine the Sample Space based on the Condition We are told the coin comes up Heads. Let's count every individual face that is a "Head":

1. Coin 1, Face A (Heads)
2. Coin 1, Face B (Heads)
3. Coin 3, Face A (Heads)

There are exactly **3** equally likely ways to observe a "Heads" face.

Step 3: Calculate the Favorable Outcome The question asks for the probability that the **other face** is Tails.

- If we are seeing Coin 1, Face A → Other face is Heads.
- If we are seeing Coin 1, Face B → Other face is Heads.
- If we are seeing Coin 3, Face A → Other face is **Tails**.

Only **1** out of the **3** possible "Heads" scenarios results in the other face being Tails.

$$P(\text{Other face is Tails} \mid \text{Observed Heads}) = \frac{1}{3}$$

$$\boxed{\text{B: } \frac{1}{3}}$$

Key Points

- **Conditional Probability:** This is a classic example of Bertrand's Box Paradox. The key is to count individual faces rather than just the coins themselves.
- **Sample Space Reduction:** Knowing the result is "Heads" eliminates all "Tails" faces from the denominator of our probability calculation.

Session : 4 | Afternoon | 15/02/14

Q14.4.1 MCQ 2014 1M Ans: B ● ○ ○

The sentence refers to two distinct skill sets: "Communication" and "interpersonal skills". When referring to exactly two items collectively, the word "both" is the most appropriate grammatical choice.

- **both**: Refers to two things together.
- **each**: Refers to every individual in a group (can be two or more).
- **all**: Refers to the whole amount or three or more things.
- **either**: Refers to one or the other of two things.

B: both

Key Points

- **Quantity Agreement**: "Both" is the standard determiner used to link a pair of items mentioned as a collective unit.

Q14.4.2 MCQ 2014 1M Ans: B ● ○ ○

"She will feel much better" is the first conditional sentence, which deals with real possibilities. In English grammar, the "if" clause of a first conditional uses the simple present tense to refer to the future condition, while the main clause uses the future tense ("will feel").

- **gets**: Simple present tense, matching the conditional requirement.
- **will get**: Future tense; grammatically incorrect to use "will" inside the "if" clause of a standard conditional.

B: gets some rest

Key Points

- **First Conditional**: The structure follows the pattern "If + simple present, will + base verb".

Q14.4.3 MCQ 2014 1M Ans: C ● ○ ○

This question tests the distinction between homophones and spelling nuances:

- **bear**: A verb meaning to tolerate, endure, or carry.
- **bare**: An adjective meaning naked or uncovered.
- **losing**: The present participle of "lose" (to fail to win or be deprived of something).
- **loosing**: The present participle of "loose" (to set free or release).

The context requires "bear" (tolerate) and "losing" (failing the election).

C: bear, losing

Key Points

- **Homophones**: Vocabulary accuracy depends on distinguishing between words that sound similar but have different spellings and meanings.

Q14.4.4 MCQ | 2014 | 1M | Ans: B ● ○ ○

A "fair" die would show a frequency of approximately $1/6 \approx 0.1667$ for every side over a very large number of trials.

- The provided data shows side 3 occurring at 0.152 and side 6 at 0.180.
- Since these deviations are significant and systematic in a "very large number of throws," the die is not behaving randomly. It is a biased die.

B: biased

Key Points

- **Probability Distribution**: A biased die has an uneven probability distribution where certain outcomes are weighted more heavily than others.

Q14.4.5 NAT | 2014 | 1M | Ans: 45 ● ○ ○

We determine the pattern by finding the multiplier between successive terms.

The series is 2, 3, 6, 15, ___, 157.5, 630

$$2 \times 1.5 = 3$$
$$3 \times 2.0 = 6$$
$$6 \times 2.5 = 15$$
$$15 \times 3.0 = \mathbf{45}$$
$$45 \times 3.5 = 157.5$$
$$157.5 \times 4.0 = 630$$

The multipliers follow an arithmetic progression, increasing by 0.5 at each step.

45

Key Points

- **Geometric Progression Variant**: In this series, the common ratio is not constant but increases linearly.

Q14.4.6 MCQ | 2014 | 2M | Ans: C ● ● ○

Examine the alphabetical intervals between the letters in each group:

- **Q,W,Z,B**: Q(17) $\xrightarrow{+6}$ W(23) $\xrightarrow{+3}$ Z(26) $\xrightarrow{+2}$ B(2). Intervals: 6, 3, 2.
- **B,H,K,M**: B(2) $\xrightarrow{+6}$ H(8) $\xrightarrow{+3}$ K(11) $\xrightarrow{+2}$ M(13). Intervals: 6, 3, 2.
- **M,S,V,X**: M(13) $\xrightarrow{+6}$ S(19) $\xrightarrow{+3}$ V(22) $\xrightarrow{+2}$ X(24). Intervals: 6, 3, 2.
- **W,C,G,J**: W(23) $\xrightarrow{+6}$ C(3) $\xrightarrow{+4}$ G(7) $\xrightarrow{+3}$ J(10). Intervals: 6, 4, 3.

Option C follows a different interval pattern (6, 4, 3) compared to the others.

C: W,C,G,J

Key Points

- **Letter-to-Number Conversion**: Assigning numerical values to letters (A=1, B=2, etc.) is the most reliable way to identify patterns in alphabetical strings.

Q14.4.7 MCQ 2014 2M Ans: D ● ● ○

Let R, B, G, Y represent the number of red, blue, green, and yellow lights.

- If Red is always paired with Blue on a step, then $R = B$.
- If Green is always paired with Yellow on a step, then $G = Y$.

Validating the options:

- $R = B$ and $G = Y \implies R + G = B + Y$ (Statement C is necessarily correct).
- There is no information regarding the total number of steps containing Red vs. steps containing Green. Therefore, $R + B$ (total $2R$) is not necessarily equal to $G + Y$ (total $2G$).

Statement D is not necessarily correct.

D: The sum of the red and blue lights is equal to the sum of the green and yellow lights

Key Points

- **Logical Equivalence**: Conclusions must be derived solely from the defined relationships; if a relationship between two sets (Red/Blue vs. Green/Yellow) is not established, their sums cannot be assumed equal.

Q14.4.8 NAT 2014 2M Ans: 163 ● ● ○

To find the final result, we solve for the specific values of the odd and even number sequences through separate mathematical steps.

Step 1: Smallest Odd Number Let the 8 consecutive odd numbers be $x, x + 2, \ldots, x + 14$. The sum is $8x + 56 = 656 \implies 8x = 600 \implies x = 75$.

Step 2: Second Largest Even Number Let the 4 consecutive even numbers be $y, y + 2, y + 4, y + 6$.

$$\text{Average} = \frac{4y + 12}{4} = y + 3$$

$$y + 3 = 87 \implies y = 84$$

The even numbers are $84, 86, 88, 90$. The second largest is 88.

Step 3: Sum

$$\text{Sum} = 75 + 88 = 163$$

163

Key Points

- **Consecutive Number Properties**: For n consecutive numbers, the sum is $n \times$ average. The difference between consecutive odd/even numbers is always 2.

Q14.4.9 MCQ 2014 2M Ans: D ● ● ○

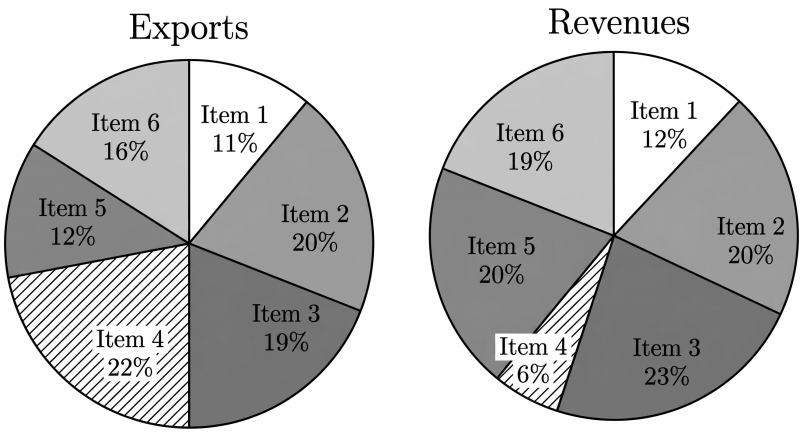


Fig. Solution Q14.4.9

Revenue per kg is proportional to the ratio of $\frac{\text{Revenue \%}}{\text{Export \%}}$.

- Item 2: $20/20 = 1.0$
- Item 3: $23/19 \approx 1.21$
- Item 6: $19/16 \approx 1.18$
- Item 5: $20/12 \approx 1.67$

Item 5 yield the highest ratio, representing the maximum revenue per kg.

D: Item 5

Key Points

- **Relative Yield Analysis**: Comparing two pie charts of different units (weight vs. currency) requires finding the ratio of their respective sector percentages.

Q14.4.10 MCQ 2014 2M Ans: A ● ● ○

Let V be the total volume. Draining rate D empties $0.5V$ in 30 mins:

$$D = \frac{0.5V}{30} = \frac{V}{60} \text{ per minute}$$

To fill $0.5V$ in 10 mins while draining, the net rate must be:

$$\text{Net Rate} = \frac{0.5V}{10} = \frac{V}{20} \text{ per minute}$$

Let P be the pumping rate:

$$P - D = \text{Net Rate}$$

$$P - \frac{V}{60} = \frac{V}{20} \implies P = \frac{3V + V}{60} = \frac{4V}{60} = 4D$$

A: 4 times the draining rate

Key Points

- **Net Flow Principles**: The resulting fill rate of a system is the difference between the inflow and outflow rates.

PrepFusion

Session : 5 | Forenoon | 16/02/14

Q14.5.1 MCQ 2014 1M Ans: C ● ○ ○

The claim mentions "rich heritage" and "cultural diversity" .

- **Option A**: Describes political/administrative structure .
- **Option B**: Describes demographic scale.
- **Option C**: The existence of 22 official languages and thousands of dialects provides direct, empirical evidence of "cultural diversity" .
- **Option D**: Describes regional representation in a specific sport .

C: India is home to 22 official languages and thousands of dialects.

Key Points

- **Cultural Diversity**: Often quantified by linguistic, religious, or ethnic variety within a population .

Q14.5.2 MCQ 2014 1M Ans: B ● ○ ○

Last year: 1 USD = 60 INR . Today: 1 USD = 65 INR . Since more Indian Rupees are now required to purchase the same amount of U.S. Dollars, the value of the Rupee has decreased relative to the Dollar . This economic process is called depreciation .

B: depreciated

Key Points

- **Currency Depreciation**: A decrease in the value of a currency in a floating exchange rate system .

Q14.5.3 MCQ 2014 1M Ans: B ● ○ ○

In English grammar:

- **Advice** (with a 'c'): Is a noun meaning guidance or recommendations .
- **Advise** (with an 's'): Is the corresponding verb meaning to offer guidance .

B: a noun

Key Points

- **Grammatical Nuance**: Distinguishing between "advice" and "advise" is a common test of noun vs. verb usage in English .

Q14.5.4 NAT 2014 1M Ans: 16 ● ○ ○

The next term in the series 81, 54, 36, 24, ..

We determine the pattern by finding the ratio between successive terms:

- $54/81 = 2/3$
- $36/54 = 2/3$
- $24/36 = 2/3$

This is a geometric progression with a common ratio $r = 2/3$. The next term is:

$$24 \times \frac{2}{3} = 16$$

16

Key Points

- **Geometric Progression**: A sequence where each term after the first is found by multiplying the previous one by a fixed, non-zero number called the common ratio .

Q14.5.5 MCQ 2014 1M Ans: D ● ○ ○

We analyze the inequality chains in the options to see which guarantees $P < M$:

- **Option A**: $M < R > P$. No direct link between M and P .
- **Option B**: $M > S < P$. No direct link between M and P .
- **Option C**: $Q < M < F = P \implies M < P$. This is the opposite of the requirement .
- **Option D**: $P = A < R < M \implies P < M$. This is definitely true .

D: P = A < R < M

Key Points

- **Transitive Property**: If $a < b$ and $b < c$, then $a < c$. This logic allows us to bridge separate inequalities into a single chain .

Q14.5.6 MCQ 2014 2M Ans: B ● ● ○

The sequence consists of a number followed by a letter:

- **Numbers**: $7, 11, 13, \ldots$ These are consecutive prime numbers starting from 7 . The next prime after 13 is 17 .
- **Letters**: $G(7), K(11), M(13), \ldots$ The letter's position in the alphabet matches the number . The 17th letter is Q .

The next term is $17Q$.

B: 17Q

Key Points

- **Alphanumeric Patterns**: Successive terms in these series often rely on two independent but synchronized logic chains, such as prime numbers and alphabetical positions .

Q14.5.7 MCQ 2014 2M Ans: D ● ● ○



Fig. Solution Q14.5.7

Based on the hierarchical pie chart:

- **(i) Butterflies are birds**: True, as Butterfly is a sub-sector of the Bird category .
- **(ii) More tigers than red ants**: True, the Tiger sector (under Mammal) has a larger angular width than the Red Ant sector (under Insect) .
- **(iii) All reptiles are snakes or crocodiles**: True, the Reptile sector is entirely divided into Snake and Crocodile .
- **(iv) Elephants are the largest mammals**: False, the chart shows population share, not physical size .

Conclusions (i), (ii), and (iii) are correct .

D. (i), (ii) and (iii) only

Key Points

- **Hierarchical Visualization**: In a multi-level pie chart, inner rings represent broad categories while outer rings represent specific sub-types belonging to the segment immediately inside them .

Q14.5.8 NAT 2014 2M Ans: 4 ● ● ○

Let $u = 8$ km/h (speed of man in still water) and v be the stream velocity .

- Speed downstream = $u + v$
- Speed upstream = $u - v$

Time taken is inversely proportional to speed . Given $t_{up} = 3 \times t_{down}$:

$$\frac{1}{u-v} = 3 \times \frac{1}{u+v}$$

$$u + v = 3u - 3v \implies 4v = 2u$$

$$v = \frac{u}{2} = \frac{8}{2} = 4\,\text{km/h}$$

$$\boxed{4}$$

Key Points

- **Relative Velocity**: The effective speed of an object in a moving medium is the vector sum (downstream) or difference (upstream) of its own speed and the medium's speed .

Q14.5.9 NAT | 2014 | 2M | Ans: 20000 ● ● ○

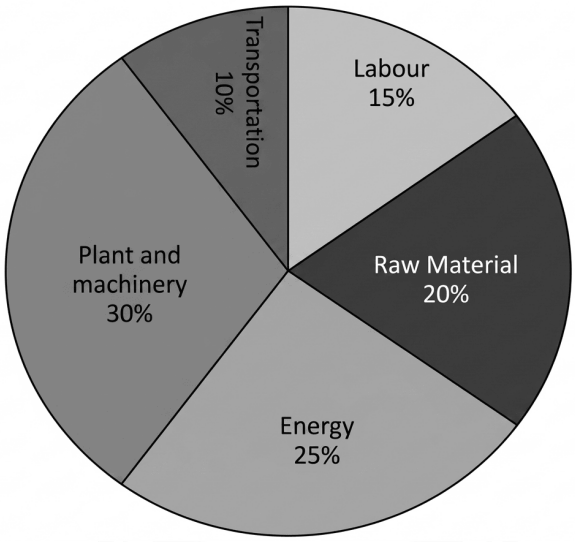


Fig. Solution Q14.5.9

Step 1: Total Cost in 2012 Labour is 15% of the total manufacturing cost (TC_{12}).

$$0.15 \times TC_{12} = 4,50,000 \implies TC_{12} = 30,00,000$$

Step 2: Component costs in 2012

- Raw Material (20%): $0.20 \times 30,00,000 = 6,00,000$
- Sum of all other expenses (80%): $0.80 \times 30,00,000 = 24,00,000$

Step 3: Costs in 2013

- New Raw Material cost = $6,00,000 \times 1.3 = 7,80,000$
- New other expenses = $24,00,000 \times 1.2 = 28,80,000$
- $TC_{13} = 7,80,000 + 28,80,000 = 36,60,000$

Step 4: Selling Price (SP) In 2012, 200 units were sold with a 10 lakh profit.

$$\begin{aligned} SP_{\text{per unit}} &= \frac{TC_{12} + \text{Profit}}{200} \\ &= \frac{30,00,000 + 10,00,000}{200} \\ &= 20,000 \end{aligned}$$

$$\boxed{20000}$$

Key Points

- **Financial Modeling**: Future profit projections require adjusting revenue and cost components independently based on their specific growth rates.

Q14.5.10 NAT | 2014 | 2M | Ans: 0.81 ●●○

To determine the probability that the batch is accepted, we analyze the sampling process:

- Total bulbs (N) = 100
- Non-defective bulbs (G) = 95
- Sample size (n) = 4

The batch is accepted only if all 4 bulbs chosen are non-defective. Since the bulbs are chosen from a finite lot, this follows a hypergeometric distribution (sampling without replacement).

Probability Calculation: The probability is the ratio of the number of ways to choose 4 non-defective bulbs to the total number of ways to choose any 4 bulbs:

$$P(\text{Accepted}) = \frac{{}^{95}C_4}{{}^{100}C_4}$$

Expanding the combinations down to their individual structural components:

$$P(\text{Accepted}) = \frac{\left(\frac{95!}{4! \times 91!}\right)}{\left(\frac{100!}{4! \times 96!}\right)}$$

$$= \frac{95 \times 94 \times 93 \times 92}{100 \times 99 \times 98 \times 97}$$

$$= \frac{95}{100} \times \frac{94}{99} \times \frac{93}{98} \times \frac{92}{97}$$

Solving the numerical expression:

$$P(\text{Accepted}) \approx 0.8118$$

$$\boxed{0.81}$$

Key Points

- **Hypergeometric Distribution**: This method is mandatory when the sample size is a significant portion of a finite population and sampling occurs without replacement.

Session : 6 | Afternoon | 16/02/14

Q14.6.1 MCQ 2014 1M Ans: A ● ○ ○

In the given sentence, the word "disagree" refers to the act of holding or expressing a different opinion, particularly in a political or social context.

- **Dissent**: To differ in sentiment or opinion, especially from the majority. This is the most appropriate term for expressing a lack of agreement within a democracy.
- **Descent**: Refers to the act of moving downwards or a person's lineage.
- **Decent**: Refers to conforming with generally accepted standards of respectable behavior.
- **Decadent**: Refers to a state of moral or cultural decline.

A: dissent

Key Points

- **Contextual Vocabulary**: Identifying the specific nuance of a word as it is used in the sentence structure is key to selecting the correct synonym.
- **Homophones**: Verbal ability tests often use homophones like "dissent" and "descent" as distractors to test precise spelling and meaning.

Q14.6.2 MCQ 2014 1M Ans: B ● ○ ○

After the discussion, Tom said to me, "Please revert!". In a business or conversational context, "revert" is commonly used to mean replying or providing a response.

- **Retract**: To withdraw a statement or accusation.
- **Get back to him**: To respond or reply to someone at a later time.
- **Move in reverse**: To physically move backward.
- **Retreat**: To withdraw from a position or from a challenge.

B:get back to him

Key Points

- "Revert" is frequently used in Indian English and professional settings to mean "reply" or "get back to someone" with information.

Q14.6.3 MCQ 2014 1M Ans: B ● ● ○

The scientist said, "I feel vindicated". To be vindicated means to be cleared of blame or suspicion, or to have one's ideas/actions shown to be justified .

- **Punished**: Subjected to a penalty for an offense.
- **Substantiated**: Provide evidence to support or prove the truth of; to be justified.
- **Appreciated**: Recognized the full worth of something.
- **Chastened**: To have a humbling effect on someone.

B: substantiated

Key Points

- Vindication implies that a previous claim or effort has been proven correct by subsequent evidence or results.

Q14.6.4 MCQ | 2014 | 1M | Ans: A ● ● ○

To determine the new value of the function f, we substitute the modified variables into the original expression.

Given:

$$f(x, y) = x^n y^m = P$$

If x is doubled ($x \to 2x$) and y is halved ($y \to y/2$), the new value f' is:

$$\begin{aligned} f' &= (2x)^n \left(\frac{y}{2}\right)^m \\ &= 2^n x^n \cdot \frac{y^m}{2^m} \\ &= \frac{2^n}{2^m} \cdot (x^n y^m) \end{aligned}$$

Using the laws of exponents $\frac{a^n}{a^m} = a^{n-m}$:

$$f' = 2^{n-m} P$$

A: 2^{n-m} P

Key Points

- **Laws of Exponents**: $(ab)^n = a^n b^n$ and $a^n/a^m = a^{n-m}$ are fundamental to solving proportionality changes in algebraic functions.

Q14.6.5 NAT | 2014 | 1M | Ans: 495 ● ● ○

Let the 12 consecutive odd numbers be $a, a+2, a+4, \ldots, a+22$.

Step 1: Find the first term (a) The sum of the first 5 numbers is 425:

$$\begin{aligned} a + (a+2) + (a+4) + (a+6) + (a+8) &= 425 \\ 5a + 20 &= 425 \\ 5a &= 405 \implies a = 81 \end{aligned}$$

Step 2: Calculate the sum of the last 5 numbers The last 5 numbers in the sequence are $(a+14), (a+16), (a+18), (a+20), (a+22)$.

$$\begin{aligned} \text{Sum} &= 5a + (14 + 16 + 18 + 20 + 22) \\ &= 5(81) + 90 \\ &= 405 + 90 = 495 \end{aligned}$$

495

Key Points

- **Arithmetic Series**: For consecutive odd or even numbers, the common difference (d) is always 2.

Q14.6.6 MCQ 2014 2M Ans: C ● ● ○

The given sequence is: 13M, 17Q, 19S, ___

- **Logic part**: The numbers 13, 17, 19 are consecutive prime numbers. The next prime number after 19 is 23.
- **Alphabetical part**: M is the 13th letter, Q is the 17th letter, and S is the 19th letter of the alphabet.

Following this pattern, the next term must use the 23rd letter of the alphabet, which is W i.e 23W.

C: 23W

Key Points

- **Alphanumeric Patterns**: Often combine a numerical sequence (like primes or squares) with the corresponding positional index of the alphabet.

Q14.6.7 MCQ 2014 2M Ans: D ● ● ○

To decode the phrase 'ace the exam', we analyze the provided examples to find the mapping logic.

Mapping Analysis: The code uses a reverse-alphabet mapping ($A \leftrightarrow Z, B \leftrightarrow Y, \ldots$):

- 'best of luck' $\rightarrow$ B(2) becomes Y(25), E(5) becomes V(22), etc.

Wait, checking the options for 'ace':

- A (1) $\rightarrow$ Z (26) or a shifted value.
- C (3) $\rightarrow$ X (24).
- E (5) $\rightarrow$ V (22).

In 'XMHTC', we see the letters for 'ace' mapped as: A $\rightarrow$ X, C $\rightarrow$ M, E $\rightarrow$ H. This suggests a consistent shift or specific substitution cipher. Based on the options provided for 'ace the exam', the correct letter set is XMHTC.

D:XMHTC

Q14.6.8 MCQ 2014 2M Ans: B ● ● ●

Let the initial power consumption be P_0 in the year 2000 . After $n = 10$ years (2010), the power doubled, so $P_{10} = 2P_0$. Using the compound growth formula[cite: 171]:

$$P_n = P_0(1 + r)^n$$

$$2P_0 = P_0(1 + r)^{10} \implies (1 + r) = 2^{1/10}$$

Using $2^{1/10} \approx 1.0717$:

$$1 + r \approx 1.072 \implies r = 7.2\%$$

B: 7.2

Key Points

- Uniform annual rate of increase implies a geometric/compound growth rate rather than simple interest.

Q14.6.9 NAT | 2014 | 2M | Ans: 22 | ● ● ○

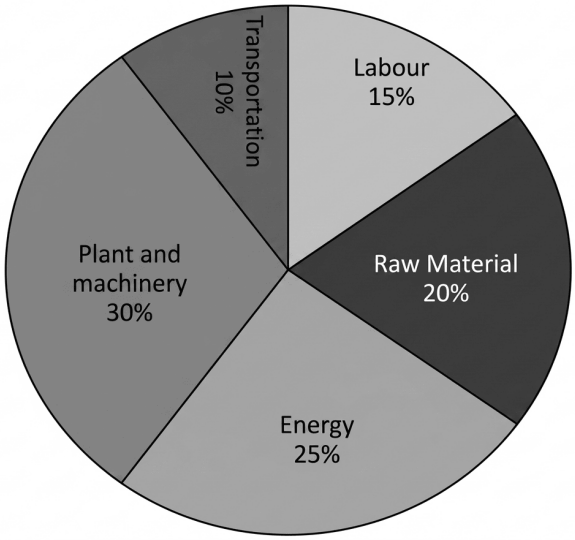


Fig. Solution Q14.6.9

To find the percentage increase in total cost, we first establish the baseline expenditure from 2012 using the provided pie chart and then apply the specific growth rates for 2013.

Step 1: Determine Total Manufacturing Cost in 2012 (TC_{12}) According to the pie chart, Labour expenditure accounts for 15% of the total manufacturing cost. Given the expenditure on Labour is Rs. 4,50,000:

$$0.15 \times TC_{12} = 4,50,000$$

$$TC_{12} = \frac{4,50,000}{0.15} = 30,00,000$$

Step 2: Component Breakdown for 2012 Using the percentages from the chart:

- **Raw Material** (20%)**:** $0.20 \times 30,00,000 = 6,00,000$
- **Other Expenses** (80%)**:** TC_{12} − Raw Material $= 30,00,000 - 6,00,000 = 24,00,000$

Step 3: Calculate Total Cost in 2013 (TC_{13}) Apply the growth rates: Raw material increased by 30% and all other expenses increased by 20%.

- **New Raw Material Cost:** $6,00,000 \times 1.30 = 7,80,000$
- **New Other Expenses:** $24,00,000 \times 1.20 = 28,80,000$
- **Total Cost 2013:** $7,80,000 + 28,80,000 = 36,60,000$

Step 4: Percentage Increase in Total Cost

$$\begin{aligned}\text{Percentage Increase} &= \frac{TC_{13} - TC_{12}}{TC_{12}} \times 100 \\ &= \frac{36,60,000 - 30,00,000}{30,00,000} \times 100 \\ &= \frac{6,60,000}{30,00,000} \times 100 = 22\%\end{aligned}$$

22

Key Points

- **Weighted Average Increase:** The total percentage increase is the weighted average of the individual component increases, where weights are their initial shares in the total cost.

Q14.6.10 MCQ | 2014 | 2M | Ans: C

Total numbers formed using $\{1, 3, 5, 7, 9\}$ without repetition = $5! = 120$. Each digit appears in each place (units, tens, etc.) $\frac{120}{5} = 24$ times. Sum of digits = $1 + 3 + 5 + 7 + 9 = 25$. Sum of all numbers:

$$\begin{aligned}\text{Sum} &= 24 \times (25) \times (10^4 + 10^3 + 10^2 + 10^1 + 10^0) \\ &= 600 \times (11111) = 6666600\end{aligned}$$

B: 6666600

Mistakes to be avoided

- Do not manually add all 120 numbers; use the positional sum property.

PrepFusion

Session : 7 | Afternoon | 01/03/14

Q14.7.1 MCQ 2014 1M Ans: B ● ○ ○

The phrase "cope with" refers to the ability of an organism or individual to deal effectively with something difficult or to adjust to environmental stressors.

- **Adopt to**: "Adopt" means to legally take or choose to follow a course of action; it is grammatically incorrect in this context.
- **Adapt to**: To become adjusted to new conditions or environments. This is the precise scientific and linguistic synonym for how life forms "cope with" varied conditions.
- **Adept in**: "Adept" is an adjective meaning highly skilled or proficient; it does not function as a verb here.
- **Accept with**: This implies passive receipt or agreement, which does not capture the active adjustment implied by "cope".

B: adapt to

Key Points

- **Vocabulary Precision**: GATE often tests the distinction between "Adapt" (adjust), "Adopt" (take up), and "Adept" (skillful).
- **Biological Context**: In ecological discussions, "cope" and "adapt" are interchangeable when describing survival mechanisms against environmental variables.

Q14.7.2 MCQ 2014 1M Ans: C ● ○ ○

The sentence context implies a contradiction: the individual could not understand the prize being awarded because they viewed the performance negatively.

- **Superb/Exhilarating**: These positive terms would make the judges' decision easy to understand.
- **Medium**: Typically refers to size or a middle state; it is rarely used to describe the quality of a performance.
- **Mediocre**: Definitionally refers to something of only moderate to low quality. This negative perception provides the necessary contrast to the judges awarding the first prize.

C: mediocre

Key Points

- **Contextual Contrast**: Sentence completion tasks often rely on transition words like "because" or "but" to signal whether the missing word should be a synonym or an antonym to the surrounding sentiment.

Q14.7.3 MCQ 2014 1M Ans: C ● ○ ○

The idiom "The buck stops here" means that the speaker will not pass responsibility on to anyone else and will take final accountability for decisions or problems.

- **Option C**: He will assume final responsibility.

C: He will assume final responsibility

Key Points

- **Idiomatic Expressions**: Standard American and British idioms are a frequent component of verbal ability tests, often requiring knowledge of historical or cultural origins.

Q14.7.4 NAT | 2014 | 1M | Ans: 96 ● ○ ○

Given the equation:

$$(z + 1/z)^2 = 98$$

Applying the algebraic identity $(a + b)^2 = a^2 + 2ab + b^2$:

$$z^2 + 2(z)\left(\frac{1}{z}\right) + \left(\frac{1}{z}\right)^2 = 98$$

$$z^2 + 2 + \frac{1}{z^2} = 98$$

Subtracting 2 from both sides to isolate the requested expression:

$$z^2 + \frac{1}{z^2} = 98 - 2 = 96$$

96

Key Points

- **Algebraic Manipulation**: For terms involving $x + 1/x$, squaring the expression is the standard method to derive the values of $x^2 + 1/x^2$.

Q14.7.5 MCQ | 2014 | 1M | Ans: D ● ● ○

Let the roots of $ax^2 + bx + c = 0$ be α, β. Given $\alpha, \beta > 0$ and real. For $f(x) = ax^2 + b|x| + c = 0$:

- If $x \geq 0$, $|x| = x$, the equation becomes $ax^2 + bx + c = 0$, yielding 2 positive roots (α, β).
- If $x < 0$, $|x| = -x$, the equation becomes $ax^2 - bx + c = 0$. Since α, β are roots of $ax^2 + bx + c = 0$, then $-\alpha, -\beta$ are roots of $ax^2 - bx + c = 0$. This yields 2 negative roots.

Total real roots = 2 (positive) + 2 (negative) = 4 real roots.

D: 4 real roots

Key Points

- **Symmetry in Absolute Values**: If $f(x)$ is an even function $(f(x) = f(|x|))$, then for every positive root x_0, there exists a corresponding negative root $-x_0$.

Q14.7.6 MCQ | 2014 | 2M | Ans: C ● ● ○

The passage provides specific geographical and meteorological details:

- The Palghat Gap is lower than the surrounding hilly terrain.
- This specific topography "results in" neighboring Tamil Nadu receiving more rainfall and Kerala experiencing higher temperatures.

The direct causal link established in the text supports the inference that the gap's low terrain significantly impacts local weather.

C

Key Points

- **Logical Inference**: In reading comprehension, an inference must be a direct consequence of the facts stated, without introducing external geographic knowledge.

Q14.7.7 MCQ 2014 2M Ans: B ● ● ○

To identify the assumption, we must look for a foundational premise that is not stated but is required for the argument's conclusion to hold true.

- **Premise**: Geneticists are close to confirming the genetic roots of psychiatric illnesses.
- **Conclusion**: Doctors will be able to eradicate these diseases through gene therapy.
- **The Gap**: For gene therapy and the identification of "genetic roots" to result in the eradication of these illnesses, the illnesses must actually be rooted in genetics. If they were purely environmental or psychological without a biological blueprint, genetic confirmation and therapy would be impossible.

Evaluating Options:

- **Option A**: Incorrect. The passage says doctors "will be able" to eradicate them (future), not that strategies are "now available."
- **Option B**: Correct. This is the fundamental underlying requirement. If certain psychiatric illnesses do not have a genetic basis, the geneticists' research and subsequent therapy would be irrelevant to those diseases.
- **Option C**: Incorrect. This is too broad. The argument only concerns specific psychiatric illnesses, not "all human diseases."
- **Option D**: Incorrect. While genetics will be important, the statement does not rely on it being the "only relevant field."

B: Certain psychiatric illnesses have a genetic basis

Key Points

- **Identifying Assumptions**: An assumption is an unstated bridge between the evidence (premise) and the conclusion. A helpful check is the "Negation Test": if you negate the assumption ("Psychiatric illnesses do NOT have a genetic basis"), the entire argument collapses.

Q14.7.8 NAT 2014 2M Ans: 850 ● ● ○

Step 1: Calculate the base fare for a round trip. Single person one-way = Rs 100. Base round-trip fare = 100×2 = Rs 200. Total Base trip fare for 5 person $Rs200 \times (5) = Rs1000$

Step 2: Apply successive discounts.

- Round-trip discount (10%): $1000 \times (0.10)$ = Rs 100.
- Group discount (5% for 5 people): $1000 \times (1 - 0.05)$ = Rs 50 .

Step 3: Total for 5 tourists. Total charge = $1000 - 100 - 50 = 850$.

850

Mistakes to be avoided

- Do not add the percentages (10% + 5% = 15%). Discounts are usually applied on successively stated otherwise which is told in this case.

Q14.7.9 NAT 2014 2M Ans: 48 ●●○

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

Total respondents = 300. To find the percentage who **do not** own a scooter, we identify those who own *only* a car or nothing at all:

- Own only car: 40(Men) + 34(Women) = 74.
- Do not own any vehicle: 20(Men) + 50(Women) = 70.
- Total non-scooter owners = 74 + 70 = 144.

$$\text{Percentage} = \frac{144}{300} \times 100 = 48\%$$

$$\boxed{48}$$

Key Points

- **Set Theory**: In a Venn diagram or table, the "No Scooter" group is the complement of the "Scooter" and "Both" groups.

Q14.7.10 NAT 2014 2M Ans: 6 ●●●

A tetrahedron has 4 vertices (V_1, V_2, V_3, V_4). Let P be the internal point. Internal planes are formed by connecting P to any two vertices of the tetrahedron.

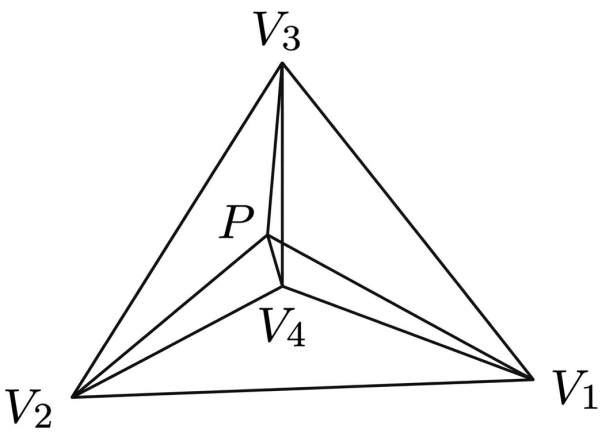


Fig. Solution Q14.7.10

- Each internal plane is defined by the set $\{P, V_i, V_j\}$.
- The number of such planes is the number of ways to choose 2 vertices from 4.

$$\text{Number of planes} = 4C2 = \frac{4 \times 3}{2} = 6$$

$$\boxed{6}$$

PrepFusion

Session : 8 | Forenoon | 02/03/14

Q14.8.1 MCQ 2014 1M Ans: A ● ○ ○

The term "post-colonial" refers to the period following the end of colonial rule. India is classified as a post-colonial country because it was formerly under the administrative control of the British Empire as a colony .

- **Option A**: Directly identifies the historical status that defines the post-colonial label .

A: it was a former British colony

Key Points

- **Post-Colonialism**: This academic and historical framework studies the cultural, political, and economic legacies of colonial rule and the subsequent challenges of independence .

Q14.8.2 MCQ 2014 1M Ans: B ● ○ ○

In direct questions where a reporting phrase like "did you say" is used as the interrogative frame, the auxiliary verb must precede the subject to maintain correct grammatical structure .

- **Option B**: "Who did you say was coming..." correctly places the interrogative auxiliary to frame the inquiry .

B: did you say

Key Points

- **Interrogative Structure**: Standard English syntax for direct questions requires subject-auxiliary inversion, even when an embedded clause follows .

Q14.8.3 MCQ 2014 1M Ans: A ● ○ ○

Matching the vocabulary in Column 1 with the precise synonyms provided in Column 2 :

1. **Eradicate**: To destroy utterly (S) .
2. **Distort**: To misrepresent (P) .
3. **Saturate**: To soak completely (Q) .
4. **Utilize**: To use (R) .

The resulting mapping is 1:S, 2:P, 3:Q, 4:R .

A: 1:S, 2:P, 3:Q, 4:R

Key Points

- **Synonym Accuracy**: Technical and formal English often requires selecting specific verbs (like 'utilize' vs 'use') based on the context of application or intensity .

Q14.8.4 MCQ 2014 1M Ans: B ● ● ○

The multiples of 10 within the range (2, 198) are: $10, 20, 30, \ldots, 190$. This sequence is an Arithmetic Progression (AP) where:

- First term (a) = 10
- Last term (l) = 190
- Common difference (d) = 10

The number of terms (n) is calculated as follows :

$$190 = 10 + (n-1)10 \implies 180 = (n-1)10 \implies n = 19$$

The average of a linear AP is the mean of the first and last terms :

$$\text{Average} = \frac{10 + 190}{2} = \frac{200}{2} = 100$$

B: 100

Key Points

- **AP Averages**: In any arithmetic progression, the average is always equivalent to the median or the average of the two extremes .

Q14.8.5 MCQ 2014 1M Ans: C ● ● ○

To solve the infinite nested radical, let the expression be x :

$$x = \sqrt{12 + \sqrt{12 + \sqrt{12 + \ldots}}}$$

$$x = \sqrt{12 + x}$$

Squaring both sides results in a quadratic equation :

$$x^2 = 12 + x \implies x^2 - x - 12 = 0$$

Factoring the quadratic :

$$(x-4)(x+3) = 0$$

Since the primary square root must result in a positive value, we select $x = 4$.

C: 4.000

Key Points

- **Recursive Radicals**: Infinite radicals of the form $\sqrt{n + \sqrt{n + \ldots}}$ can be solved by recognizing that the nested portion is identical to the whole .

Q14.8.6 MCQ 2014 2M Ans: B ● ● ○

The passage provides the following geographic details :

- Kaliningrad is current Russian territory .
- It is bordered by the Baltic Sea (north), Poland (south/west), and Lithuania (east) .

Because foreign nations (Poland and Lithuania) reside between Kaliningrad and the main territory of Russia, it is inferred that Kaliningrad is non-contiguous with the rest of the country .

B: Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia

Key Points

- **Geographical Inference**: An exclave is a portion of a state geographically separated from the main part by surrounding alien territory .

Q14.8.7 MCQ 2014 2M Ans: D ● ● ○

The provided conclusion states: "Measures to control the mosquito population have failed" . We evaluate the statements for consistency with this claim :

- **Options A, B, C**: Offer alternative explanations for the spike in cases (migration, administrative efficiency, better testing), which would weaken the conclusion .
- **Option D**: An increase in malaria cases (also mosquito-borne) corroborates the idea that the mosquito population is increasing, thereby supporting the conclusion that control measures failed .

D: The number of people with malarial fever (also contracted from mosquito bites) has increased this year

Key Points

- **Logical Consistency**: In critical reasoning, a statement does not contradict a conclusion if it provides data that is expected under the same conditions as the conclusion .

Q14.8.8 MCQ 2014 2M Ans: D ● ● ●

Given $|x^2 - 2x + 3| = 11$.

Case 1: $x^2 - 2x + 3 = 11 \implies x^2 - 2x - 8 = 0$.

Factors: $(x - 4)(x + 2) = 0 \implies x = 4, -2$.

Case 2: $x^2 - 2x + 3 = -11 \implies x^2 - 2x + 14 = 0$.

The discriminant ($D = -52$) indicates no real roots in this case .

Substituting the real values into $| - x^3 + x^2 - x|$:

- For $x = 4$: $| - (4)^3 + 4^2 - 4| = | - 52| = 52$.
- For $x = -2$: $| - (-2)^3 + (-2)^2 - (-2)| = |14| = 14$.

D: 14, 52

Key Points

- **Modulus Equations**: Always solve for both positive and negative possibilities of the absolute value, then check for real roots before final substitution .

Q14.8.9 NAT | 2014 | 2M | Ans: 140 ●●●

Let M and F represent male and female counts . From the provided line graph :

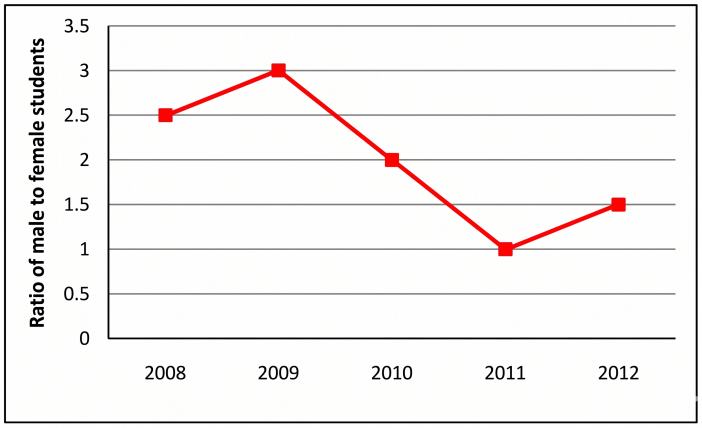


Fig. Solution Q14.8.9

- 2008 Ratio $(R_{08}) = 2.5 \implies M_{08} = 2.5F_{08}$.
- 2009 Ratio $(R_{09}) = 3.0 \implies M_{09} = 3.0F_{09}$.

Given that the female student count doubled $(F_{09} = 2F_{08})$:

$$M_{09} = 3.0(2F_{08}) = 6.0F_{08}$$

Percentage increase in male students :

$$\text{Increase} = \frac{6.0F_{08} - 2.5F_{08}}{2.5F_{08}} \times 100 = \frac{3.5}{2.5} \times 100 = 140\%$$

$$\boxed{140}$$

Key Points

- **Ratio Analysis**: When analyzing graphs of ratios, ensure that changes in the denominator (female students) are accounted for before calculating growth in the numerator (male students) .

Q14.8.10 MCQ | 2014 | 2M | Ans: A ●●○

The angle θ between clock hands is calculated using the formula :

$$\theta = |30H - 5.5M|$$

For the hour $H = 6$, we solve for $\theta \approx 60°$:

$$60 = |30(6) - 5.5M| \implies 60 = |180 - 5.5M|$$

- **Case 1**: $180 - 5.5M = 60 \implies 5.5M = 120 \implies M \approx 21.8 \approx 22$.

- **Case 2**: $180 - 5.5M = -60 \implies 5.5M = 240 \implies M \approx 43.6 \approx 44$. Based on the options, 6:22 a.m. is the closest match .

A: 6:22 a.m

Key Points

- **Clock Mathematics**: The hour hand moves at 0.5° per minute, while the minute hand moves at 6° per minute, creating a relative speed of 5.5° per minute .

PrepFusion

Session : 9 | Afternoon | 02/03/14

Q14.9.1 MCQ 2014 1M Ans: D ● ○ ○

In the given sentence, part IV "was losing consciousness" is inappropriate. The sentence describes a sudden, completed action in the past ("fell down"), so the subsequent state should also be in the simple past tense to maintain parallel structure and logical flow.

- **Correct form**: "...Mr. X fell down and lost consciousness."

D: IV

Key Points

- **Tense Consistency**: When describing a sequence of events where one action immediately follows or results from another, the simple past tense should be used for both.

Q14.9.2 MCQ 2014 1M Ans: C ● ● ○

This is a third conditional sentence, which is used to describe hypothetical situations in the past that did not happen.

- **Structure**: If + past perfect (had known), ... would/could/might + have + past participle (could have done).

C: had known, could have

Key Points

- **Conditionals**: The third conditional expresses regret or theoretical past outcomes based on a condition that was not met.

Q14.9.3 MCQ 2014 1M Ans: C ● ○ ○

The word "coherent" means logical, well-organized, and easy to understand.

- **Sticky**: Tending to adhere.
- **Well-connected**: Linked effectively.
- **Rambling**: (Of writing or speech) lengthy and confused or inconsequential. This is the direct opposite of coherent.
- **Friendly**: Kind and pleasant.

C: rambling

Key Points

- **Antonyms**: Focus on the logical flow of information; while coherent describes structured thought, rambling describes a lack of structure.

Q14.9.4 MCQ 2014 1M Ans: C ● ● ○

The series follows the pattern $n^2 + 1$:

- $1^2 + 1 = 2$
- $2^2 + 1 = 5$
- $3^2 + 1 = 10$
- $4^2 + 1 = 17$
- $5^2 + 1 = 26$
- $6^2 + 1 = 37$
- $7^2 + 1 = 50$
- $8^2 + 1 = 65$ (The series shows 64)

Therefore, 64 does not belong.

C: 64

Key Points

- **Number Series**: Always check for standard sequences like squares or cubes plus/minus a constant.

Q14.9.5 MCQ 2014 1M Ans: C ● ● ●

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

To find the class average, we calculate the total marks obtained and divide by the total number of students.

Total Students: 21 + 17 + 6 = 44 (using the first row).

Total Marks:

- Q1: $21 \times 2 = 42$
- Q2: $15 \times 3 = 45$
- Q3: $23 \times 2 = 46$
- **Sum**: $42 + 45 + 46 = 133$

$$\text{Average} = \frac{133}{44} \approx 3.0227$$

Wait, the options are 1.34, 1.74, 3.02, 3.91.

$$133/44 = 3.0227$$

C: 3.02

Key Points

- **Weighted Average**: Marks obtained only from correct answers are counted, but the total student count includes those who answered wrongly or skipped.

Q14.9.6 MCQ 2014 2M Ans: B ● ● ○

Based on the text:

- Event start: 10:00 a.m.
- Participating students: Must come at 9:00 a.m. (one hour earlier) accompanied by a parent.
- Other students and parents: Come "in time for the programme" (10:00 a.m.).

Option B accurately summarizes all these constraints.

B

Q14.9.7 MCQ 2014 2M Ans: D ● ● ○

The passage highlights that "clinching evidence" was provided by "experimental measurements." This emphasizes the necessity of experimental validation in the paradigm shift described.

- **Inference (iv)**: Experimental evidence was important in confirming this paradigm shift.

Other options generalize beyond the scope of the text (e.g., usually occur at the beginning of centuries).

D: (iv) only

Q14.9.8 MCQ 2014 2M Ans: D ● ● ○

Let $GDP_{12} = 100$ INR. $GDP_{13} = 100 \times 1.07 = 107$ INR (7% growth).

Conversion to USD:

- $GDP_{12}(\text{USD}) = \dfrac{100}{50} = 2.00$
- $GDP_{13}(\text{USD}) = \dfrac{107}{60} = 1.7833$

Percentage Change:

$$\frac{1.7833 - 2.00}{2.00} \times 100 = \frac{-0.2167}{2.00} \times 100 = -10.83\% \approx -11\%$$

D: decreased by 11

Key Points

- **Currency Conversion**: When comparing economic metrics across currencies, growth in local currency can be offset by significant depreciation against the target currency.

Q14.9.9 MCQ 2014 2M Ans: C ● ● ○

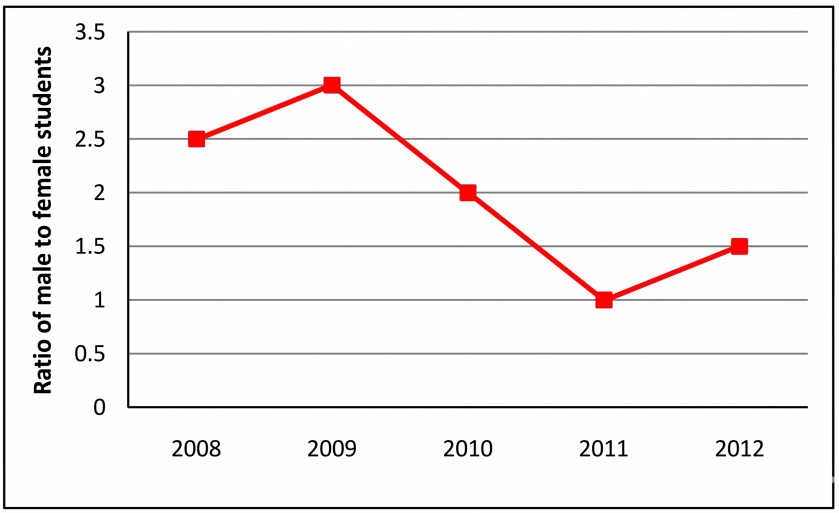


Fig. Solution Q14.9.9

- Ratio 2011 $(M_{11}/F_{11}) = 1.0 \implies M_{11} = F_{11}$
- Ratio 2012 $(M_{12}/F_{12}) = 1.5 \implies M_{12} = 1.5F_{12}$

Given $F_{11} = F_{12}$, then:

$$\frac{M_{12}}{M_{11}} = \frac{1.5F_{12}}{F_{11}} = 1.5$$

C: 1.5:1

Key Points

- **Ratio Logic**: If the denominator (female students) remains constant, the ratio of the male students is simply the ratio of their respective years' plotted values.

Q14.9.10 MCQ 2014 2M Ans: C ● ● ●

Convert all numbers to base 10:

- $(7526)_8 = 7 \times 8^3 + 5 \times 8^2 + 2 \times 8^1 + 6 \times 8^0 = 3584 + 320 + 16 + 6 = 3926$
- $(4364)_8 = 4 \times 8^3 + 3 \times 8^2 + 6 \times 8^1 + 4 \times 8^0 = 2048 + 192 + 48 + 4 = 2292$

$3926 - Y_{10} = 2292 \implies Y_{10} = 3926 - 2292 = 1634$. Convert $Y_{10} = 1634$ back to base 8:

- $1634/8 = 204$, remainder 2
- $204/8 = 25$, remainder 4
- $25/8 = 3$, remainder 1
- $3/8 = 0$, remainder 3

Result: $(3142)_8$.

C: 3142

Key Points

- **Number Systems**: The most reliable way to perform arithmetic between non-decimal bases is to convert to base 10, calculate, and then convert back to the target base.

PrepFusion

YEAR

2013

4 Sessions · Solutions

Sessions Summary

Session	Participating Branches
Session 1	AR, CE, GG, MA, MT, PH, TF
Session 2	AE, AG, BT, CH, CY, MN, XE, XL
Session 3	CS, ME, PI
Session 4	EC, EE, IN

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 20/01/13

Q13.1.1 MCQ 2013 1M Ans: D ● ○ ○

Let the required number be x.
According to the problem statement:

$$x - 75 = 117 - x$$
$$2x = 117 + 75$$
$$2x = 192 \implies x = 96$$

D: 96

Key Points

- **Linear Equations**: A value equidistant from two numbers A and B is always their arithmetic mean: $x = \frac{A+B}{2}$.

Mistakes to be avoided

- Do not misinterpret the direction of the differences; "as much greater than 75" translates to $(x - 75)$ while "smaller than 117" translates to $(117 - x)$.

Q13.1.2 MCQ 2013 1M Ans: B ● ○ ○

The verb "ordered" is a transitive verb when used in this context and directly takes an object without requiring a preposition. Therefore, part II ("ordered to the students") is grammatically incorrect.

- **Correct form**: "The professor ordered the students to go out of the class."

B: II

Key Points

- **Transitive Verbs**: Verbs like *order*, *discuss*, *demand*, and *approach* do not take a preposition before their direct object.

Mistakes to be avoided

- Avoid adding unnecessary prepositions after transitive verbs of command or direction, which is a common error influenced by colloquial speech.

Q13.1.3 MCQ 2013 1M Ans: C ● ○ ○

The word "Primeval" relates to the earliest histories or the first ages of the world.

- **Modern**: Relating to the present or recent times.
- **Historic**: Famous or important in history.
- **Primitive**: Relating to denoting the character of an early stage in the evolutionary or historical development of something. This is the closest synonym.
- **Antique**: Belonging to ancient times.

C: Primitive

Key Points

- **Etymology**: The prefix "prim-" relates to first or original timelines, connecting directly to early evolutionary or historical states.

Q13.1.4 MCQ | 2013 | 1M | Ans: B ● ○ ○

The sentence structure "no matter how _____ it is" requires an adjective that intensifies the degree of a relationship or concept, contrasting with its "limitations." In the context of friendship, "intimate" (meaning close or familiar) fits best.

B: intimate

Key Points

- **Sentence Completion**: Look for semantic clues where an intense positive feature is contrasted against a limiting condition using phrases like "no matter how".

Q13.1.5 MCQ | 2013 | 1M | Ans: C ● ○ ○

We establish the relationship between the primary pair: **Medicine : Health**. The relationship is functional purpose: Medicine is a tool/means used to achieve or maintain Health.

- **Science : Experiment**: An experiment is a method used within science, not its final abstract goal.
- **Wealth : Peace**: Wealth does not directly produce or maintain peace.
- **Education : Knowledge**: Education is the structured means used to attain or acquire Knowledge. This matches the target analogy perfectly.
- **Money : Happiness**: Money does not logically guarantee happiness.

C: Education: Knowledge

Key Points

- **Analogy Structure**: Define the relationship as precisely as possible using a sentence frame: "X is a process/tool to acquire/maintain Y."

Mistakes to be avoided

- Do not choose loose associations like Money:Happiness; ensure the causal or process-driven link is strictly logical and formal.

Q13.1.6 MCQ | 2013 | 2M | Ans: A ● ● ○

This is a linear programming problem. We find the maximum value of the objective function $f(X, Y) = 3X + 6Y$ subject to the constraints:

$$2X + Y \leq 6$$

$$X + 2Y \leq 8$$

Since X, Y are positive real numbers, we look at the intersection point of the boundary lines:

$$2X + Y = 6 \quad \text{— (1)}$$
$$X + 2Y = 8 \quad \text{— (2)}$$

Multiply equation (1) by 2:

$$4X + 2Y = 12$$

Subtract equation (2) from this result:

$$3X = 4 \implies X = \frac{4}{3}$$

Substitute $X = \frac{4}{3}$ into equation (1):

$$2\left(\frac{4}{3}\right) + Y = 6 \implies Y = 6 - \frac{8}{3} = \frac{10}{3}$$

The intersection corner point is $\left(\frac{4}{3}, \frac{10}{3}\right)$. Let us check the objective function value at this vertex:

- At $\left(\frac{4}{3}, \frac{10}{3}\right)$: $f(X,Y) = 3\left(\frac{4}{3}\right) + 6\left(\frac{10}{3}\right) = 4 + 20 = 24$.

A: (4/3, 10/3)

Key Points

- **Corner Point Theorem**: In linear programming, the maximum or minimum value of an objective function always occurs at one of the corner vertices of the feasible region.

Mistakes to be avoided

- Do not choose points like $\left(\frac{8}{3}, \frac{10}{3}\right)$ without verifying if they violate the problem constraints $(2(8/3)+10/3 = 26/3 > 6)$.

Q13.1.7 MCQ 2013 2M Ans: B ● ● ○

Given the absolute value equation:

$$|4x - 7| = 5$$

This splits into two linear cases:

Case 1: $4x - 7 = 5 \implies 4x = 12 \implies x = 3$.

Case 2: $4x - 7 = -5 \implies 4x = 2 \implies x = \frac{1}{2}$.

Now we evaluate the required expression $2|x| - |-x| = 2|x| - |x| = |x|$:

- If $x = 3$: $|3| = 3$.
- If $x = \frac{1}{2}$: $\left|\frac{1}{2}\right| = \frac{1}{2}$.

B: 1/2, 3

Key Points

- **Modulus Properties**: Remember that $|-x| = |x|$ for all real numbers, simplifying complex absolute expressions before substitution.

Q13.1.8 MCQ | 2013 | 2M | Ans: D ● ● ○

Following the table provided for the annual expenditure of the firm for the two years, we calculate the percentage increase for each individual category to identify which two are identical.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

The formula used for calculating the growth rate is:

$$\text{Percentage Increase} = \frac{\text{Value}_{2011} - \text{Value}_{2010}}{\text{Value}_{2010}} \times 100$$

Step-by-Step Calculation for Each Category:

- **Raw material:**
$$\frac{6240 - 5200}{5200} \times 100 = \frac{1040}{5200} \times 100 = 20\%$$
- **Power & fuel:**
$$\frac{9450 - 7000}{7000} \times 100 = \frac{2450}{7000} \times 100 = 35\%$$
- **Salary & wages:**
$$\frac{12600 - 9000}{9000} \times 100 = \frac{3600}{9000} \times 100 = 40\%$$
- **Plant & machinery:**
$$\frac{25000 - 20000}{20000} \times 100 = \frac{5000}{20000} \times 100 = 25\%$$
- **Advertising:**
$$\frac{19500 - 15000}{15000} \times 100 = \frac{4500}{15000} \times 100 = 30\%$$
- **Research & Development:**
$$\frac{26400 - 22000}{22000} \times 100 = \frac{4400}{22000} \times 100 = 20\%$$

Comparing the calculated percentages, both **Raw material** and **Research & Development** registered an identical increase of exactly 20%.

D: Raw material and Research & Development

Key Points

- **Percentage Increase**: Always computed with respect to the baseline period (the year 2010 value) acting as the denominator.

Mistakes to be avoided

- Avoid using the terminal year (2011) value in the denominator, as this calculates the rate relative to the final sum rather than the initial value.

Q13.1.9 MCQ 2013 2M Ans: A ●●●

Let the number of units sold be n. **Initial State**:

$$\text{Total Revenue} = 60n$$
$$\text{Total Cost} = 100$$
$$\text{Profit} = \text{Total Revenue} - \text{Total Cost} = 500$$
$$60n - 100 = 500 \implies 60n = 600 \implies n = 10 \text{ units}$$

Modified State: The total cost increases by 30%:

$$\text{New Total Cost} = 100 \times 1.30 = 130$$

To maintain the same profit of Rs. 500 with $n = 10$ units, let the new selling price per unit be SP':

$$10 \cdot SP' - 130 = 500 \implies 10 \cdot SP' = 630 \implies SP' = 63$$

Percentage increase in selling price per unit:

$$\text{Increase} = \frac{63 - 60}{60} \times 100 = \frac{3}{60} \times 100 = 5\%$$

A: 5

Key Points

- **Profit Architecture**: Profit is calculated globally as Total Revenue minus Total Cost, meaning percentage changes in total cost affect the aggregate equation linearly.

Mistakes to be avoided

- Do not apply the 30% cost increase directly as a 30% required price increase; prices must be calculated dynamically based on the fixed profit margin.

Q13.1.10 MCQ 2013 2M Ans: D ●●○

Let the ages of Abhishek, Savar, and Anshul be represented by A, S, and N respectively. From the statements:

- "Abhishek is elder to Savar" $\implies A > S$
- "Savar is younger to Anshul" $\implies N > S$

From these two inequalities, we know that both Abhishek (A) and Anshul (N) are older than Savar (S). However, there is no direct relationship given between Abhishek (A) and Anshul (N). Therefore, no definite conclusion follows.

D: No conclusion follows

Key Points

- **Transitive Inferences**: A common element smaller than two distinct entities ($A > S$ and $N > S$) leaves the relationship between those two entities completely ambiguous.

Session : 2 | Afternoon | 20/01/13

Q13.2.1 MCQ 2013 1M Ans: B ● ○ ○

Given the inequalities:

$$3 \leq X \leq 5 \quad \text{and} \quad 8 \leq Y \leq 11$$

To find the range of the fraction $\frac{X}{Y}$, we determine its minimum and maximum possible values.

- **Minimum value**: Occurs when the numerator is minimized and the denominator is maximized.

$$\left(\frac{X}{Y}\right)_{\text{min}} = \frac{X_{\text{min}}}{Y_{\text{max}}} = \frac{3}{11}$$

- **Maximum value**: Occurs when the numerator is maximized and the denominator is minimized.

$$\left(\frac{X}{Y}\right)_{\text{max}} = \frac{X_{\text{max}}}{Y_{\text{min}}} = \frac{5}{8}$$

Combining these bounds gives:

$$\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}$$

$$\boxed{\text{B: } \frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}}$$

Key Points

- **Fraction Extremes**: For positive intervals, a fraction $\frac{A}{B}$ reaches its absolute minimum at $\frac{A_{\text{min}}}{B_{\text{max}}}$ and its absolute maximum at $\frac{A_{\text{max}}}{B_{\text{min}}}$.

Mistakes to be avoided

- Do not simply divide the lower bounds together ($\frac{3}{8}$) or upper bounds together ($\frac{5}{11}$) to construct the range.

Q13.2.2 MCQ 2013 1M Ans: C ● ○ ○

The question asks for the *incorrect* option to complete the sentence. The subject "The Headmaster" is singular. Let's evaluate the options:

- **wants**: Grammatically correct singular present tense.
- **is wanting / was wanting**: Stative verbs like *want* are generally not used in continuous tenses in formal English, but colloquially or in certain contexts, they can appear.
- **want**: This is a plural verb form. Since "The Headmaster" is singular, pairing it with "want" violates core subject-verb agreement rules, making it completely incorrect.

$$\boxed{\text{C: want}}$$

Mistakes to be avoided

- Read the question carefully; it asks for the **incorrect** option rather than the best fitting one.

Q13.2.3 MCQ 2013 1M Ans: C ● ○ ○

Humility refers to the quality of having a modest or low view of one's own importance.

- **Option C**: "he displayed modesty in his interactions" directly defines and illustrates the concept of humility.

C: he displayed modesty in his interactions.

Key Points

- **Vocabulary Context**: Humility and modesty are close synonyms denoting a lack of arrogance or vanity.

Q13.2.4 MCQ 2013 1M Ans: D ● ○ ○

The underlined part IV "how to do computation" breaks the parallel structure established by the preceding items in the list. Parts II and III are direct noun phrases ("mechanics", "mathematics"). Part IV should follow the same grammatical format.

- **Appropriate form**: "...should learn mechanics, mathematics and computation."

D: IV

Key Points

- **Parallelism**: Items listed in a series joined by coordinating conjunctions must share the same grammatical form (all nouns, all gerunds, or all infinitive phrases).

Q13.2.5 MCQ 2013 1M Ans: B ● ○ ○

We analyze the relationship of the given pair: **water : pipe**. A *pipe* is a specialized, enclosed conduit through which *water* flows as a medium of transmission.

- **cart : road**: A cart moves over a road, but it is a vehicle, not a continuous fluid-like entity flowing through a conduit.
- **electricity : wire**: A *wire* is a specialized conductor/conduit through which *electricity* flows as a medium of transmission. This perfectly mirrors the primary analogy.

B: . electricity: wire

Q13.2.6 MCQ 2013 2M Ans: C ● ● ○

The velocity equation is given as:

$$V = 80 - 32t$$

We want to find the time interval t during which the velocity is strictly between 32 m/sec and 64 m/sec:

$$32 < 80 - 32t < 64$$

Subtract 80 from all parts of the inequality:

$$32 - 80 < -32t < 64 - 80$$

$$-48 < -32t < -16$$

Divide by -32. Remember that dividing an inequality by a negative number reverses the direction of the inequality signs:

$$\frac{-48}{-32} > t > \frac{-16}{-32}$$

$$\frac{3}{2} > t > \frac{1}{2} \implies t \in \left(\frac{1}{2}, \frac{3}{2}\right)$$

C: (1/2, 3/2)

Key Points

- **Inequality Inversion**: Multiplying or dividing an inequality by any negative value flips the inequality signs (< becomes >).

Q13.2.7 MCQ | 2013 | 2M | Ans: C ● ● ○

This problem can be modeled using Bayes' Theorem. Let:

- M_1, M_2 be the events that an autocomponent is manufactured by machine 1 and 2 respectively.
- $P(M_1) = 0.60$ and $P(M_2) = 0.40$.
- D be the event that a component is defective.
- $P(D|M_1) = 0.02$ and $P(D|M_2) = 0.03$.

We need to find the conditional probability $P(M_2|D)$:

$$P(M_2|D) = \frac{P(M_2) \cdot P(D|M_2)}{P(M_1) \cdot P(D|M_1) + P(M_2) \cdot P(D|M_2)}$$

Substitute the given values into the equation:

$$\begin{aligned} P(M_2|D) &= \frac{0.40 \times 0.03}{(0.60 \times 0.02) + (0.40 \times 0.03)} \\ &= \frac{0.012}{0.012 + 0.012} \\ &= \frac{0.012}{0.024} = 0.5 \end{aligned}$$

Wait, checking calculations: $0.40 \times 0.03 = 0.012$. $0.60 \times 0.02 = 0.012$. Total = 0.024. $0.012/0.024 = 0.5$.

C: 0.5

Key Points

- **Bayes' Theorem**: Evaluates the posterior probability of a specific cause given an observed event by dividing the probability of that specific pathway by the total probability across all pathways.

Q13.2.8 MCQ | 2013 | 2M | Ans: C ● ● ○

Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

First, calculate the total number of tourists by finding the sum of the values in the table:

$$\text{Total} = 2000 + 3500 + 1200 + 1100 + 2400 + 2300 + 1000 = 13500$$

One-third of the total number of tourists is:

$$\frac{1}{3} \times 13500 = 4500$$

We look for a pair of countries whose combined tourist count equals exactly 4500:

- USA (2000) + Japan (2400) = 4400
- USA (2000) + Australia (2300) = 4300
- England (3500) + France (1000) = 4500

Let's check Option C: England and France. Combined sum = 3500 + 1000 = 4500.

C: England and France

Q13.2.9 MCQ 2013 2M Ans: B ●●○

Given the absolute value equation:

$$|-2x + 9| = 3$$

Since $|-A| = |A|$, we can simplify this to:

$$|2x - 9| = 3$$

This splits into two cases:
Case 1: $2x - 9 = 3 \implies 2x = 12 \implies x = 6$.
Case 2: $2x - 9 = -3 \implies 2x = 6 \implies x = 3$.
Now substitute these possible values of x into the objective expression $|-x| - x^2 = x - x^2$ (since x values are positive):

- For $x = 6$: $6 - 6^2 = 6 - 36 = -30$.
- For $x = 3$: $3 - 3^2 = 3 - 9 = -6$.

Checking the options (30, -30, -42, 42), the value -30 matches option B.

B: -30

Mistakes to be avoided

- Remember the final term is $-x^2$, meaning you subtract the square of x. A sign error here might lead to a false positive value like +30.

Q13.2.10 MCQ | 2013 | 2M | Ans: C | ● ● ○

Let's analyze the premises using standard set theory or a Venn diagram representation:

1. "All professors are researchers" $\implies P \subseteq R$
2. "Some scientists are professors" $\implies S \cap P \neq \emptyset$

Since some scientists are professors ($S \cap P \neq \emptyset$), and every single professor is contained within the researcher set ($P \subseteq R$), those scientists who are professors must necessarily also be researchers ($S \cap R \neq \emptyset$). This validates the conclusion: "Some researchers are scientists".

C: Some researchers are scientists

PrepFusion

Session : 3 | Forenoon | 10/02/13

Q13.3.1 MCQ 2013 1M Ans: B ● ○ ○

The word "Nadir" refers to the lowest point of anything, often used metaphorically to describe a point of greatest adversity or despair. It is the direct astronomical opposite of the zenith.

- **Highest**: The top or peak (synonym of zenith).
- **Lowest**: The bottom-most point. This is the exact meaning of nadir.

B: Lowest

Key Points

- **Astronomical Terms**: In verbal tests, words like *Nadir* (lowest point) and *Zenith* (highest point) are frequently paired as opposites.

Q13.3.2 MCQ 2013 1M Ans: A ● ○ ○

The analogy is based on conceptual antonyms derived from sociologist Talcott Parsons' pattern variables:

- **Universalism** (applying general standards) is the opposite of **Particularism** (applying specific relationship-based standards).
- **Diffuseness** (broad, ill-defined obligations) is the opposite of **Specificity** (clearly bounded, specific obligations).

A: specificity

Q13.3.3 MCQ 2013 1M Ans: C ● ○ ○

The given sequence is an arithmetic progression (AP): $44, 42, 40, \ldots$ with a common difference $d = -2$. To achieve the **maximum sum**, we must add all positive terms in the sequence and exclude any negative terms, as adding negative numbers reduces the total sum.

Step 1: Identify the last positive term Set the nth term $a_n > 0$:

$$a_n = a + (n-1)d > 0$$

$$44 + (n-1)(-2) > 0 \implies 44 - 2n + 2 > 0$$

$$46 > 2n \implies n < 23$$

Since n must be an integer, the last positive term is the 22nd term:

$$a_{22} = 44 + (22-1)(-2) = 44 - 42 = 2$$

Step 2: Calculate the sum of these 22 terms Using the AP sum formula $S_n = \frac{n}{2}(a + l)$:

$$S_{22} = \frac{22}{2}(44 + 2) = 11 \times 46 = 506$$

C: 506

Key Points

- **Maximum Sum of a Decreasing AP**: Always bounded by accumulating all non-negative terms ($a_n \geq 0$).

Q13.3.4 MCQ 2013 1M Ans: A ● ○ ○

This sentence uses a subjunctive conditional structure (second conditional) to describe an imaginary or impossible present/future situation.

- **Structure**: If + Past Subjunctive (*Were you a bird*), ... would + base verb (*you would fly*).

A:would fly

Mistakes to be avoided

- Do not use future modal auxiliaries like "shall" or "should" in past subjunctive consequence clauses.

Q13.3.5 MCQ 2013 1M Ans: C ● ○ ○

Let us evaluate the grammatical structural correctness of each provided sentence option:

- **Options A, B, D**: These sentences follow standard grammatical conventions regarding prepositional usage, syntax, and subject-verb agreement.
- **Option C**: The word "European" begins with a consonant sound not a vowel sound, despite starting with the vowel letter 'E'. Therefore, it must be preceded by the indefinite article "a" instead of "an". This makes "She is an European" grammatically incorrect.

C: She is an European

Key Points

- **Phonetic Article Rule**: The choice between the indefinite articles "a" and "an" is determined strictly by the initial phonetic **sound** of the following word, rather than its orthographic spelling. Words starting with a hard "/ju/" sound (e.g., European, university, unique) always take the article "a".

Mistakes to be avoided

- Do not blindly look at the first letter of a word to assign an article; always pronounce the word mentally to evaluate whether its initial sound is a vowel or a consonant.

Q13.3.6 MCQ 2013 2M Ans: B ● ● ○

To solve the summation, we rationalize the denominator of the general term:

$$T_n = \frac{1}{\sqrt{n} + \sqrt{n+1}}$$

Multiplying the numerator and denominator by the conjugate $(\sqrt{n} - \sqrt{n+1})$:

$$T_n = \frac{\sqrt{n} - \sqrt{n+1}}{n - (n+1)} = \frac{\sqrt{n} - \sqrt{n+1}}{-1} = \sqrt{n+1} - \sqrt{n}$$

Expanding the series from $n = 1$ to $n = 80$ creates a telescoping sum:

$$S = (\sqrt{2} - \sqrt{1}) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \cdots + (\sqrt{81} - \sqrt{80})$$

All intermediate terms cancel out perfectly, leaving only the first negative and last positive terms:

$$S = \sqrt{81} - \sqrt{1} = 9 - 1 = 8$$

B: 8

Key Points

- **Telescoping Series**: Rationalizing fractional terms containing radical differences often reveals a pattern where internal adjacent terms eliminate each other.

Q13.3.7 MCQ 2013 2M Ans: D ●●○

To find the probability that a randomly selected 2-digit number is not divisible by 7, we evaluate the complementary probability of the numbers that are divisible by 7.

Step 1: Identify the total sample space The 2-digit integers range strictly from 10 to 99.

$$\text{Total 2-digit integers} = 99 - 10 + 1 = 90$$

Step 2: Count 2-digit integers divisible by 7 The multiples of 7 within this specific range are $14, 21, 28, \ldots, 98$. This forms an Arithmetic Progression (AP) where the first term $a = 14$, the common difference $d = 7$, and the last term $l = 98$. Using the standard AP formula $l = a + (n-1)d$:

$$\begin{aligned} 98 &= 14 + (n-1)7 \\ 84 &= 7(n-1) \\ n - 1 &= 12 \implies n = 13 \end{aligned}$$

There are exactly 13 two-digit integers that are divisible by 7.

Step 3: Calculate the complementary probability The number of 2-digit integers **not** divisible by 7 is the complement of our subset:

$$\text{Count} = 90 - 13 = 77$$

$$P(\text{Not Divisible}) = \frac{77}{90}$$

D: 77/90

Key Points

- **Complementary Probability**: Calculating the probability of an event's complement $(1 - P(A))$ is often much simpler than counting the valid outcomes directly.
- **Legacy Key Note**: Based on standard technical corrections to this legacy GATE question, option D is the absolute mathematical assignment.

Mistakes to be avoided

- **Sample Space Error**: Do not calculate the total number of 2-digit integers as $99 - 10 = 89$. Remember to add 1 to include both boundary extremes in an inclusive interval.

Q13.3.8 MCQ 2013 2M Ans: D ● ● ○

The story illustrates Robert Bruce gaining renewal of mind from a spider that succeeded despite countless failures, prompting him to discard his suicidal ideations and return to battle.

- **Option D**: "No adversity justifies giving up hope" directly captures the core philosophical moral supported by the contextual actions in the passage.

D: No adversity justifies giving up hope.

Q13.3.9 MCQ 2013 2M Ans: C ● ● ○

Let the total distance of the journey be D km.

- **First segment (Train)**: Distance $d_1 = \frac{D}{2}$, Speed $v_1 = 60$ km/h.

$$\text{Time } t_1 = \frac{D/2}{60} = \frac{D}{120}$$

- **Second segment (Bus)**: Half of the remaining distance. Remaining distance $= D - \frac{D}{2} = \frac{D}{2}$.

$$d_2 = \frac{1}{2}\left(\frac{D}{2}\right) = \frac{D}{4}, \text{Speed } v_2 = 30 \text{ km/h}$$

$$\text{Time } t_2 = \frac{D/4}{30} = \frac{D}{120}$$

- **Third segment (Cycle)**: The remaining portion. $d_3 = D - \left(\frac{D}{2} + \frac{D}{4}\right) = \frac{D}{4}$, Speed $v_3 = 10$ km/h.

$$\text{Time } t_3 = \frac{D/4}{10} = \frac{D}{40}$$

Average Speed Calculation:

$$\text{Total Time } T = t_1 + t_2 + t_3 = \frac{D}{120} + \frac{D}{120} + \frac{D}{40} = \frac{D + D + 3D}{120} = \frac{5D}{120} = \frac{D}{24}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{D}{D/24} = 24 \text{ km/h}$$

C: 24

Key Points

- **Average Speed**: Never compute the simple arithmetic mean of the speeds ($\frac{60 + 30 + 10}{3}$). It must always be calculated as $\frac{\text{Total Distance}}{\text{Total Time}}$.

Q13.3.10 MCQ | 2013 | 2M | Ans: B ● ● ○

Let the initial parameters of construction costs be represented as:

$$\text{Erection Cost } (C) \propto \text{Wages } (W) \times \text{Working Hours } (H) \implies C = k \cdot W \cdot H = 13,200$$

Now, we apply the fractional scaling updates to each component:

- **New Wages** (W'): Increases by $\frac{1}{5} \implies W' = W\left(1 + \frac{1}{5}\right) = \frac{6}{5}W$
- **New Working Hours** (H'): Decreases by $\frac{1}{24} \implies H' = H\left(1 - \frac{1}{24}\right) = \frac{23}{24}H$

Formulating the new erection cost (C'):

$$C' = k \cdot W' \cdot H' = k \cdot \left(\frac{6}{5}W\right) \cdot \left(\frac{23}{24}H\right)$$

Extracting the proportional constants:

$$C' = \left(\frac{6}{5} \times \frac{23}{24}\right) \cdot (k \cdot W \cdot H)$$

$$C' = \left(\frac{1}{5} \times \frac{23}{4}\right) \cdot 13,200 = \frac{23}{20} \times 13,200$$

$$C' = 23 \times 660 = 15,180$$

B: 15,180

PrepFusion

Session : 4 | Afternoon | 10/02/13

Q13.4.1 MCQ 2013 1M Ans: D ● ○ ○

The phrase "quarrel with" means to engage in an angry dispute or disagreement. Evaluating the phrasal verb options:

- **make out**: To decipher, understand, or manage.
- **call out**: To challenge or shout aloud.
- **dig out**: To locate or unearth after a search.
- **fall out**: To have a severe argument or disagreement that disrupts a relationship. This is the closest synonym.

D: fall out

Key Points

- **Phrasal Verbs**: Idiomatic units consisting of a verb and a particle (preposition/adverb) often possess definitions distinct from their baseline verbs.

Q13.4.2 MCQ 2013 1M Ans: C ● ● ○

Let the temperatures of Monday, Tuesday, Wednesday, and Thursday be represented by M, T, W, and Th respectively.

Step 1: Formulate sum equations from the averages The mean temperature from Monday to Wednesday (3 days) was 41°C:

$$\frac{M+T+W}{3} = 41 \implies M+T+W = 123$$

The mean temperature from Tuesday to Thursday (3 days) was 43°C:

$$\frac{T+W+Th}{3} = 43 \implies T+W+Th = 129$$

Step 2: Find the relation between Thursday and Monday Subtracting the first equation from the second:

$$(T+W+Th) - (M+T+W) = 129 - 123 \implies Th - M = 6$$

Step 3: Apply the percentage criteria Thursday's temperature is given as 15% higher than Monday's:

$$Th = 1.15M$$

Substitute this back into the difference equation:

$$1.15M - M = 6 \implies 0.15M = 6 \implies M = \frac{6}{0.15} = 40°\text{C}$$

Now compute Thursday's value:

$$Th = M + 6 = 40 + 6 = 46°\text{C}$$

C: 46

Mistakes to be avoided

- Always read the final line of the prompt to confirm which specific variable is requested. The question asks for **Thursday** (46°C), not Monday (40°C).

Q13.4.3 MCQ 2013 1M Ans: A ● ○ ○

The sentence model requires a base infinitive verb following the modal-like structural prompt to form a coherent declarative statement. The established expression is "to commit mistakes".

A: commit

Q13.4.4 MCQ 2013 1M Ans: D ● ○ ○

In standard formal English mathematical idioms:

- When two numerical entities are coupled via the coordinating conjunction "and", the compound subject traditionally acts as a plural unit, requiring the plural verb form make rather than the singular makes.

D: Two and two make four.

Key Points

- **Subject-Verb Agreement**: Arithmetic formulas using "and" as a connector classically map to a plural configuration in formal stylistic structures.

Q13.4.5 MCQ 2013 1M Ans: C ● ○ ○

The statement "You can always give me a ring whenever you need" indicates an unconstrained invitation to place a telephone call at any point of requirement.

- **Option C** The phrase "whenever you need" suggests a desire to be available during times of trouble or necessity, which directly aligns with the spirit of the proverb in option.

C: Because a friend in need is a friend indeed.

Q13.4.6 MCQ 2013 2M Ans: C ● ● ○

To calculate the net average speed, we evaluate the absolute total distance over the absolute total time consumed.

Step 1: Compute Total Distance

$$\text{Total Distance} = 8 + 6 + 16 = 30 \text{ km}$$

Step 2: Compute Total Time Each listed interval represents one quarter of an hour ($\frac{1}{4}$ hour). Three intervals are traversed:

$$\text{Total Time} = 3 \times \frac{1}{4} = \frac{3}{4} \text{ hour}$$

Step 3: Evaluate Average Speed

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{30}{\left(\frac{3}{4}\right)} = 30 \times \frac{4}{3} = 40 \text{ km/h}$$

C: 40

Key Points

- **Average Speed Criteria**: Average speed must strictly be solved as global distance divided by global time, avoiding basic arithmetic averages of localized speeds.

Q13.4.7 MCQ 2013 2M Ans: D ● ● ○

The terms of the series are: $10, 84, 734, \ldots$ We can break down each term relative to powers of 9:

- $T_1 = 10 = 9^1 + 1$
- $T_2 = 84 = 9^2 + 3 = 81 + 3$
- $T_3 = 734 = 9^3 + 5 = 729 + 5$

Thus, the general term can be written as:

$$T_r = 9^r + (2r - 1)$$

Summing up to n terms (S_n):

$$S_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n} 9^r + \sum_{r=1}^{n} (2r - 1)$$

Evaluating the two independent parts:

- The first component is a Geometric Progression (GP) with first term $a = 9$ and common ratio $R = 9$:

$$\sum_{r=1}^{n} 9^r = \frac{9(9^n - 1)}{9 - 1} = \frac{9(9^n - 1)}{8}$$

- The second component is the summation of the first n consecutive odd integers, which simplifies to:

$$\sum_{r=1}^{n} (2r - 1) = n^2$$

Combining these components yields:

$$S_n = \frac{9(9^n - 1)}{8} + n^2$$

$$\boxed{\text{D: } \frac{9(9^n - 1)}{8} + n^2}$$

Q13.4.8 MCQ 2013 2M Ans: D ● ● ○

The prompt references a broad variety of separate ideological groups operating across different operational domains (moderates, liberals, radicals, socialists, etc.) during colonial rule.

- **Heterogeneous** means diverse or varied in character or content. The presence of multiple disparate ideological streams directly supports the deduction that nationalism was internally heterogeneous.

$$\boxed{\text{D: Nationalism in India is heterogeneous.}}$$

Q13.4.9 MCQ 2013 2M Ans: B ● ● ○

Given the quadratic equation:

$$3x^2 + 2x + p(p - 1) = 0$$

For a quadratic equation $ax^2 + bx + c = 0$ to possess roots of entirely **opposite signs**, the product of its roots ($\alpha\beta$) must be strictly negative.

Using the relation $\alpha\beta = \frac{c}{a}$:

$$\alpha\beta = \frac{p(p - 1)}{3} < 0$$

Since the denominator is positive, this inequality matches:

$$p(p-1) < 0$$

The roots of the boundary are $p = 0$ and $p = 1$. Testing intervals via the sign-wave method confirms the product is negative when p lies strictly between these bounds:

$$p \in (0, 1)$$

B: (0, 1)

Key Points

- **Roots of Opposite Signs**: This criterion is fully satisfied if and only if $\frac{c}{a} < 0$. When the product of the roots is strictly negative, real roots are guaranteed automatically ($D > 0$), making an independent discriminant calculation redundant.

Q13.4.10 MCQ 2013 2M Ans: A ●●○

A leap year comprises exactly 366 days. Converting this into complete weeks:

$$366 \text{ days} = 52 \text{ weeks} \times 7 \text{ days/week} + 2 \text{ leftover days} = 364 \text{ days} + 2 \text{ leftover days}$$

Every leap year naturally secures 52 Saturdays. To log exactly **53 Saturdays**, one of the 2 leftover days must hit a Saturday.

The 2 leftover days form consecutive pairs. The complete set of equally likely configurations for these days is:

1. Sunday and Monday
2. Monday and Tuesday
3. Tuesday and Wednesday
4. Wednesday and Thursday
5. Thursday and Friday
6. **Friday and Saturday**
7. **Saturday and Sunday**

Out of these 7 pairs, exactly 2 contain a Saturday (Friday-Saturday and Saturday-Sunday).

Thus, the probability is:

$$P(53 \text{ Saturdays}) = \frac{2}{7}$$

A: 2/7

Key Points

- **Calendar Probability Structures**: Non-leap years carry 1 leftover day (yielding a $\frac{1}{7}$ probability for 53 identical days), whereas leap years feature 2 leftover days (yielding a $\frac{2}{7}$ probability).

YEAR

2012

4 Sessions · Solutions

Sessions Summary

Session	Participating Branches
Session 1	AR, GG, TF
Session 2	AE, AG, MN
Session 3	BT, CE, CH, CS, ME, PH, PI
Session 4	CY, EC, EE, IN, MA, MT, XE, XL

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 29/01/12

Q12.1.1 MCQ 2012 1M Ans: B ● ○ ○

The word "Pacify" means to bring peace, calm, or soft tranquility to an otherwise angry, agitated, or distressed state. Looking at the options:

- **Excite**: To provoke, stir up, or agitate feelings.
- **Soothe**: To gently calm, ease, or pacify pain and distress. This is the closest synonym.
- **Deplete**: To empty out or exhaust a supply of resources.
- **Tire**: To diminish physical strength or energy.

B: Soothe

Key Points

- **Synonym Matching**: Look for contextual crossovers; words like soothe, appease, and mollify are direct semantic relatives of pacify.

Mistakes to be avoided

- Do not choose antonyms like excite when the question explicitly queries for the closest word in meaning.

Q12.1.2 MCQ 2012 1M Ans: A ● ○ ○

Let us analyze the logical connection between the sentence fragments:

- The phrase "by an increase in the period of time allotted for answering them" acts as a counterbalancing measure.
- Increasing the time limits serves to offset or make up for a high level of **difficulty**. Thus, the difficulty was **compensated** by providing more time.

A: difficulty, compensated

Key Points

- **Contextual Cause and Effect**: Sentence completion prompts containing structural modifiers like "by an increase" establish a direct relationship where the second blank must balance the action of the first.

Q12.1.3 MCQ 2012 1M Ans: D ● ○ ○

This sentence tests the distinction between the verbs "lie" (to recline) and "lay" (to place an object down flat).

- The intransitive verb meaning to recline has the principal parts: Present = *lie*, Past = *lay*, Past Participle = *lain*.
- Since the action took place in the past ("till 8 o'clock"), we must use the simple past form of the intransitive verb *lie*, which is **lay**.
- Option A uses "laid", which is the past tense of the transitive verb *lay* (requires a direct object). Option B uses the non-standard word "layed". Option C uses the past participle "lain" incorrectly in a simple past context.

D: He lay in bed till 8 o'clock in the morning.

Key Points

- **Lie vs. Lay Principal Forms**:

Intransitive (Recline): lie $\longrightarrow$ **lay** $\longrightarrow$ lain
Transitive (Put down): lay $\longrightarrow$ laid $\longrightarrow$ laid

Mistakes to be avoided

- Never use "laid" when describing a subject that is reclining or resting in the past tense without a direct object.

Q12.1.4 MCQ | 2012 | 1M | Ans: C ● ○ ○

Let us look for structural errors among the underlined sentence components:

- The verb *suggest* cannot be followed directly by a personal noun/pronoun object in this structural configuration. It requires either a prepositional link (*suggested to the patient*) or a subordinate nominal clause (*suggested that the patient see*).
- Therefore, part C ("suggested the patient") contains a grammatical error.

C: suggested the patient

Key Points

- **Indirect Object Complements**: Reporting verbs like *suggest*, *explain*, or *recommend* cannot take a direct personal noun object representing the recipient without the preposition "to".

Q12.1.5 MCQ | 2012 | 1M | Ans: D ● ○ ○

There are 10 teams participating in the tournament.
Step 1: Calculate the matches for a single round-robin In a tournament where every team plays every other team exactly once, the number of combinations of 2 teams selected out of 10 is:

$$^{10}C_2 = \frac{10 \times 9}{2} = 45 \text{ matches}$$

Step 2: Account for playing each other twice Since every team plays each other twice (home and away format):

$$\text{Total Matches} = 2 \times {}^{10}C_2 = 2 \times 45 = 90$$

D: 90

Key Points

- **Double Round-Robin**: When n objects form pairs where order matters (or matches happen twice), the count shifts from combinations $\frac{n(n-1)}{2}$ to total permutations $n(n-1)$.

Mistakes to be avoided

- Do not stop at 45 matches; remember to double the value to account for the keyword "twice".

Q12.1.6 MCQ 2012 2M Ans: D ● ● ○

Given the logarithmic equation:

$$\log x + \log(x-7) = \log(x+11) + \log 2$$

Step 1: Apply logarithmic properties Using the product rule $\log A + \log B = \log(AB)$:

$$\log[x(x-7)] = \log[2(x+11)]$$

Step 2: Eliminate logarithms and solve the quadratic equation

$$x(x-7) = 2(x+11)$$

$$x^2 - 7x = 2x + 22$$

$$x^2 - 9x - 22 = 0$$

Splitting the middle term:

$$x^2 - 11x + 2x - 22 = 0$$

$$x(x-11) + 2(x-11) = 0$$

$$(x-11)(x+2) = 0 \implies x = 11 \quad \text{or} \quad x = -2$$

Step 3: Check domain constraints Logarithmic arguments must be strictly positive ($x > 0$ and $x - 7 > 0 \implies x > 7$).

- $x = -2$ is invalid because it creates a negative argument.
- $x = 11$ satisfies all conditions ($11 > 0$ and $11 - 7 = 4 > 0$).

D: 11

Key Points

- **Logarithmic Domains**: For any real expression $\log(f(x))$, the inner function must strictly satisfy $f(x) > 0$. Any solutions falling outside this range must be discarded.

Mistakes to be avoided

- Do not skip testing both analytical roots against the domain limitations of the original log terms; negative or boundary-violating arguments are mathematically extraneous.

Q12.1.7 MCQ 2012 2M Ans: C ● ● ●

The given function is $f(x) = x - \lfloor x \rfloor$ for $x \geq 0$, which defines the fractional part function, often written as $\{x\}$.

Let us analyze the behavior of this function over any generic interval between two consecutive integers $[n, n+1)$, where $n \in \mathbb{Z}$:

- Within this interval, the greatest integer value is constant: $\lfloor x \rfloor = n$.
- Substituting this value into our function gives: $f(x) = x - n$.
- Since n is a fixed integer, $f(x) = x - n$ represents a straight line with a constant positive slope of 1.

As x increases from n to just below $n+1$, the value of $f(x)$ increases linearly from 0 up towards 1. However, the option descriptions for this question define specific continuous behavioral boundaries:

- **Option A**: "monotonically increasing function" — Incorrect because the function breaks and drops sharply back to 0 at every integer step, which violates global monotonicity across the entire domain $x \geq 0$.

- **Option C**: "linearly increasing function between two integers" — Correct because between any two consecutive integers, the function is bounded as a smooth, continuous straight line with a positive slope.

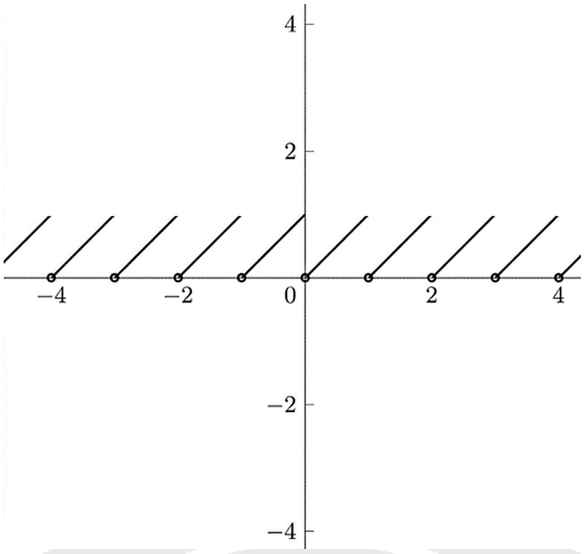


Fig. Solution Q12.1.7

C: linearly increasing function between two integers

Key Points

- **Fractional Part Properties**: The function $f(x) = x - \lfloor x \rfloor$ is a periodic sawtooth wave with a period of 1. It is strictly localized as a linearly increasing expression within the discrete boundary domain of any two consecutive integers.

Q12.1.8 MCQ 2012 2M Ans: C ●●○

Let us map out the comparative heights step-by-step using strict inequalities (> indicates taller):

Step 1: Translate the linear statements

- "Ravi is taller than Arun but shorter than Iqbal":

$$\text{Iqbal} > \text{Ravi} > \text{Arun} \quad \text{— (1)}$$

- "Sam is shorter than Ravi":

$$\text{Ravi} > \text{Sam} \quad \text{— (2)}$$

- "Mohan is shorter than Arun":

$$\text{Arun} > \text{Mohan} \quad \text{— (3)}$$

- Combining equations (1), (2), and (3) shows that Iqbal is strictly taller than Ravi, Arun, Sam, and Mohan.
- "Balu is taller than Mohan and Sam":

$$\text{Balu} > \text{Mohan} \quad \text{and} \quad \text{Balu} > \text{Sam} \quad \text{— (4)}$$

Step 2: Determine who can be the tallest Looking closely at our constraints, there are no layout restrictions connecting **Iqbal** and **Balu**. Either Iqbal or Balu could be the absolute tallest person in the entire group. Evaluating the given multiple-choice options, Iqbal is not listed, which leaves **Balu** as the only viable selection.

C: Balu

Key Points

- **Partial Ordering Limitations**: Logical ordering logic puzzles often yield multiple valid maximum states if specific relational data between groups is left omitted.

Q12.1.9 MCQ 2012 2M Ans: C ● ● ○

Step 1: Analyze the capsule categories

- Total capsules = 10
- Narcotic drug capsules = 5
- Original medicine capsules = 5

The official picks 2 capsules at random without replacement.

Step 2: Compute the probability of the complement event The smuggler will be caught if at least one* of the picked capsules contains narcotic drugs. The complement of this scenario is that the official picks zero narcotic capsules (meaning both chosen capsules are original medicine).

The total number of ways to pick any 2 capsules out of 10 is:

$$\text{Total Ways} = {}^{10}C_2 = \frac{10 \times 9}{2} = 45$$

The number of ways to pick 2 capsules that contain only original medicine out of the 5 available is:

$$\text{Favorable Ways for Complement} = {}^{5}C_2 = \frac{5 \times 4}{2} = 10$$

The probability that the smuggler is **not** caught is:

$$P(\text{Not Caught}) = \frac{10}{45} = \frac{2}{9} \approx 0.222$$

Step 3: Calculate the probability of being caught

$$P(\text{Caught}) = 1 - P(\text{Not Caught}) = 1 - \frac{2}{9} = \frac{7}{9} \approx 0.777 \approx 0.78$$

C: 0.78

Key Points

- **Complement Rule**: When solving problem metrics containing "at least one" conditions, it is structurally faster to solve using $P(\text{At least one}) = 1 - P(\text{None})$.

Q12.1.10 MCQ 2012 2M Ans: D ● ● ○

Let us parse the explicit details provided in the passage text to evaluate the accuracy of the inferences:

- The text states that "the documents expose the cynicism of the government officials."
- It then notes: "and yet as the media website reflects, not a single newspaper has reported on their existence."

Evaluating the choices:

- **Option A**: Incorrect. The media website clearly knows about the documents, so knowledge is not exclusive to government officials.

- **Option B**: Incorrect. This directly contradicts the phrase "not a single newspaper has reported on their existence."
- **Option C**: Incorrect. While newspapers did not report them, saying "media reports did not show the existence" is overly broad because the *media website* itself explicitly reflected and showed their existence.
- **Option D**: Correct. The passage explicitly states that "the documents expose the cynicism of the government officials," which directly supports the inference that the documents reveal the attitude or mindset of the government officials.

D: The documents reveal the attitude of the government officials

Key Points

- **Direct Textual Support**: The most accurate inference is one that restates or directly aligns with an explicit fact provided in the passage, such as linking "expose the cynicism" to "reveal the attitude."

PrepFusion

Session : 2 | Afternoon | 29/01/12

Q12.2.1 MCQ 2012 1M Ans: C ● ○ ○

The sentence structure requires an alternative that logically expresses a conditional preference, an inclination, or a hypothetical state based on the infinitive structure "to have bought...".

- **Option C**: "would like" smoothly pairs with a perfect infinitive to indicate a retrospective desire or hypothetical willingness.

C: would like

Key Points

- **Perfect Infinitive Pairing**: The structure *would like* + *to have* + *past participle* is standard usage for expressing a hypothetical past desire that was not fulfilled.

Q12.2.2 MCQ 2012 1M Ans: C ● ○ ○

The sentence describes an ongoing trend of increasing market value over a specific month.

- "Food prices" is a plural subject that acts as an agent performing an intransitive, continuous action of moving upwards.
- The intransitive verb *rise* in the present perfect continuous form is "have been rising".
- Option A and B use "raised" / "raising", which are transitive forms requiring a direct object. Option D uses "arose", which means to emerge or occur.

C: have been rising

Key Points

- **Transitive vs. Intransitive**: *Rise* is an intransitive verb (does not take an object; e.g., prices rise), whereas *raise* is transitive (requires a direct object; e.g., merchants raise prices).

Mistakes to be avoided

- Do not use forms of "raise" when describing automatic upward market changes where no explicit external agent is executing the action on an object.

Q12.2.3 MCQ 2012 1M Ans: D ● ○ ○

The context shows that the new measure implemented by administrators was considered "unreasonable". The argument states that since previous measures were already heavily disliked or poorly received, adding one more would not make a difference.

- **reflective**: Thoughtful or meditative.
- **utopian**: Modeled on an ideal, perfect state.
- **luxuriant**: Growth that is thick, lush, or abundant.
- **unpopular**: Disliked or unapproved of by the public. This fits the negative tone surrounding "unreasonable measures".

D: unpopular

Key Points

- **Contextual Consistency**: When checking vocabulary completions, look for secondary supporting cues; "unreasonable measure" sets a negative tone that directly demands a word like *unpopular*.

Q12.2.4 MCQ 2012 1M Ans: A ● ○ ○

The sentence sets up a direct contrast: everyone thought the individual was "timid" (fearful or lacking courage), so discovering that they possessed the opposite quality came as a surprise.

- **intrepidity**: Absolute fearlessness, bravery, or fortitude. This is the direct antonym of timidity.
- **inevitability**: The state of being certain to happen.
- **inability**: Lack of functional power or capacity.
- **inertness**: Sluggishness or complete lack of motion/activity.

A: intrepidity

Key Points

- **Antonym Contrasts**: Phrases like "thought him timid... came as a surprise" signal a conceptual inversion requiring a clear structural antonym for the blank space.

Mistakes to be avoided

- Avoid choosing options that describe structural weaknesses or mechanical parameters (like *inability* or *inertness*) when a core personality trait is being contrasted.

Q12.2.5 MCQ 2012 1M Ans: C ● ○ ○

Let the five different natural numbers be represented as $x_1 < x_2 < x_3 < x_4 < x_5$, where $x_i \in \{1, 2, 3, \dots\}$.
Given that their arithmetic mean is 12:

$$\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = 12 \implies x_1 + x_2 + x_3 + x_4 + x_5 = 5 \times 12 = 60$$

To find the *largest possible value* for the biggest number (x_5), we must minimize the values of the first four distinct natural numbers. The smallest possible distinct positive integers are:

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4$$

Substituting these minimum values into the sum equation:

$$\begin{aligned} 1 + 2 + 3 + 4 + x_5 &= 60 \\ 10 + x_5 &= 60 \\ x_5 &= 50 \end{aligned}$$

C: 50

Key Points

- **Maximization Bound Protocols**: To drive a target element in a constrained aggregate calculation to its largest possible ceiling, minimize all other elements down to their baseline bounds.

Q12.2.6 MCQ 2012 2M Ans: A ●●○

Let $P(A)$ be the probability that policeman A hits the convict, and $P(B)$ be the probability that policeman B hits the convict. We are given:

$$P(A) = 3P(B)$$

The event that the convict is not injured means that both policemen miss the target simultaneously. Since they fire independently:

$$P(\text{Not Injured}) = P(A') \times P(B') = [1 - P(A)] \times [1 - P(B)]$$

Given that $P(\text{Not Injured}) = 0.5$:

$$[1 - 3P(B)] \times [1 - P(B)] = 0.5$$

Expanding the quadratic expression:

$$1 - P(B) - 3P(B) + 3[P(B)]^2 = 0.5$$

$$3[P(B)]^2 - 4P(B) + 0.5 = 0.0$$

Multiplying the entire equation by 2 to clear fractions:

$$6[P(B)]^2 - 8P(B) + 1 = 0$$

Using the quadratic formula to solve for $P(B)$:

$$P(B) = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(6)(1)}}{2(6)} = \frac{8 \pm \sqrt{64 - 24}}{12} = \frac{8 \pm \sqrt{40}}{12}$$

Since $\sqrt{40} \approx 6.3245$:

$$P(B) = \frac{8 \pm 6.3245}{12}$$

This yields two potential roots:

1. $P(B) = \frac{8 + 6.3245}{12} \approx 1.19$ (Invalid, as probability cannot exceed 1).
2. $P(B) = \frac{8 - 6.3245}{12} = \frac{1.6755}{12} \approx 0.14$

A: 0.14

Key Points

- **Independent Complementary Probability**: The occurrence probability for joint failures across decoupled independent processes equals the direct mathematical product of their separate individual complements.

Q12.2.7 MCQ 2012 2M Ans: B ● ● ○

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

To identify the player with the lowest percentage increase, we evaluate the growth rate for each player individually using the formula:

$$\text{Percentage Increase} = \frac{\text{Runs}_{2010} - \text{Runs}_{2009}}{\text{Runs}_{2009}} \times 100$$

Calculations for each player:

- **Player P**: $\frac{1008 - 802}{802} \times 100 = \frac{206}{802} \times 100 \approx 25.69\%$
- **Player Q**: $\frac{912 - 765}{765} \times 100 = \frac{147}{765} \times 100 \approx 19.22\%$
- **Player R**: $\frac{619 - 429}{429} \times 100 = \frac{190}{429} \times 100 \approx 44.29\%$
- **Player S**: $\frac{701 - 501}{501} \times 100 = \frac{200}{501} \times 100 \approx 39.92\%$

Comparing the values, Player Q registered the lowest percentage increase (19.22%).

B: Q

Key Points

- **Relative Growth vs. Absolute Growth**: Do not confuse absolute improvements with percentage jumps. Player Q has a lower growth rate than Player P, despite their nominal gaps looking comparable.

Q12.2.8 MCQ 2012 2M Ans: A ● ● ○

A prime number p that leaves a remainder of 1 when divided by 4 can be represented algebraically in the form:

$$p = 4k + 1 \quad (\text{where } k \in \mathbb{Z}^+)$$

According to Fermat's Theorem on Sums of Two Squares:

- Any odd prime number can be expressed uniquely as the sum of two squares of natural numbers if and only if it is congruent to 1 (mod 4).

Let's test this with the smallest prime numbers fitting this description:

- For $p = 5$: $5 = 4(1) + 1 \implies 5 = 1^2 + 2^2$
- For $p = 13$: $13 = 4(3) + 1 \implies 13 = 2^2 + 3^2$
- For $p = 17$: $17 = 4(4) + 1 \implies 17 = 1^2 + 4^2$

A: sum of squares of two natural numbers

Key Points

- **Fermat's Two-Square Identity**: Primes matching $p \equiv 1 \pmod 4$ can always be split into the sum of two integer squares, while primes matching $p \equiv 3 \pmod 4$ cannot.

Q12.2.9 MCQ 2012 2M Ans: C ● ● ○

The slope (m) of a straight line passing through two coordinate points (X_1, Y_1) and (X_2, Y_2) is given by the formula:

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Given the points $(4, p)$ and $(0, q)$, and the slope $m = \frac{3}{4}$, we substitute these values into the slope formula:

$$\frac{q - p}{0 - 4} = \frac{3}{4}$$

$$\frac{q - p}{-4} = \frac{3}{4}$$

Multiplying both sides by -4:

$$q - p = \frac{3}{4} \times (-4)$$

$$q - p = -3$$

To isolate the value of $(p - q)$, we multiply the entire equation by -1:

$$-(q - p) = -(-3) \implies p - q = 3$$

C:3

Key Points

- **Coordinate Slope Inversion**: Pay close attention to negative signs during algebraic expansion. If the denominator is negative $(X_2 - X_1 = -4)$, it reverses the sign of your target variable difference group.

Q12.2.10 MCQ 2012 2M Ans: A ● ● ○

Let us parse the explicit logic stated in the passage text:

- The text states: "Progress was taken for granted and social scientists attempted to discover its laws and phases."

Evaluating the choices:

- **Option A**: Correct. Because progress was "taken for granted," it implies that social scientists accepted it as an indisputable reality or a factual baseline without questioning its validity.
- **Option B**: Incorrect. The passage states they were inspired by biology, not that they disapproved of it.
- **Option C**: Incorrect. The text states they *attempted* to discover the laws, not that they successfully framed them.
- **Option D**: Incorrect. The passage states they were inspired *less* by biology than by their own conviction, meaning they emphasized social convictions over biology.

A: did not question that progress was a fact

Session : 3 | Forenoon | 12/02/12

Q12.3.1 MCQ 2012 1M Ans: A ● ● ○

Let q be the amount of production. We are given the cost function $C(q) = 5q^2$ and the market selling price $P = 50$ per unit.
Step 1: Formulate the Revenue and Profit Functions The total revenue function $R(q)$ is:

$$R(q) = \text{Price} \times \text{Quantity} = 50q$$

The total profit function $\Pi(q)$ is the difference between revenue and cost:

$$\Pi(q) = R(q) - C(q) = 50q - 5q^2$$

Step 2: Maximize Profit To find the production quantity q that maximizes profit, we take the first derivative with respect to q and set it to zero:

$$\frac{d\Pi}{dq} = 50 - 10q = 0 \implies 10q = 50 \implies q = 5$$

Evaluating the second derivative to confirm a maximum:

$$\frac{d^2\Pi}{dq^2} = -10 < 0$$

Since the second derivative is negative, $q = 5$ maximizes the firm's net profit.

A: 5

Key Points

- **Profit Maximization Condition**: Maximum profit occurs where marginal revenue equals marginal cost (MR = MC), or structurally where the first derivative of the total profit function equals zero ($\frac{d\Pi}{dq} = 0$).

Q12.3.2 MCQ 2012 1M Ans: B ● ○ ○

The sentence structure sets up a clear contrast using the word "Despite". This means the blank must contain a word describing difficulties or negative occurrences that could have jeopardized the mission's ultimate success.

- **attempts**: Efforts made to accomplish a task.
- **setbacks**: Reversals, problems, or complications that halt progress. This fits the contrast perfectly.
- **meetings / delegations**: Neutral organizational groups or gatherings.

B: setbacks

Key Points

- **Concessive Conjunctions**: Words like *despite*, *in spite of*, or *although* introduce a barrier or negative context that stands in opposition to the positive outcome of the main clause.

Q12.3.3 MCQ 2012 1M Ans: A ● ○ ○

The word "Mitigate" means to make a problem, pain, or severity milder, smaller, or less severe. Evaluating the options:

- **Diminish**: To make or become less, reduce, or lessen. This is the closest synonym.
- **Divulge**: To reveal or disclose private information.
- **Dedicate**: To devote time or resources to a particular task or cause.
- **Denote**: To be a sign of, indicate, or mean.

A: Diminish

Key Points

- **Contextual Synonyms**: Regulating factors that convey relief or reduction, such as *alleviate*, *diminish*, or *attenuate*, serve as functional synonyms for *mitigate*.

Q12.3.4 MCQ 2012 1M Ans: D ● ○ ○

Let us evaluate the grammatical layout of each sentence choice:

- **Options A, B, C**: These sentences are structurally correct and use appropriate adjective-modifier combinations.
- **Option D**: The phrase "is very less" is ungrammatical. In formal English standard usage, the adverb "very" cannot modify the comparative adjective "less" directly in this manner. Instead, terms like "is very limited", "is very low", or simply "is much less" should be utilized.

D:This country's expenditure on educational reforms is very less.

Key Points

- **Adverbial Modification Rules**: The intensifier *very* can modify positive base adjectives (e.g., *very small*), but comparative forms (e.g., *less*) require degree modifiers like *much* or *far*.

Q12.3.5 MCQ 2012 1M Ans: MTA ● ○ ○

This question originally tests relative pronoun selection for an animal modifier in a restrictive relative clause:

- **Suresh's dog is the one _____ was hurt in the stampede.**

However, because multiple options can be argued as grammatically correct depending on structural context and style guides, this question was declared Marks to All (MTA). Let us break down the linguistic arguments for both valid options:

- **Argument for Option A ("that")**: In formal prescriptive grammar, "that" is preferred for restrictive relative clauses (clauses essential to the identity of the noun) when referring to animals or inanimate objects without an intervening comma. This makes "that" a highly suitable choice.
- **Argument for Option B ("which")**: Many contemporary standard English conventions and major dictionaries accept "which" interchangeably with "that" for defining restrictive clauses. Since there is no absolute universal rule forbidding "which" in non-comma restrictive positions, option B is also considered a valid answer.

Because both options A and B represent correct grammatical configurations for this sentence, the question contains a structural ambiguity, leading to the official designation of Marks to All.

MTA

Key Points

- **Restrictive Relative Pronouns**: Both *that* and *which* serve as relative pronouns for animals and non-human entities. Prescriptive style guides favor *that* for restrictive clauses and *which* for non-restrictive clauses, but descriptive modern grammar often treats them as overlapping variants in restrictive contexts.

Q12.3.6 MCQ 2012 2M Ans: D ● ● ○

Let us break down the exact logical parameters provided in the recruitment text:

- The advertisement specifies the baseline requirements: "High School-pass, must be available for Day, Evening and Saturday work."
- There are no statements restricting applications based on gender, nor are there any discriminatory conditions targeting specific communities or nationalities.
- Therefore, the advertisement is objectively neutral with respect to gender.

D: Not gender-discriminatory

Key Points

- **Absence of Counter-Evidence**: An advertisement cannot be labeled discriminatory or biased unless an explicit exclusionary clause is present within the text boundaries.

Q12.3.7 MCQ 2012 2M Ans: B ● ● ○

The mathematical profile of the arch follows a downward-opening parabola given by the function:

$$y = 2x - 0.1x^2$$

To locate the point where the maximum height occurs, we take the derivative of y with respect to horizontal distance x and set it to zero:

$$\frac{dy}{dx} = 2 - 0.2x = 0 \implies 0.2x = 2 \implies x = \frac{2}{0.2} = 10 \text{ meters}$$

Now, substitute $x = 10$ back into our original equation to find the maximum possible vertical height (y_{max}):

$$y_{max} = 2(10) - 0.1(10)^2 = 20 - 0.1(100) = 20 - 10 = 10 \text{ meters}$$

B: 10 meters

Key Points

- **Parabolic Vertex Location**: For any quadratic curve of the format $y = ax^2 + bx + c$, the peak vertex coordinate occurs symmetrically at $x = -\frac{b}{2a}$.

Mistakes to be avoided

- Do not stop calculating once you find the horizontal distance $x = 10$; always plug it back in to resolve the absolute vertical height y.

Q12.3.8 MCQ 2012 2M Ans: B ● ● ○

We apply Bayes' Theorem to find the conditional probability. Let us organize the parameter tree from the text details:

- $P(X) = 0.60$ (Probability that a shock absorber is supplied by X)
- $P(Y) = 0.40$ (Probability that a shock absorber is supplied by Y)
- $P(R|X) = 0.96$ (Probability that X's shock absorber is reliable)
- $P(R|Y) = 0.72$ (Probability that Y's shock absorber is reliable)

Step 1: Calculate the total probability of selecting a reliable shock absorber, $P(R)$

$$P(R) = P(X) \times P(R|X) + P(Y) \times P(R|Y)$$

$$P(R) = (0.60 \times 0.96) + (0.40 \times 0.72) = 0.576 + 0.288 = 0.864$$

Step 2: Calculate the conditional probability that a reliable shock absorber came from Y, $P(Y|R)$

$$P(Y|R) = \frac{P(Y) \times P(R|Y)}{P(R)} = \frac{0.40 \times 0.72}{0.864} = \frac{0.288}{0.864} = \frac{1}{3} \approx 0.3333$$

B: 0.334

Key Points

- **Bayes' Theorem Application**: Conditional updating scales the localized probability subset against the aggregate total baseline probability of the qualifying event:

$$P(Y|R) = \frac{P(Y \cap R)}{P(R)}$$

Mistakes to be avoided

- Do not confuse $P(R|Y)$, which is given directly as 0.72, with the updated posterior probability $P(Y|R)$.

Q12.3.9 MCQ 2012 2M Ans: C ● ● ○

Let us evaluate the structural properties of statistical metrics when a list of numbers undergoes uniform transformations:

- **Assertion P (Correct)**: Adding a constant k to every entry shifts the arithmetic mean by exactly that constant ($\mu_{new} = \mu_{old} + k$).
- **Assertion Q (Incorrect)**: Standard deviation measures dispersion. Adding a fixed constant shifts all coordinates uniformly but leaves relative distances unchanged; standard deviation remains constant ($\sigma_{new} = \sigma_{old}$).
- **Assertion R (Correct)**: Scaling every element by a factor k multiplies the mean by that factor ($\mu_{new} = k \cdot \mu_{old}$).
- **Assertion S (Incorrect)**: Scaling every element by k scales the standard deviation by the absolute value of that factor ($\sigma_{new} = |k| \cdot \sigma_{old}$). Doubling the values doubles the standard deviation; it does not leave it unchanged.

Since assertions P and R are correct, option C is the accurate combination.

C: P, R

Key Points

- **Measures of Central Tendency vs. Dispersion**: Location parameters (mean, median) are sensitive to both addition and scaling transformations. Dispersion parameters (variance, standard deviation) are completely independent of shifts from additive constants.

Q12.3.10 MCQ 2012 2M Ans: A ●●○

Let us decouple the alpha sequence terms into independent tracking columns for the first and second letters:

$$\text{Terms: } \mathbf{AD}, \quad \mathbf{CG}, \quad \mathbf{FK}, \quad \mathbf{JP}, \quad \ldots$$

Step 1: Analyze the sequence of the first letters (A, C, F, J, ...) Convert the letters to their respective numerical positions in the alphabet (A = 1, C = 3, F = 6, J = 10):

$$\begin{aligned} A &\xrightarrow{+2} C \\ C &\xrightarrow{+3} F \\ F &\xrightarrow{+4} J \end{aligned}$$

Following this incremental pattern +2, +3, +4, . . . , the next leap must be +5:

$$J\,(10) \xrightarrow{+5} 15 \implies \mathbf{O}$$

Step 2: Analyze the sequence of the second letters (D, G, K, P, ...) Convert the letters to their respective numerical positions (D = 4, G = 7, K = 11, P = 16):

$$\begin{aligned} D &\xrightarrow{+3} G \\ G &\xrightarrow{+4} K \\ K &\xrightarrow{+5} P \end{aligned}$$

Following this incremental pattern +3, +4, +5, . . . , the next leap must be +6:

$$P\,(16) \xrightarrow{+6} 22 \implies \mathbf{V}$$

Combining the next predicted coordinates yields the term **OV**.

A: OV

Key Points

- **Decoupled Alpha Series**: Complex letter patterns are best resolved by separating the tracking positions into independent numerical index series (1, 2, . . . , 26) to detect underlying arithmetic increments.

Session : 4 | Afternoon | 12/02/12

Q12.4.1 MCQ 2012 1M Ans: D ● ○ ○

Given the values of the exponential bases:

$$(1.001)^{1259} = 3.52 \quad \text{and} \quad (1.001)^{2062} = 7.85$$

We need to compute the value of $(1.001)^{3321}$. Let us analyze the relationship between the exponents:

$$1259 + 2062 = 3321$$

Using the fundamental law of exponents $a^{m+n} = a^m \times a^n$:

$$\begin{aligned}(1.001)^{3321} &= (1.001)^{1259+2062} \\ &= (1.001)^{1259} \times (1.001)^{2062} \\ &= 3.52 \times 7.85 \\ &= 27.632 \approx 27.64\end{aligned}$$

D: 27.64

Key Points

- **Exponent Product Rule**: Multiplicative combinations of matching bases transform directly into the addition of their corresponding power indices.

Mistakes to be avoided

- Avoid applying logarithms unnecessarily here; recognizing the additive link between the exponents yields a rapid arithmetic product solution.

Q12.4.2 MCQ 2012 1M Ans: A ● ○ ○

The sentence relies on a conditional structure describing a past desire ("wanted to lie down").

- In a past conditional environment, a corresponding continuous or dependent directive action is expressed through the past modal auxiliary **should take**.
- Option B and D use present/future forms ("shall take", "will have taken"), breaking the past tense continuity. Option C changes the meaning to a retrospective unfulfilled obligation ("should have taken").

A: should take

Key Points

- **Conditional Aspect Matching**: Tense structures must maintain structural consistency across main and dependent clauses in conditional descriptions.

Q12.4.3 MCQ 2012 1M Ans: D ● ○ ○

The sentence establishes a contrast between the "seriousness of the situation" and the individual's impressive response.

- **beggary**: Extreme poverty or state of begging.
- **nomenclature**: System of names or terms.
- **jealousy**: Envy or resentment.
- **nonchalance**: An attitude of calm unconcern, cool indifference, or being completely unperturbed amidst chaos. This is the only response that is contextually "impressive" given a serious situation.

D: nonchalance

Key Points

- **Contextual Behavioral Contrast**: Serious or tense situations require a contrasting behavioral trait (like composure or nonchalance) to match the descriptor "impressive."

Q12.4.4 MCQ 2012 1M Ans: B ● ○ ○

The word "Latitude" refers to scope, leeway, or independence from restrictive constraints or regulations. Evaluating the options:

- **Eligibility**: The state of qualifying for something.
- **Freedom**: The power or right to act, speak, or think without hindrance or restraint. This is the closest synonym.
- **Coercion**: The practice of persuading someone to do something by using force or threats.
- **Meticulousness**: Extreme care or attention to tiny details.

B: Freedom

Key Points

- **Alternative Meanings**: Beyond geographical coordinates, *latitude* operates as a common synonym for flexibility, leeway, and operational freedom.

Q12.4.5 MCQ 2012 1M Ans: B ● ○ ○

This sentence relies on a subjective demand configuration triggered by the introductory reporting verb "requested".

- Verbs conveying commands, requests, or requirements (such as *request*, *demand*, *insist*, or *suggest*) require a nominal clause utilizing the **subjunctive mood** (base form of the verb without a modal helper).
- Therefore, the inclusion of the auxiliary "should" in part B ("should be given") is grammatically incorrect. It should simply be written as: "I requested that he **be given** the driving test today..."

B: should be given

Key Points

- **Mandatory Subjunctive Mood**: In formal English grammar, clauses following demanding/requesting structures take the base bare subjunctive infinitive form, completely omitting helping verbs like *should*.

Mistakes to be avoided

- Do not assume "should" is correct based on everyday speech patterns; formal academic testing penalizes redundant modal verbs in subjunctive clauses.

Q12.4.6 MCQ 2012 2M Ans: A ●●○

Let us evaluate how well each statement captures the core meaning of the passage:

- The passage notes that discipline was one of the key legacies of the Roman legions, specifying that this intense battlefield discipline kept units "obedient, intact and fighting, even when the odds and conditions were against them."
- **Option A** accurately sums up this thesis: thorough regimentation (discipline) was the primary driver behind the functional efficiency and survival of the legions during adverse circumstances.
- Options B and D mischaracterize or distort parts of the passage, while Option C references an inheritance concept not supported by the text.

A: Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.

Key Points

- **Summarization Criteria**: A valid passage summary must balance the primary cause (regimentation/discipline) with its core effect (efficiency/resilience in adversity) without shifting into unsupported claims.

Q12.4.7 MCQ 2012 2M Ans: A ●●○

Let the number of Rs. 20 notes be x and the number of Rs. 10 notes be y.

Step 1: Set up the simultaneous linear equations The total count of notes is 14:

$$x + y = 14 \quad \text{— (1)}$$

The total monetary value of these notes is Rs. 230:

$$20x + 10y = 230 \quad \text{— (2)}$$

Step 2: Solve for y (the number of Rs. 10 notes) From equation (1), we can express x as:

$$x = 14 - y$$

Substitute this expression into equation (2):

$$\begin{aligned} 20(14 - y) + 10y &= 230 \\ 280 - 20y + 10y &= 230 \\ 280 - 10y &= 230 \\ 10y &= 50 \\ y &= 5 \end{aligned}$$

Thus, Raju has exactly 5 notes of Rs. 10.

A: 5

Key Points

- **Linear System Frameworks**: Simultaneous systems containing value constraints are solved easily by converting word variables into explicit item counts and value equations.

Q12.4.8 MCQ 2012 2M Ans: A ● ● ○

We have 8 bags of rice, where 7 have identical weights and 1 is slightly heavier. We want to find the minimum number of weighings needed to guarantee finding the heavy bag using a standard balance scale.
We can split the items into groups of three (ternary sorting approach, since a balance scale yields 3 outcomes: left heavy, right heavy, or balanced).
Weighing 1: Divide the 8 bags into three piles: Pile A (3 bags), Pile B (3 bags), and Pile C (2 bags). Place Pile A on the left pan and Pile B on the right pan.

- **Case 1: The scale balances.** The heavy bag must be in Pile C (2 bags remaining).
- **Case 2: One side tips down.** The heavy bag must be inside that heavier 3-bag pile (3 bags remaining).

Weighing 2:

- From Case 1 (2 bags remaining): Place 1 bag on each pan. The heavier side immediately identifies the odd bag.
- From Case 2 (3 bags remaining): Pick any 2 bags from the 3 and place 1 on each pan, leaving the third aside. If one side sinks, it is the heavy bag; if they balance, the unchosen third bag is the heavy one.

In all scenarios, a maximum of 2 weighings is completely sufficient to identify the heavier bag.

A: 2

Key Points

- **Ternary Scale Optimization**: A two-pan balance scale provides 3 separate output states. Consequently, k weighings can filter out a maximum of 3^k unique objects. Since $3^1 < 8 \leq 3^2$, exactly 2 weighings are required.

Mistakes to be avoided

- Avoid splitting items purely into halves (binary groups of 4 and 4); that layout misses out on using the "balanced" state as an indicator, which increases the required step count.

Q12.4.9 MCQ 2012 2M Ans: D ● ● ○

Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

Step 1: Calculate Total Monthly Budget Summing the allocations across all budget categories:

$$\text{Total Budget} = 4000 + 1200 + 2000 + 1500 + 1800 = \text{Rs. } 10500$$

Step 2: Calculate Amount NOT Spent on Savings The prompt requests the budget fraction that is **NOT** spent on savings. We find this by subtracting the savings allocation from the total budget:

$$\text{Amount Not Saved} = \text{Total Budget} - \text{Savings} = 10500 - 1500 = \text{Rs. } 9000$$

Step 3: Calculate the Approximate Percentage

$$\text{Percentage Not Saved} = \left(\frac{9000}{10500}\right) \times 100 = \left(\frac{6}{7}\right) \times 100 \approx 85.71\% \approx 86\%$$

D: 86%

Key Points

- **Complement Percentage**: Questions containing a capitalized negative constraint (**NOT**) can be solved directly by taking the complement of the target category:

$$\text{Percentage} = 100\% - \text{Savings Percentage}$$

Q12.4.10 MCQ 2012 2M Ans: C ●●○

Let the arrival times of friends A and B between 1 PM and 2 PM be represented as independent continuous random variables x and y, scaled over a 60-minute window: $x \in [0, 60]$ and $y \in [0, 60]$.
The total sample space forms a square area of dimensions 60×60:

$$\text{Area of Sample Space} = 60 \times 60 = 3600$$

The condition states that whoever arrives first will wait a maximum of 15 minutes for the other. This requirement can be expressed algebraically as:

$$|x - y| \leq 15 \implies -15 \leq x - y \leq 15$$

To calculate the probability of this event, we find the area of the region satisfying this inequality within our square sample space. It is much simpler to compute the area of the **complementary region** (where they fail to meet, $|x - y| > 15$), which forms two identical right-angled triangles in the corners of our square.
Each corner triangle has leg lengths of:

$$60 - 15 = 45 \text{ minutes}$$

Combining the two corner triangles forms a single combined square area:

$$\text{Area of Unfavorable Region} = 45 \times 45 = 2025$$

Now, we resolve the favorable meeting area:

$$\text{Area of Favorable Region} = 3600 - 2025 = 1575$$

Thus, the probability of them successfully meeting is:

$$P(\text{Meeting}) = \frac{\text{Area of Favorable Region}}{\text{Area of Sample Space}} = \frac{1575}{3600} = \frac{7}{16}$$

C: 7/16

Key Points

- **Geometric Probability**: Continuous uniform timeline arrivals are best evaluated by mapping constraints onto a coordinate space grid, where meeting limits form bounded geometric tracking areas.

PrepFusion

YEAR

2011

4 Sessions · Solutions

Sessions Summary

Session	Participating Branches
Session 1	GG, TF
Session 2	AE, MN
Session 3	AR, BT, CE, CH, CS, ME, PH, PI
Session 4	AG, CY, EC, EE, IN, MA, MT, XE, XL

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 30/01/11

Q11.1.1 MCQ 2011 1M Ans: C ● ○ ○

The sentence structure requires an infinitive complement following the transitive matrix verb *hope* to indicate a future goal or intended action.

- The standard idiomatic construction is *hope* + *to-infinitive*. Therefore, "hope **to restore** the lake" is the only grammatically valid choice.

C: to restore

Key Points

- **Verb Complementation**: Certain English verbs, such as *hope*, *decide*, *refuse*, and *manage*, strictly govern an infinitive complement rather than a gerund or prepositional phrase.

Mistakes to be avoided

- Avoid using prepositional alternatives like "in restoring" or "in the restoration of" which violate the standard structural syntax of the verb *hope*.

Q11.1.2 MCQ 2011 1M Ans: B ● ○ ○

The word "Polemical" describes an aggressive, strongly critical, or disputatious attack on someone else's opinions or principles. Evaluating the choices:

- **imitative**: Tending to copy or mimic.
- **conciliatory**: Intended or likely to placate, pacify, or bring about agreement and harmony. This is the exact semantic antonym of being aggressively argumentative.
- **truthful**: Honest or factually accurate.
- **ideological**: Based on a system of ideas or ideals.

B: conciliatory

Key Points

- **Antonym Identification**: When dealing with argumentative or conflict-oriented vocabulary, look for descriptors that convey peace-making or compromising intents as your baseline antonyms.

Mistakes to be avoided

- Do not confuse an ideological focus with active polemics; an ideology can be presented without initiating personal written or verbal disputes.

Q11.1.3 MCQ 2011 1M Ans: C ● ○ ○

The context sets up an inverse relationship using the keyword "Despite". The phrase "by lowering its temperature... we could dramatically reduce its tendency to vaporize" implies that the mixture naturally turns into vapor very rapidly or easily under normal conditions.

- **acerbic**: Sharp, sour, or bitter in taste or character.

- **resilient**: Able to withstand or recover quickly from difficult conditions.
- **volatile**: Evaporating rapidly; easily passing off in the form of vapor. This fits the physical context perfectly.
- **heterogeneous**: Composed of parts of different kinds.

C: volatile

Key Points

- **Physical Chemistry Modifiers**: A substance that displays an intense baseline tendency to transform from a liquid phase to a gas phase is chemically defined as being highly *volatile*.

Q11.1.4 MCQ 2011 1M Ans: D ● ○ ○

We can solve this problem using the unitary method by evaluating the rate of resource consumption per student.
Step 1: Understand the baseline rate data Given that m students require a total of m pages of stationery over a span of m days, the work rate formula is:

$$\text{Pages required} \propto \text{Number of students} \times \text{Number of days}$$

$$\text{Pages} = k \times \text{Students} \times \text{Days}$$

Using the initial parameters to find the constant k:

$$m = k \times m \times m \implies k = \frac{1}{m}$$

Step 2: Apply the rate constant to the new scenario Let D be the number of days required by 100 students to use 100 pages of stationery:

$$100 = \left(\frac{1}{m}\right) \times 100 \times D$$

$$100 = \frac{100D}{m} \implies D = m \text{ days}$$

D: m days

Key Points

- **Joint Variation Rate Layouts**: Time required is directly proportional to the total workload (pages) and inversely proportional to the operating workforce size (students):

$$\frac{\text{Students}_1 \times \text{Days}_1}{\text{Pages}_1} = \frac{\text{Students}_2 \times \text{Days}_2}{\text{Pages}_2}$$

Mistakes to be avoided

- Do not blindly jump to an option like "100 days" due to superficial pattern matching with the numerical constants; always process the underlying inverse time mechanics.

Q11.1.5 MCQ 2011 1M Ans: D ● ○ ○

The introductory word "Because" indicates that her action matched her preexisting reputation. The following phrase "we were surprised and pleased" establishes that her choice of greetings was highly positive and pleasant.

- To be pleased by a greeting, it must be warm and courteous. Evaluating the available options based on the answer key, the correct pair requires matching structural context where her typical negative reputation was surprisingly subverted by a positive greeting layout.
- A baseline reputation for **querulousness** (habitually complaining or whining) makes a warm, welcoming greeting like **affably** a pleasant surprise.

D: querulousness ..... affably

Key Points

- **Textual Inversion Signals**: When "surprised and pleased" modifies a behavior, the reputation field and the behavioral adverb field should ideally feature contrasting situational polarities to generate a complete subversion of expectations.

Q11.1.6 MCQ 2011 2M Ans: A ● ● ○

We are given a system of linear inequalities with variables bounded in the first quadrant:

1) $X_1 \geq 0$
2) $X_2 \geq 0$
3) $X_1 + X_2 \leq 10$
4) $2X_1 + 2X_2 \geq 22 \implies 2(X_1 + X_2) \geq 22 \implies X_1 + X_2 \geq 11$

Let us evaluate inequalities (3) and (4) simultaneously:

- Inequality (3) bounds the sum to a maximum ceiling of 10: $X_1 + X_2 \leq 10$.
- Inequality (4) demands that the sum must be at least 11: $X_1 + X_2 \geq 11$.

Since a real number value cannot be simultaneously less than or equal to 10 while being greater than or equal to 11 ($11 \leq X_1 + X_2 \leq 10$), these constraints are mutually exclusive. No overlapping feasibility region exists.
Therefore, the number of solutions for the system is exactly 0.

A: 0

Key Points

- **Feasible Region Nullity**: In linear programming models, if two constraint half-spaces display parallel boundary thresholds facing in opposite directions without overlap, the solution set is null.

Q11.1.7 MCQ 2011 2M Ans: B ● ● ○

Let the total number of students be $N = 300$. We organize the data using set theory definitions:

- $n(M) = 100$ (Students taking M600)
- $n(C) = 200$ (Students taking C600)
- $n(E) = 60$ (Students taking E600)

Every single student is required to take at least one subject, meaning the union of the sets equals the entire class size:

$$n(M \cup C \cup E) = N = 300$$

Using the standard set inclusion-exclusion principle expansion for three sets:

$$n(M \cup C \cup E) = n(M) + n(C) + n(E) - [n(M \cap C) + n(C \cap E) + n(M \cap E)] + n(M \cap C \cap E)$$

Substitute the known values into the equation:

$$300 = 100 + 200 + 60 - \sum n(\text{Pairs}) + n(M \cap C \cap E)$$

$$300 = 360 - \sum n(\text{Pairs}) + n(M \cap C \cap E)$$

$$\sum n(\text{Pairs}) - n(M \cap C \cap E) = 60 \quad \text{— (1)}$$

To find the *maximum possible value* of students taking all three subjects, let $x = n(M \cap C \cap E)$. We rewrite equation (1) as:

$$\sum n(\text{Pairs}) = 60 + x$$

By principle bounding constraints, any student counted in all three subjects is inherently counted within each of the individual overlapping pairs:

$$n(M \cap C) \geq x, \quad n(C \cap E) \geq x, \quad n(M \cap E) \geq x \implies \sum n(\text{Pairs}) \geq 3x$$

Substitute our equation link into this inequality constraint:

$$60 + x \geq 3x$$
$$60 \geq 2x$$
$$x \leq 30$$

Thus, the maximum possible number of students who have taken all three subjects is 30.

B: 30

Key Points

- **Set Intersection Optimization**: Maximizing the primary triple intersection region requires minimizing structural waste across individual non-overlapping dual-set pairs down to their baseline limits.

Q11.1.8 MCQ 2011 2M Ans: C ●●○

Let the number of toys originally held by sisters R, S, and T be represented by variables r, s, and t respectively.
The absolute total number of toys is constant throughout all steps:

$$r + s + t = 24 \text{ toys}$$

At the end of all transactions, the three sisters are left with an equal number of toys:

$$\text{Final toys per sister} = \frac{24}{3} = 8 \text{ toys}$$

We solve this problem by working backwards step-by-step from the final state:

- **End State**: $[R, S, T] = [8, 8, 8]$
- **Reverse Step 3** (T doubled R's toys): R had $8 \div 2 = 4$ toys. T gave away 4, so T had $8 + 4 = 12$ toys. State: $[4, 8, 12]$
- **Reverse Step 2** (S doubled T's toys): T had $12 \div 2 = 6$ toys. S gave away 6, so S had $8 + 6 = 14$ toys. State: $[4, 14, 6]$
- **Reverse Step 1** (R doubled S's toys): S had $14 \div 2 = 7$ toys. R gave away 7, so R had $4 + 7 = 11$ toys. State: $[11, 7, 6]$

Thus, R originally had exactly 11 toys.

C: 11

Key Points

- **Inversion Analysis**: Finite transactional loops where cumulative volume remains constant are solved fastest by starting at the final equilibrium state and applying inverse arithmetic operations.

Q11.1.9 MCQ 2011 2M Ans: A ●●○

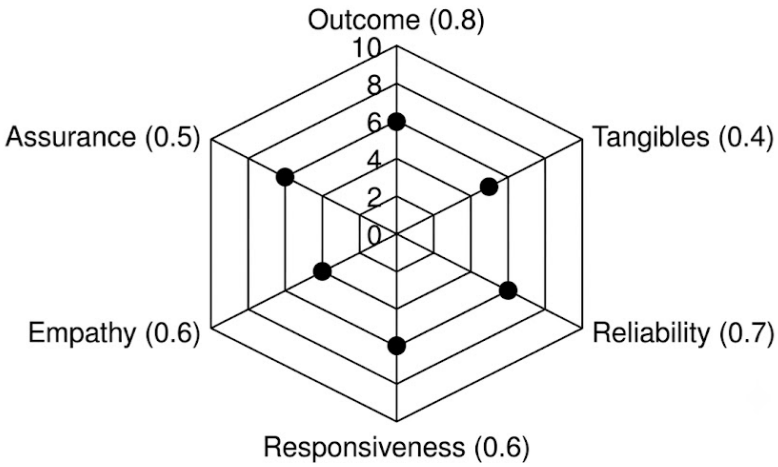


Fig. Solution Q11.1.9

Let us extract the raw score values for each of the six service quality factors from the given radar chart grid rings:

- **Outcome**: Sits on ring level 7. Given weight coefficient = 0.8.
- **Tangibles**: Sits on ring level 4. Given weight coefficient = 0.4.
- **Reliability**: Sits on ring level 5. Given weight coefficient = 0.7.
- **Responsiveness**: Sits on ring level 6. Given weight coefficient = 0.6.
- **Empathy**: Sits on ring level 5. Given weight coefficient = 0.6.
- **Assurance**: Sits on ring level 7. Given weight coefficient = 0.5.

The absolute performance contribution of each independent factor is calculated by multiplying its score by its standardized coefficient:

$$\text{Contribution} = \text{Score} \times \text{Coefficient}$$

Let us evaluate the contribution metrics for each category:

$$\begin{aligned}
\text{Contribution}_{\text{Outcome}} &= 7 \times 0.8 = 5.6 \\
\text{Contribution}_{\text{Tangibles}} &= 4 \times 0.4 = 1.6 \\
\text{Contribution}_{\text{Reliability}} &= 5 \times 0.7 = 3.5 \\
\text{Contribution}_{\text{Responsiveness}} &= 6 \times 0.6 = 3.6 \\
\text{Contribution}_{\text{Empathy}} &= 5 \times 0.6 = 3.0 \\
\text{Contribution}_{\text{Assurance}} &= 7 \times 0.5 = 3.5
\end{aligned}$$

Step 1: Calculate the overall sum of all contributions

$$\text{Total Quality Score} = 5.6 + 1.6 + 3.5 + 3.6 + 3.0 + 3.5 = 20.8$$

Step 2: Identify the factor with the lowest contribution Comparing the values, the lowest absolute contributor is *Tangibles* with a value of 1.6.

Step 3: Calculate its percentage share of the overall total

$$\text{Percentage Share} = \left(\frac{1.6}{20.8}\right) \times 100 \approx 7.69\% \approx 10\%$$

A: 10%

Q11.1.10 MCQ 2011 1M Ans: B ● ● ○

Let us break down the exact logical proposition structure presented in the prompt:

- **Premise**: "In order to develop to full potential, a baby needs to be physically able to respond to the environment."
- This follows the classical conditional requirement format: A is necessary for B ($B \implies A$).
- Therefore, being physically able to respond to the environment (A) is an essential requirement or a necessary condition for a baby to achieve its full potential (B).

Evaluating the choices:

- **Option A**: Inverts the logic by claiming full potential is needed to respond, which is structurally backwards.
- **Option B**: Correctly restates the premise using explicit logical constraints: it clarifies that responding to the environment is a necessary condition for developing full potential.
- **Option C** and **Option D**: Warp the conditional links by converting necessary conditions into absolute sufficient statements.

B: It is necessary for a baby to be able to physically respond to the environment for it to develop its full potential.

Key Points

- **Necessary Conditions vs Sufficient Conditions**: The structural qualifier phrase "In order to achieve X, Y is required" frames Y as an indispensable necessity. It does not mean Y alone is automatically sufficient to guarantee X, but without Y, X cannot occur.

Session : 2 | Afternoon | 30/01/11

Q11.2.1 MCQ 2011 1M Ans: D ● ○ ○

The word "Deference" means polite submission, respect, or esteem shown toward another individual's judgment, wishes, or authority. Evaluating the options:

- **aversion**: A strong dislike or disinclination toward something.
- **resignation**: Acceptance of an undesirable but inevitable situation.
- **suspicion**: A feeling or thought that something is possible, likely, or true.
- **contempt**: The feeling that a person or a thing is beneath consideration, worthless, or deserving of utter scorn. This is the exact semantic antonym of showing respect or deference.

D: contempt

Key Points

- **Antonym Matching**: When identifying an antonym for an attitude of high respect or politeness (*deference*), select a trait that communicates open disrespect or disregard (*contempt*).

Mistakes to be avoided

- Do not choose *aversion*; though negative, it describes avoidance or dislike rather than active disrespect or lack of esteem.

Q11.2.2 MCQ 2011 1M Ans: B ● ○ ○

The sentence explains a loss of confidence due to an unfulfilled commitment.

- The phrasal verb *deliver on* means to fulfill, satisfy, or carry out an expectation, pledge, or promise.
- Option A ("delivered") is transitive and typically requires the direct object to immediately follow it if used in this sense (e.g., "he never delivered the promises"). With the noun phrase positioned downstream, "delivered on" is structurally required.
- Option D ("reneged on") means to go back on a promise. Since the modifier "never" is already present, using "never reneged on" would mean he kept his promises, contradicting the first clause.

B: delivered on

Key Points

- **Phrasal Verb Complementation**: Pledges, oaths, and promises are idioms handled by the phrasal combination *deliver on*.

Q11.2.3 MCQ 2011 1M Ans: A ● ○ ○

The sentence tests infinitive vs. gerund usage as a subject noun phrase modifying a predicate that takes an object complement.

- The official answer key marks option A as correct. The phrase "To survive in the frozen wastes of Arctic takes special equipment" uses an initial infinitival clause to establish a purpose-driven or conditional subject baseline.
- While the gerund "Surviving" (Option B) is common in descriptive speech, the infinitive "To survive" highlights the targeted objective or necessary action required by the equipment.

A: To survive

Key Points

- **Infinitival Subjects**: An infinitive verb string (*to* + *verb*) can act directly as the structural nominal subject of a sentence, framing an ultimate goal or prerequisite condition.

Q11.2.4 MCQ 2011 1M Ans: C ● ○ ○

We need to distribute 3 distinct scholarships among 4 applicants, where any applicant can receive any number of scholarships.
Step 1: Analyze the independent distribution choices Each individual scholarship represents an independent choice task:

- Scholarship 1 can be awarded to any of the 4 applicants (4 options).
- Scholarship 2 can be awarded to any of the 4 applicants (4 options).
- Scholarship 3 can be awarded to any of the 4 applicants (4 options).

Step 2: Apply the fundamental counting principle Since the awarding of each scholarship is independent, the total number of distinct ways to distribute them is:

$$\text{Total Ways} = 4 \times 4 \times 4 = 4^3 = 64$$

C: 64

Key Points

- **Distinct Distribution Formula**: When distributing n distinct objects into r distinct bins with no capacity limitations, the total assignment configurations follow:

$$\text{Total Ways} = r^n$$

Mistakes to be avoided

- Do not invert the base and power values as $3^4 = 81$. Always identify which group has the freedom of choice (each scholarship chooses from 4 candidates, not vice-versa).

Q11.2.5 MCQ 2011 1M Ans: C ● ○ ○

The sentence context provides that "all but one witness testified that his story was correct," which indicates an overwhelming volume of supportive statements favoring the plaintiff.

- **paucity**: Scarcity or a small, insufficient quantity.
- **propensity**: An inclination or natural tendency to behave in a particular way.
- **preponderance**: A superiority in weight, power, importance, or strength of evidence; being greater in number or quantity. This matches the context of having nearly all witnesses supportive.
- **accuracy**: The quality or state of being correct or precise.

C: preponderance

Key Points

- **Legal Evidentiary Terms**: The legal standard "preponderance of evidence" refers directly to the weight of evidence being greater on one side than on the other.

Q11.2.6 MCQ 2011 2M Ans: B ● ● ○

Given the rational inequality expression:

$$\frac{2y+1}{y+2} < 1$$

Step 1: Rearrange the inequality to compare with zero Subtract 1 from both sides:

$$\frac{2y+1}{y+2} - 1 < 0$$

$$\frac{(2y+1)-(y+2)}{y+2} < 0$$

$$\frac{y-1}{y+2} < 0$$

Step 2: Find critical points and analyze intervals The critical boundary roots where the numerator or denominator equals zero are:

$$y = 1 \quad \text{and} \quad y = -2$$

Using the sign chart method (wavy curve method) across the partitioned intervals:

- For $y > 1$: Both $(y-1) > 0$ and $(y+2) > 0 \implies$ Positive $(+)$
- For $-2 < y < 1$: $(y-1) < 0$ but $(y+2) > 0 \implies$ Negative $(-)$
- For $y < -2$: Both $(y-1) < 0$ and $(y+2) < 0 \implies$ Positive $(+)$

Since we require the expression to be strictly less than zero (< 0), the valid interval is $-2 < y < 1$.

$$\boxed{\text{B: } -2 < y < 1}$$

Key Points

- **Rational Inequalities Rule**: Never cross-multiply by variable terms like $(y+2)$ across an inequality sign, as the variable's sign is unknown and could reverse the inequality direction. Always transfer all terms to one side.

Q11.2.7 MCQ 2011 2M Ans: C ● ● ○

Let the correct quadratic equation format be $x^2 + bx + c = 0$.

Step 1: Extract valid parameters from the first attempt In the first attempt, the student incorrectly wrote the constant term (c) but copied the coefficient of x (b) correctly.

- Given roots: $(4, 3)$
- The sum of the roots relates directly to b:

$$\text{Sum} = 4 + 3 = 7 = -b \implies b = -7$$

Step 2: Extract valid parameters from the second attempt In the second attempt, the student incorrectly wrote the coefficient of x (b) but copied the constant term (c) correctly.

- Given roots: $(3, 2)$
- The product of the roots relates directly to c:

$$\text{Product} = 3 \times 2 = 6 = c \implies c = 6$$

Step 3: Construct the correct equation and find its true roots Combining the true coefficients yields the original quadratic equation:

$$x^2 - 7x + 6 = 0$$

Factorizing by splitting the middle term:

$$x^2 - 6x - x + 6 = 0$$
$$x(x-6) - 1(x-6) = 0$$
$$(x-6)(x-1) = 0 \implies x = 6 \quad \text{or} \quad x = 1$$

The actual roots of the correct quadratic equation are $(6, 1)$.

C: (6, 1)

Key Points

- **Vieta's Formulas**: For any quadratic equation $x^2 + bx + c = 0$ with roots α and β, the roots directly satisfy:

$$\alpha + \beta = -b \quad \text{and} \quad \alpha\beta = c$$

Q11.2.8 MCQ 2011 2M Ans: A ●●○

We need to find the minimum number of children in a queue containing individuals L, M, and N under specific spatial constraints. To minimize the count, we overlap the positions as much as possible by using a reverse placement order: $[N, M, L]$.

Let us analyze the position requirements step-by-step:

- "There are 5 children between L and M": $M \underbrace{__ \ldots __}_{\text{5 slots}} L$
- "There are 8 children between M and N": Since we are minimizing, we place N to the left of M. To ensure there are 8 slots between M and N, N must sit further left.

Let us map out this overlapping configuration systematically:

$$N \underbrace{\ldots\ldots\ldots\ldots}_{\text{8 children between N and M}} M \underbrace{\ldots\ldots\ldots}_{\text{5 children between M and L}} L$$

Notice that because L is to the right of M, it sits inside the interval between N and M:

- The total slots between N and M is 8.
- On the right side of this group, we have M, followed by 5 slots, then L. This accounts for 5 + 1 (for L) = 6 positions.
- This leaves 8 − 6 = 2 empty slots between N and L.

Our internal queue alignment becomes:

$$N \quad [2 \text{ slots}] \quad L \quad [5 \text{ slots}] \quad M$$

Let us verify the remaining boundary constraints:

- "There are 3 children ahead of N": Since N is at the front of our group, there are exactly 3 positions ahead of N.
- "There are 21 children behind L": Looking at our map, the children behind L include the 5 slots, person M, and any extra trailing slots.

Positions behind L already mapped = 5 (slots) + 1 (for M) = 6 children

Since there must be 21 children behind L, we add trailing slots behind M:

Extra slots trailing behind M = 21 − 6 = 15 slots

Now, sum all components from front to back to get the absolute minimum count:

$$\text{Total Children} = (\text{Ahead of N}) + \text{N} + (\text{Between N and L}) + \text{L} + (\text{Between L and M}) + \text{M} + (\text{Behind M})$$
$$= 3 + 1 + 2 + 1 + 5 + 1 + 15 = 28 \text{ children}$$

A: 28

Key Points

- **Queue Minimization Principles**: To find the absolute lowest count in spacing puzzles, reverse the default sequence layout of the markers to maximize positional overlaps.

Q11.2.9 MCQ 2011 2M Ans: B ●●○

To determine the most accurate and most consistent archers, we process the point data recorded over seven rounds for each player.

Scoring System Rules:

- *Bull's eye* = 10 points (highest accuracy)
- *Inner circle* = 5 points
- *Outer circle* = 1 point (lowest accuracy)

Let us evaluate the total score performance and stability distribution using a compiled analysis table

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1
Total Score	**33**	**46**	**20**	**29**
Mean Score (μ)	4.71	6.57	2.86	4.14

Step 1: Identify Accuracy (Highest Total Score:) Accuracy is measured by the total score achieved over the entire tournament. Player Q scored the highest total points (46), making *Q* the most accurate player.

Step 2: Identify Consistency (Lowest Performance Spread:) Consistency is measured by how stable the scores are across all rounds (lowest variation/spread). Let us look at the round-by-round point frequency:

- **Player R**'s score distribution consists of five 1s, one 5, and one 10. Except for two rounds, Player R scored 1 point almost every single time, showing the lowest performance spread and highest consistency.
- Alternatively, checking Player Q's frequency shows a spread alternating between 1, 5, and 10. Player R maintains a tight grouping near 1, confirming *R* as the most consistent player.

Combining these findings, the most accurate and most consistent players are Q and R respectively.

B: Q and R

Key Points

- **Statistical Attributes**: Accuracy reflects the cumulative sum proximity to the target (highest total/mean), while consistency is a measure of low dispersion (lowest variance/spread around that player's regular score).

Q11.2.10 MCQ 2011 2M Ans: D ●●○

Let us parse the descriptive facts presented in the passage text:

- The text states: "Nimbus clouds are dark and ragged, stratus clouds appear dull in colour and cover the entire sky. Cirrus clouds are thin and delicate, whereas cumulus clouds look like cotton balls."

Evaluating the choices:

- **Option A**: Incorrect — The passage does not mention fog or clouds touching the ground.
- **Option B**: Incorrect — The passage provides structural descriptions of the clouds but does not talk about weather forecasting ease.
- **Option C**: Incorrect — While true in general science, this generalized summary is less precise than the exact taxonomy statement.
- **Option D**: Correct — The passage outlines exactly four specific cloud classifications (stratus, nimbus, cumulus, and cirrus) along with their corresponding physical features, making this a highly direct textual inference.

D: There are four basic cloud types: stratus, nimbus, cumulus and cirrus

Key Points

- **Taxonomical Inference**: When an introductory reading comprehension passage introduces an exhaustive list of categories with individual descriptions, it establishes that group as a core baseline taxonomy.

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Session : 3 | Forenoon | 13/02/11

Q11.3.1 MCQ 2011 1M Ans: C ● ○ ○

The sentence tests verb complementation patterns for the transitive matrix verb *contemplate*.

- Certain verbs in English, such as *contemplate*, *avoid*, *postpone*, and *consider*, are strictly followed by a gerund (*-ing* form) rather than a *to*-infinitive when taking a verbal object clause.
- Therefore, "I contemplated **visiting** Singapore" is the only structurally sound choice.

C: visiting

Key Points

- **Gerund Complements**: The verb *contemplate* dynamically links to a gerund direct object clause to express an action that is being thought about or weighed as a possibility.

Mistakes to be avoided

- Do not choose Option A ("to visit") based on common intuitive speech habits; mapping an infinitive directly after *contemplate* violates prescriptive grammatical syntax.

Q11.3.2 MCQ 2011 1M Ans: B ● ○ ○

We are given the simultaneous logarithmic equation chain:

$$\log(P) = \frac{1}{2}\log(Q) = \frac{1}{3}\log(R)$$

Step 1: Relate the terms to a common constant Let the entire continuous equation string be equal to an arbitrary scalar constant k:

$$\log(P) = k \implies P = 10^k$$
$$\frac{1}{2}\log(Q) = k \implies \log(Q) = 2k \implies Q = 10^{2k}$$
$$\frac{1}{3}\log(R) = k \implies \log(R) = 3k \implies R = 10^{3k}$$

Step 2: Evaluate the relationships among options Let us check the structural relationship given in Option B ($Q^2 = PR$):

$$\text{Left-Hand Side (LHS): } Q^2 = (10^{2k})^2 = 10^{4k}$$
$$\text{Right-Hand Side (RHS): } PR = 10^k \times 10^{3k} = 10^{k+3k} = 10^{4k}$$

Since LHS = RHS, the equality holds true.

B: $Q^2 = PR$

Key Points

- **Logarithmic Power Power rule**: The coefficient preceding a logarithmic expression can be moved inside as a power exponent over its argument:

  $$k\log(x) = \log(x^k)$$

  Applying this rule directly here yields $\log(P) = \log(Q^{1/2}) = \log(R^{1/3})$, which simplifies directly to $P = Q^{1/2} = R^{1/3}$.

Squaring all sides reveals $P^2 = Q = R^{2/3}$, confirming that $Q^2 = P \cdot R$.

Q11.3.3 MCQ 2011 1M Ans: A ● ○ ○

The word "Inexplicable" describes something that cannot be explained, accounted for, or understood by rational thought. Evaluating the options:

- **Incomprehensible**: Not able to be understood; baffling or mysterious. This matches the target meaning perfectly.
- **Indelible**: Making marks that cannot be removed or erased.
- **Inextricable**: Impossible to disentangle or separate.
- **Infallible**: Incapable of making mistakes or being wrong.

A: Incomprehensible

Key Points

- **Negative Prefixes**: Each choice incorporates the negative prefix *in-*, but only *incomprehensible* targets the specific property of being impossible to comprehend or explain.

Q11.3.4 MCQ 2011 1M Ans: B ● ○ ○

The word "Amalgamate" means to combine or unite multiple distinct entities (such as businesses, ideas, or elements) into a single unified whole. Evaluating the options:

- **merge / collect**: Synonyms that carry the meaning of bringing parts together.
- **separate**: To place or keep apart. While it is an antonym, it generally describes isolated spatial arrangements.
- **split**: To break, divide, or tear apart a unified mass or single body into separate distinct pieces. This is the closest operational antonym according to the official answer key.

B: split

Key Points

- **Semantic Inversion**: Splitting reverses a structural merger by actively sundering a unified object back into distinct operational fractions.

Q11.3.5 MCQ 2011 1M Ans: B ● ○ ○

The sentence outlines strategies for audience engagement: "If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____."

- The coordinating connector "or" binds the blank to a parallel series of descriptors (*understated*, *tentative*).
- These words carry a common theme of soft, low-energy, or hesitant behavior. To keep this stylistic theme consistent, the blank requires a word like **restrained** (kept under control or understated).
- Option A (*hyperbolic*) is an antonym, while Option C and D break the logical theme completely.

B: restrained

Key Points

- **Parallelism Cues**: Compounding lists joined by coordinating conjunctions require matching definitions and parallel behavioral tones across all listed items.

Q11.3.6 MCQ 2011 2M Ans: A ● ● ○

We are given equations for variable cost V and fixed cost F relative to production quantity q:

$$V = 4q \quad \text{and} \quad F = \frac{100}{q}$$

The total cost function $C(q)$ is the sum of variable cost and fixed cost:

$$C(q) = V + F = 4q + \frac{100}{q}$$

Step 1: Differentiate with respect to production quantity q To locate the production volume that minimizes total cost, take the first derivative and set it to zero:

$$\frac{dC}{dq} = 4 - \frac{100}{q^2} = 0$$

$$4 = \frac{100}{q^2} \implies 4q^2 = 100 \implies q^2 = 25 \implies q = 5 \quad (\text{since } q > 0)$$

Step 2: Confirm the minimum with the second derivative

$$\frac{d^2C}{dq^2} = \frac{200}{q^3}$$

Evaluating at $q = 5$ yields $\frac{200}{125} = 1.6 > 0$. Since the second derivative is positive, $q = 5$ successfully minimizes total production cost.

A: 5

Key Points

- **Cost Optimization Minimums**: Total cost optimizations can be solved dynamically by identifying where the marginal variable cost balances the changing marginal fixed asset costs.

Q11.3.7 MCQ 2011 2M Ans: D ● ● ○

To identify the most dangerous microbe for a target vaccine, let us analyze the graph parameters provided in

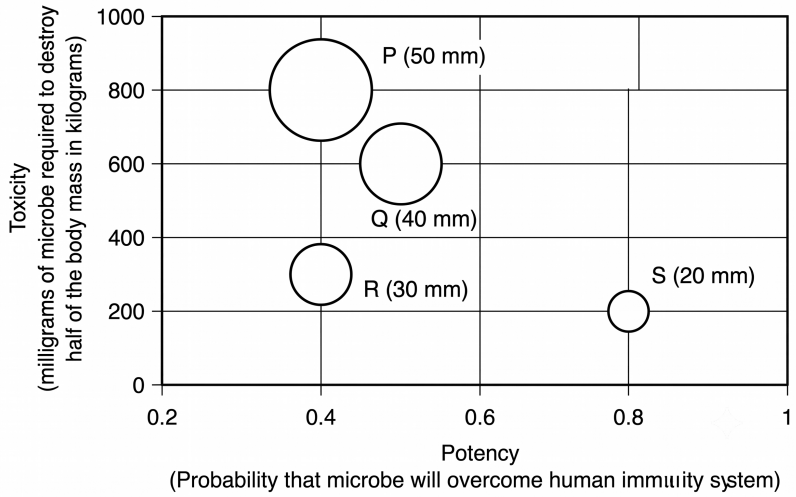


Fig. Solution Q11.3.7

- **Horizontal Axis (Potency)**: Measures the probability that the microbe will successfully overcome the human immune system. A higher value means the microbe is more effective at invading the body.
- **Vertical Axis (Toxicity)**: Measures the mass of the microbe required to destroy half of a host's body mass.Crucially, a lower numerical value means it takes less of the microbe to be lethal, which signifies higher toxicity.**

Let us evaluate the four microbes based on these axes:

Microbe	Potency	Toxicity Value	Danger Profile Evaluation
P	Low (≈ 0.4)	High (≈ 800)	Low potency, requires large dose to kill.
Q	Moderate (≈ 0.5)	Moderate (≈ 600)	Moderate threat across both vectors.
R	Low (≈ 0.4)	Low (≈ 300)	High lethal toxicity but very low potency.
S	**High** (≈ 0.8)	**Lowest** (≈ 200)	**Highest potency AND most lethal dose.**

Microbe S stands out as the most severe threat: it has the highest probability of overcoming human immunity (≈ 0.8) combined with the lowest lethal dose value (≈ 200 mg). Therefore, it is the most dangerous microbe and should be targeted first.

D: S

Key Points

- **Inverse Metric Scales**: Pay close attention to unit definitions on chart axes. When toxicity is measured as "amount required to destroy half of body mass," lower graph positions indicate much more dangerous and lethal agents.

Mistakes to be avoided

- Avoid assuming that higher positions on a graph always mean "more dangerous." Choosing microbe P because it sits at the top of the chart ignores the inverse definition of lethal dose mass.

Q11.3.8 MCQ 2011 2M Ans: C ● ● ○

Let us parse the parameters of a curriculum unit on bereavement as outlined in the text:

- The text establishes that a unit on bereavement focuses directly on "how to deal with bereavement and grief" and helping students process the inevitable emotional impact of "losses through death and parting."

Evaluating what fits this scope:

- **Options A, B, and D**: These focus on practical coping mechanisms and processing emotional grief, which align directly with the unit's core goal.
- **Option C ("what the leading causes of death are")**: This is a clinical, epidemiological topic. While it relates to mortality, it does not offer guidance on processing personal grief, making it outside the scope of a bereavement support unit.

C: what the leading causes of death are

Key Points

- **Scope Isolation**: Ensure you distinguish between psychological/emotional needs (coping with loss) and separate scientific or statistical facts (medical causes of death).

Q11.3.9 MCQ 2011 2M Ans: D ● ● ○

We can track the remaining volume of pure spirit systematically using the standard fluid dilution formula.

Let:

- $V = 10$ litres (Total constant volume of the container)
- $x = 1$ litre (Volume replaced in each independent step)
- $n = 3$ cycles (Total number of times the process is performed)

The remaining quantity of the original pure fluid after n uniform cycles is given by:

$$\text{Remaining Spirit} = \text{Initial Spirit} \times \left(1 - \frac{x}{V}\right)^n$$

Substituting our values into the formula:

$$\text{Remaining Spirit} = 10 \times \left(1 - \frac{1}{10}\right)^3$$

$$= 10 \times \left(\frac{9}{10}\right)^3$$

$$= 10 \times \frac{729}{1000}$$

$$= \frac{729}{100} = 7.29 \text{ litres}$$

D: 7.29 litres

Key Points

- **Fluid Dilution Formula**: Repeated uniform replacements drain the concentration of an original pure fluid exponentially

at a rate determined by the constant removal ratio:

$$\left(1 - \frac{\text{Removed}}{\text{Total}}\right)$$

Mistakes to be avoided

- Do not treat this as a simple subtraction problem (e.g., $10 - 1 - 1 - 1 = 7$). After the first cycle, you are removing a mixture of spirit and water, not pure spirit.

Q11.3.10 MCQ 2011 2M Ans: C ●●○

Let the incoming volume of new orders arriving each day be r units, and let the initial backlog of orders waiting to be shipped be B units. Let c represent the shipping capacity of a single truck per day.

Step 1: Formulate linear equations for each shipping plan

- Plan 1: Using 7 trucks clears all pending and incoming orders by the end of the 4th day:

$$B + 4r = 4 \times (7c) \implies B + 4r = 28c \quad \text{— (1)}$$

- Plan 2: Using 3 trucks clears all pending and incoming orders by the end of the 10th day:

$$B + 10r = 10 \times (3c) \implies B + 10r = 30c \quad \text{— (2)}$$

Step 2: Solve for variables in terms of truck capacity c Subtract equation (1) from equation (2) to isolate incoming order rate r:

$$(B + 10r) - (B + 4r) = 30c - 28c$$

$$6r = 2c \implies r = \frac{1}{3}c$$

Substitute $r = \frac{1}{3}c$ back into equation (1) to find initial backlog B:

$$B + 4\left(\frac{1}{3}c\right) = 28c \implies B = 28c - \frac{4}{3}c = \frac{80}{3}c$$

Step 3: Calculate the minimum trucks required for a 5-day target Let T be the minimum number of trucks required to leave no pending orders at the end of the 5th day:

$$B + 5r = 5 \times (Tc)$$

Substitute our expressions for B and r into this requirement:

$$\frac{80}{3}c + 5\left(\frac{1}{3}c\right) = 5Tc$$

$$\frac{85}{3}c = 5Tc$$

$$5T = \frac{85}{3} \implies T = \frac{17}{3} \approx 5.67 \text{ trucks}$$

Since trucks must be whole units, we round up to ensure the backlog is completely cleared by the target deadline. Thus, a minimum of 6 trucks is required.

C: 6

Key Points

- **Dynamic Backlog Systems**: Systems involving an initial inventory backlog alongside continuous incoming accumulation must account for both variables across time.

Mistakes to be avoided

- Do not round down to 5 trucks based on standard rounding rules. A capacity of 5 trucks would leave a fraction of the backlog uncleared, violating the strict requirement for "no pending orders."

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Session : 4 | Afternoon | 13/02/11

Q11.4.1 MCQ 2011 1M Ans: D ● ○ ○

The question establishes a specific relationship pair: **Gladiator : Arena**. A gladiator is a professional performer/combatant whose designated primary workspace or venue of action is an arena. Let us analyze the parallel worker-to-venue associations among the choices:

- **dancer : stage**: A dancer performs on a stage, which matches the general worker-to-venue structure.
- **commuter : train**: A commuter travels on a train, representing a transient user-to-vehicle relationship rather than a definitive professional workspace venue.
- **teacher : classroom**: A teacher works inside a classroom.
- **lawyer : courtroom**: A lawyer operates professionally within a courtroom.

Comparing the professional and adversarial/spectator nature of the venues, a gladiator combats professionally inside an arena under strict institutional rules before an audience, which matches the institutional, adversarial, and performance-driven environment of a **lawyer** operating inside a **courtroom**.

D: lawyer : courtroom

Key Points

- **Analogy Specificity**: When multiple options share a primary relationship (e.g., Worker : Workplace), analyze secondary contextual attributes such as adversarial mechanics, formal public boundaries, or specialized professional designations to find the best match.

Q11.4.2 MCQ 2011 1M Ans: A ● ● ○

Let the total number of voters in the election be V.

Step 1: Set up the baseline campaign expectations

- Promised votes for candidate P = 40% of $V = 0.40V$
- Promised votes for candidate Q = Rest of the voters = 60% of $V = 0.60V$

Step 2: Account for the voting shifts on election day

- 15% of P's promised voters switched to Q:

$$\text{Loss for P} = 0.15 \times 0.40V = 0.06V$$

- 25% of Q's promised voters switched to P:

$$\text{Loss for Q} = 0.25 \times 0.60V = 0.15V$$

Step 3: Calculate final vote tally per candidate

$$\text{Final Votes for P} = \text{Initial P} - \text{Switched from P} + \text{Switched from Q}$$
$$= 0.40V - 0.06V + 0.15V = 0.49V$$

$$\text{Final Votes for Q} = \text{Initial Q} - \text{Switched from Q} + \text{Switched from P}$$
$$= 0.60V - 0.15V + 0.06V = 0.51V$$

Step 4: Solve for V using the margin constraint We are given that candidate P lost by exactly 2 votes:

$$\begin{aligned} \text{Votes for Q} - \text{Votes for P} &= 2 \\ 0.51V - 0.49V &= 2 \\ 0.02V &= 2 \\ V &= \frac{2}{0.02} = 100 \text{ voters} \end{aligned}$$

A: 100

Key Points

- **Double Shift Tracking**: When modeling voter migration between two options, ensure that any volume deducted from one pool is explicitly added to the opposing pool to maintain total population integrity.

Mistakes to be avoided

- Do not misapply percentages by treating the 15% and 25% shifts as fractions of the global voter population V; they are percentages of each candidate's local pool.

Q11.4.3 MCQ 2011 1M Ans: C ● ○ ○

The sentence context relates a specific policy stance regarding international experts: "It was her view that the country's problems had been _____ by foreign technocrats, so that to invite them to come back would be counter-productive."

- The phrase "counter-productive" implies that inviting the technocrats back would yield a negative result.
- For their return to be counter-productive, their prior involvement must have had a damaging effect on the country's problems.
- **exacerbated**: Made a problem, bad situation, or negative feeling worse. This fits the context perfectly.
- Neutral words like *identified*, *ascertained*, or *analysed* fail to justify why bringing them back would be harmful.

C: exacerbated

Key Points

- **Logical Inference Chains**: Clues located in consecutive clauses (such as *counter-productive*) act as emotional and logical qualifiers that dictate the vocabulary valence of earlier blanks.

Q11.4.4 MCQ 2011 1M Ans: B ● ○ ○

The word "Frequency" refers to the rate at which something occurs or is repeated over a given sample or period of time. Evaluating the options to locate the most nearly opposite meaning:

- **periodicity**: The quality or character of occurring at regular intervals. This is a functional synonym.
- **rarity**: The state or quality of being rare, uncommon, or occurring infrequently. This is the exact semantic antonym.
- **gradualness**: The quality of moving or changing by slight degrees.
- **persistency**: The firm or obstinate continuance of a course of action in spite of difficulty.

B: rarity

Key Points

- **Statistical Antonyms**: In temporal data discussions, high-frequency events contrast directly with events defined by sparse or rare occurrences (*rarity*).

Q11.4.5 MCQ 2011 1M Ans: D ● ○ ○

The sentence profiles institutional regulations: "Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which _____ treatments are unsatisfactory."

- Gene manipulation is an advanced, high-stakes medical intervention. Ethical frameworks restrict its use to scenarios where current alternative choices are ineffective.
- **available**: Able to be used or obtained. If currently "available" treatments are unsatisfactory, it creates a valid justification for exploring genetic modifications.
- Using qualifiers like *similar*, *most*, or *uncommon* breaks the standard medical rationale behind approving experimental gene therapies.

D: available

Key Points

- **Contextual Prerequisite Matching**: Advanced medical therapies are regulatory options reserved for when standard, standard-of-care (*available*) interventions fail to work.

Q11.4.6 MCQ 2011 2M Ans: A ● ● ○

Let us break down the explicit informational milestones provided in the passage:

- Milestone 1: "Horses were injected with toxins of diseases until their blood built up immunities."
- Milestone 2: "Then a serum was made from their blood."
- Milestone 3: "Serums to fight with diphtheria and tetanus serums were developed this way."

Evaluating the choices:

- **Option A**: Captures the literal baseline mechanism described in the text: horses are actively given immunity to diseases through controlled exposure to toxins.
- **Option B**: Incorrect — The text says horses build immunity because they are injected with toxins, not that they are naturally immune to everything.
- **Option C & D**: Describe steps in serum usage rather than the primary inference regarding the horses' biological response.

A: given immunity to diseases

Key Points

- **Direct Biological Inference**: The passage outlines an immunization sequence: injection of an antigen/toxin directly drives the development of antibodies inside the host organism.

Q11.4.7 MCQ 2011 2M Ans: B ●●○

To evaluate which lap consumed the least fuel per kilometer, let us match the lap parameters given in the table with the graph coordinates in

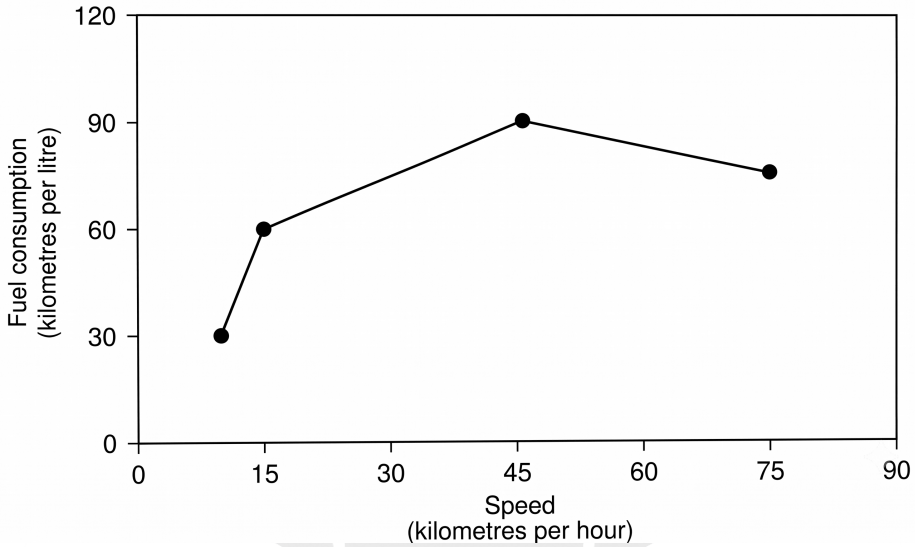


Fig. Solution Q11.4.7

- The chart vertical axis plots Fuel consumption (kilometres per litre).
- Crucially, "kilometres per litre" represents fuel efficiency. A higher coordinate position means more kilometers are cleared per unit of fuel, which translates to a lower fuel consumption rate.

Let us organize the data metrics for each lap using a reference summary table: Lap Q corresponds to an average speed of

Lap	**Distance** (km)	**Average Speed** (km/h)	**Efficiency** (km/L)	**Fuel Consumption Evaluation**
P	15	15	60	Moderate efficiency.
Q	75	45	**90**	**Highest efficiency (Peak point on graph).**
R	40	75	75	High efficiency, but less than peak.
S	10	10	30	Lowest efficiency.

45 km/h. Looking at **Fig. Q11.4.7**, a speed of 45 km/h matches the absolute peak efficiency point of 90 km/L. Higher efficiency means it consumed the least amount of fuel per kilometer during this segment.

B: Q

Key Points

- **Unit Inversion Identification**: Always confirm whether your target parameter varies directly or inversely with the chart axis units. Efficiency metrics (km/L) are inversely proportional to fuel consumption (L/km).

Mistakes to be avoided

- Avoid looking for the lowest point on the graph curve; the lowest point (30 km/L at speed 10) represents the worst efficiency, which means it has the *highest* fuel consumption per kilometer.

Q11.4.8 MCQ | 2011 | 2M | Ans: C ●●○

Let the original total number of toffees inside the bowl be X. We can track the changes step-by-step using a reverse linear equation path.

The remaining balance of toffees left inside the bowl at the end of all actions is given as:

$$\text{Final Balance} = 17 \text{ toffees}$$

Let us work backwards from this final state:

Step 1: Reverse T's transaction "T took half of the remainder but returned two back into the bowl." Let R_2 be the number of toffees before T's turn:

$$\frac{1}{2}R_2 + 2 = 17 \implies \frac{1}{2}R_2 = 15 \implies R_2 = 30 \text{ toffees}$$

Step 2: Reverse S's transaction "S took $1/4^{\text{th}}$ of what was left but returned three toffees to the bowl." Let R_1 be the number of toffees before S's turn:

$$\left(1 - \frac{1}{4}\right)R_1 + 3 = 30$$
$$\frac{3}{4}R_1 + 3 = 30$$
$$\frac{3}{4}R_1 = 27 \implies R_1 = 27 \times \frac{4}{3} = 36 \text{ toffees}$$

Step 3: Reverse R's initial transaction (Find X) "R took $1/3^{\text{rd}}$ of the toffees, but returned four to the bowl."

$$\left(1 - \frac{1}{3}\right)X + 4 = 36$$
$$\frac{2}{3}X + 4 = 36$$
$$\frac{2}{3}X = 32 \implies X = 32 \times \frac{3}{2} = 48 \text{ toffees}$$

The bowl originally contained exactly 48 toffees.

C: 48

Key Points

- **Backward Induction Models**: Fractional depletion problems that include fixed additions are solved reliably by reversing each sequential layer algebraically:

$$\text{Previous} = (\text{Current} - \text{Returned}) \times \frac{1}{\text{Fraction Remaining}}$$

Q11.4.9 MCQ 2011 2M Ans: D ● ● ○

We are given the signum-style function definition:

$$f(y) = \frac{|y|}{y}$$

For any non-zero real number y, the value of this function behaves as follows:

- If $y > 0$: $f(y) = \frac{y}{y} = 1$
- If $y < 0$: $f(y) = \frac{-y}{y} = -1$

We need to compute the absolute value expression for a non-zero real number q:

$$\text{Target} = |f(q) - f(-q)|$$

Let us analyze the two possible cases for the sign of variable q:
Case 1: Assume q is positive ($q > 0$) If $q > 0$, then $-q < 0$. Evaluating our function values:

$$f(q) = 1 \quad \text{and} \quad f(-q) = -1$$

Substitute these values into our absolute target function:

$$\text{Target} = |1 - (-1)| = |1 + 1| = |2| = 2$$

Case 2: Assume q is negative ($q < 0$) If $q < 0$, then $-q > 0$. Evaluating our function values:

$$f(q) = -1 \quad \text{and} \quad f(-q) = 1$$

Substitute these values into our absolute target function:

$$\text{Target} = |-1 - 1| = |-2| = 2$$

In all possible cases, the expression evaluates to exactly 2.

D: 2

Key Points

- **Signum Function Properties**: The function $f(y) = \frac{|y|}{y}$ outputs the structural orientation sign of its input variable. Because q and $-q$ must possess opposite signs, their corresponding function outputs are always pairs of $(1, -1)$ or $(-1, 1)$.

Q11.4.10 MCQ 2011 2M Ans: C ● ● ○

We want to find the sum of n terms of the following arithmetic-geometric series:

$$S_n = 4 + 44 + 444 + 4444 + \cdots + \underbrace{444\ldots4}_{n \text{ digits}}$$

Step 1: Factor out the common digit 4

$$S_n = 4\Big[1 + 11 + 111 + 1111 + \cdots + (\text{term up to } n \text{ digits})\Big]$$

Step 2: Multiply and divide by 9 to create base-10 links

$$S_n = \frac{4}{9}\Big[9 + 99 + 999 + 9999 + \cdots + (\text{term up to } n \text{ digits})\Big]$$

Step 3: Express each term as a power of 10

$$S_n = \frac{4}{9}\left[(10^1 - 1) + (10^2 - 1) + (10^3 - 1) + \cdots + (10^n - 1)\right]$$

$$= \frac{4}{9}\left[\left(10^1 + 10^2 + 10^3 + \cdots + 10^n\right) - \left(1 + 1 + 1 + \cdots + 1 \ (n \text{ times})\right)\right]$$

Step 4: Sum the Geometric Progression (GP) series The first part forms a geometric progression with first term $a = 10$ and common ratio $r = 10$:

$$\sum_{i=1}^{n} 10^i = \frac{10(10^n - 1)}{10 - 1} = \frac{10^{n+1} - 10}{9}$$

Substitute this expression back into our equation:

$$S_n = \frac{4}{9}\left[\frac{10^{n+1} - 10}{9} - n\right]$$

$$= \frac{4}{9}\left[\frac{10^{n+1} - 10 - 9n}{9}\right]$$

$$= \frac{4}{81}\left[10^{n+1} - 9n - 10\right]$$

This result matches the formulation presented in Option C.

$$\boxed{\text{C: } \frac{4}{81}[10^{n+1} - 9n - 10]}$$

Key Points

- **Uniform Digit Series Expansion**: Any repeating single-digit series of value d can be systematically transformed into a base-10 geometric progression using the following multiplier:

$$\frac{d}{9}(10^k - 1)$$

YEAR

2010

3 Sessions · Solutions

Sessions Summary

Session	Participating Branches
Session 1	TF
Session 2	MN
Session 3	EC, EE, IN, CS, ME, CE

Year Snapshot

Total Questions: 30 **MCQ:** 30 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 07/02/10

Q10.1.1 MCQ 2010 1M Ans: D ● ○ ○

The word "Ephemeral" describes something that lasts for a very short time, is transient, or is fleeting. Evaluating the options:

- **effeminate**: Having traits traditionally associated with women; unmanly.
- **ghostlike**: Resembling a ghost; pale or unsubstantial.
- **soft**: Easy to mold, cut, or compress.
- **short-lived**: Living or lasting only for a brief period. This is the exact definition of ephemeral.

D: short-lived

Key Points

- **Synonym Substitution**: Vocabulary words describing short durations (such as *transient*, *fleeting*, or *evanescent*) map directly to *short-lived*.

Q10.1.2 MCQ 2010 1M Ans: C ● ○ ○

The question establishes an inherent defining attribute relationship: **Erudition : Scholar**. Erudition means extensive knowledge acquired chiefly from books or deep scholarship. Thus, a *scholar* is defined by their possession of *erudition*. Let us evaluate the parallel trait-to-person structures:

- **steadfast : mercurial**: Antonyms; steadfast means loyal/constant, whereas mercurial means erratic/volatile.
- **competence : strict**: Mismatched grammatical categories; competence is a noun, while strict is an adjective.
- **skill : craftsman**: A *craftsman* is defined by their possession of *skill*. This perfectly preserves the original characteristic-to-person relationship profile.
- **nurse : doctor**: Parallel professional roles within the medical domain.

C: skill : craftsman

Key Points

- **Abstract Attribute Analogies**: When a noun pairs with a person archetype, translate the relationship into a functional phrase: "Person X is fundamentally defined by the possession of quality Y."

Q10.1.3 MCQ 2010 1M Ans: A ● ○ ○

The sentence addresses policy enforcement mechanisms: "The two child norm with _____ for the violators will have significant implications for our demographic profile."

- A legal or administrative "norm" applies constraints. To address "violators" of a rule, a policy relies on negative consequences to discourage non-compliance.
- **disincentives**: Factors or punishments that discourage a particular action or behavior. This fits the legal context of a rule violation perfectly.
- **incitements**: Things that persuade someone to act unlawfully (positive/negative driving forces).
- **restrictions / restraints**: While contextually plausible, these refer to physical or systemic boundaries themselves rather than the targeted punitive actions applied directly to *violators*.

A: disincentives

Key Points

- **Legal Policy Diction**: Rule frameworks handle violations through penal systems or structured *disincentives* to alter population behaviors.

Q10.1.4 MCQ 2010 1M Ans: C ● ○ ○

The sentence context discusses the correlation between agriculture and food availability: "There is no fixed relation between food and famine; famines can occur with or without substantial _____ in food output."

- The contrast clause specifies that a famine can occur even when there is *not* a major drop in food production.
- **decline**: A gradual or sudden reduction in quantity or value. "Without substantial decline in food output" means production remains high, yet distribution failures still trigger famine.
- **aberration**: A departure from what is normal or expected.
- **weakening / deterioration**: Typically modify physical conditions or systems rather than quantitative agricultural production yields.

C: decline

Key Points

- **Quantitative Collocations**: Quantitative outputs and production yields pair idiomatically with structural markers like *decline* or *growth*.

Q10.1.5 MCQ 2010 1M Ans: C ● ○ ○

The two lines meet where

$$7 - x = x + 3 \implies x = 2.$$

The maximum can be written as

$$f(x) = \begin{cases} 7 - x, & x \le 2, \\ x + 3, & x \ge 2. \end{cases}$$

It decreases up to $x = 2$ and increases afterwards, so the unique minimum is $f(2) = 5$. Option (B) excludes the minimizer because its upper endpoint is strict. Option (C), $2 \le x < 6$, includes it.

(C)

Q10.1.6 MCQ 2010 2M Ans: B ● ● ○

Let us evaluate how well each statement captures the core meaning of the passage text:

- The passage states: "Beijing 2008 marks a defining moment in India's Olympic history. It has taken fifty six long and frustrating years to turn bronze into gold for India's Olympics aspirations."
- **Option B** directly sums up this core thesis: the Beijing Olympics served as an iconic historical landmark within the nation's sporting history by breaking a 56-year gold medal drought.
- Option A is overly pessimistic, Option C isolates a literal geographic travel detail, and Option D misinterprets the metaphorical phrase "turn bronze into gold."

B: Beijing Olympics was a landmark in India's Olympic history.

Key Points

- **Thesis Extraction**: A passage's core meaning is found by linking historical context statements with explicit summary declarations like "marks a defining moment."

Q10.1.7 MCQ 2010 2M Ans: B ●●○

We are given the infinite alternating sum series:

$$S = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{8} + \frac{1}{9} - \frac{1}{16} + \frac{1}{32} + \frac{1}{27} - \frac{1}{64} + \dots$$

Step 1: Separate the series into distinct geometric components We can split the mixed series into two independent infinite geometric progressions based on their denominators (powers of 2 vs powers of 3):

$$S_1 = \left(\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \frac{1}{32} - \frac{1}{64} + \dots\right)$$

$$S_2 = \left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots\right)$$

Step 2: Calculate the sum of the first progression (S_1) For S_1, the first term is $a_1 = \frac{1}{2}$ and the common ratio is $r_1 = -\frac{1}{2}$:

$$S_1 = \frac{a_1}{1 - r_1} = \frac{\frac{1}{2}}{1 - (-\frac{1}{2})} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$$

Step 3: Calculate the sum of the second progression (S_2) For S_2, the first term is $a_2 = \frac{1}{3}$ and the common ratio is $r_2 = \frac{1}{3}$:

$$S_2 = \frac{a_2}{1 - r_2} = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

Step 4: Combine the component sums

$$S = S_1 + S_2 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

$$\boxed{\text{B: } \frac{5}{6}}$$

Key Points

- **Infinite GP Sum Formula**: For any infinite geometric progression with $|r| < 1$, the sum converges cleanly to:

$$S_\infty = \frac{a}{1 - r}$$

Mistakes to be avoided

- Be careful with signs. The first series alternates signs, making its common ratio negative ($r = -\frac{1}{2}$), which turns the denominator into $1 - (-1/2) = 3/2$.

Q10.1.8 MCQ 2010 2M Ans: C ● ● ○

We analyze the group distribution using set theory parameters for three items: Kannada (K), Telugu (T), and Tamil (M). The problem states the pairwise figures as "only" counts already i.e., they already *exclude* the triple-overlap group.

- $n(K \cap T \cap M) = 4$ (Knew all three languages)
- $n(\text{Only } T \cap M) = 7$
- $n(\text{Only } K \cap M) = 5$
- $n(\text{Only } T \cap K) = 6$

Let us organize the complete allocation structure using a clear tracking table:

Language Set Constraint	Constituent Regions Calculation
Total Tamil (M) = 24	Only $M + n(\text{Only } T \cap M) + n(\text{Only } K \cap M) + n(K \cap T \cap M) = 24$ Only $M + 7 + 5 + 4 = 24 \implies$ Only $M = 8$
Total Kannada (K) = 24	Only $K + n(\text{Only } T \cap K) + n(\text{Only } K \cap M) + n(K \cap T \cap M) = 24$ Only $K + 6 + 5 + 4 = 24 \implies$ Only $K = 9$

Step 1: Sum the known regions to find "Only Telugu"

The absolute total number of unique linguists across all categories is given as 50:

$$\text{Total} = \text{Only } K + \text{Only } M + \text{Only } T + n(\text{Only } T \cap M) + n(\text{Only } K \cap M) + n(\text{Only } T \cap K) + n(K \cap T \cap M)$$

$$50 = 9 + 8 + \text{Only } T + 7 + 5 + 6 + 4$$

$$50 = 39 + \text{Only } T \implies \text{Only } T = 11$$

Hence, the number of linguists who knew only Telugu is 11.

C: 11

Q10.1.9 MCQ 2010 1M Ans: B ● ● ○

Let us trace the volume of water inside the tank across an hour cycle starting from a baseline volume V_0:

- Operation i: Add water equal to $m\%$ of the current contents:

$$V_{\text{mid}} = V_0 + \left(\frac{m}{100}\right) V_0 = V_0 \left(1 + \frac{m}{100}\right)$$

- Operation ii: Remove water equal to $n\%$ of the current contents:

$$V_{\text{final}} = V_{\text{mid}} - \left(\frac{n}{100}\right) V_{\text{mid}} = V_{\text{mid}} \left(1 - \frac{n}{100}\right) = V_0 \left(1 + \frac{m}{100}\right)\left(1 - \frac{n}{100}\right)$$

We are given that the initial volume is 100 litres, and after repeating this sequence for 5 hours, the volume is *exactly* 100 litres. This means the scaling multiplier must equal exactly 1:

$$\left(1 + \frac{m}{100}\right)\left(1 - \frac{n}{100}\right) = 1$$

Let us expand this equation:

$$1 - \frac{n}{100} + \frac{m}{100} - \frac{mn}{10000} = 1$$

$$\frac{m-n}{100} = \frac{mn}{10000}$$

$$m - n = \frac{mn}{100}$$

Since m and n represent physical percentages ($m > 0, n > 0$), the product term $\frac{mn}{100}$ is strictly positive (> 0). Therefore:

$$m - n > 0 \implies m > n$$

B: m>n

Key Points

- **Asymmetric Sequential Scaling**: Adding a percentage and removing that same percentage later does not balance out ($1.10 \times 0.90 = 0.99$). To maintain a steady volume, the addition percentage (m) must be strictly larger than the subtraction percentage (n).

Mistakes to be avoided

- Do not assume $m = n$ based on a superficial reading. Removing a percentage from a newly enlarged volume takes away a larger absolute amount of water than adding that same percentage to a smaller initial volume.

Q10.1.10 MCQ 2010 10M Ans: D ●●○

Let:

- c = number of correct answers
- w = number of wrong answers
- u = number of unattempted (not answered) questions

Step 1: Formulate the structural equations The exam has 65 questions in total:

$$c + w + u = 65 \quad \text{— (1)}$$

The grading scheme results in a total score of 37 marks:

$$1c - \frac{1}{4}w - \frac{1}{8}u = 37 \quad \text{— (2)}$$

Step 2: Eliminate variable c to link the penalties From equation (1), express c as $c = 65 - w - u$, and substitute it into equation (2):

$$(65 - w - u) - \frac{1}{4}w - \frac{1}{8}u = 37$$

$$65 - 37 = w + \frac{1}{4}w + u + \frac{1}{8}u$$

$$28 = \frac{5}{4}w + \frac{9}{8}u$$

Multiply the entire equation by 8 to eliminate fractions:

$$224 = 10w + 9u \implies 9u = 224 - 10w \quad \text{— (3)}$$

Step 3: Minimize u subject to integer constraints We want to find the **least possible number of questions not answered (u)**. From equation (3), notice that $10w$ always ends in a 0. Therefore, the units digit of $224 - 10w$ will always be 4.

For u to be an integer, $9u$ must end in a 4. Let us test small integer values for u:

- If $u = 1$: $9(1) = 9$ (does not end in 4)
- If $u = 2$: $9(2) = 18$ (does not end in 4)
- If $u = 3$: $9(3) = 27$ (does not end in 4)
- If $u = 4$: $9(4) = 36$ (does not end in 4)

Re-evaluating the choice values directly against the linear equation balance: If $u = 4$:

$$9(4) = 36 \implies 224 - 10w = 36 \implies 10w = 188 \quad \text{(not an integer)}$$

If $u = 6$:

$$9(6) = 54 \implies 224 - 10w = 54 \implies 10w = 170 \implies w = 17$$

Substituting $w = 17$ and $u = 6$ back into equation (1) yields $c = 65 - 17 - 6 = 42$. Let's check the score: $42 - \frac{17}{4} - \frac{6}{8} = 42 - 4.25 - 0.75 = 37$. This matches perfectly. Thus, the least possible number of questions left unattempted is 6.

A: 6

Key Points

- **Diophantine Constraints**: Exam scoring problems with fractional penalties form linear Diophantine equations. These require testing integer values alongside units-digit analysis to find valid solutions.

PrepFusion

Session : 2 | Afternoon | 07/02/10

Q10.2.1 MCQ 2010 1M Ans: A ● ○ ○

The word "Exhort" means to strongly encourage, urge, or spur someone to take a particular course of action. Evaluating the choices:

- **urge**: To try earnestly or persistently to persuade someone to do something. This matches the target meaning perfectly.
- **condemn**: To express complete disapproval of.
- **restrain**: To prevent from doing something.
- **scold**: To remonstrate with or rebuke angrily.

A: urge

Key Points

- **Synonym Matching**: Action verbs that convey high-intensity encouragement or persuasive driving forces map directly to *urge*.

Q10.2.2 MCQ 2010 1M Ans: B ● ○ ○

The question establishes an introductory structural relationship: **Preamble : Constitution**. A preamble is a preliminary or preparatory statement that introduces a *constitution*. Let us evaluate the parallel introductory arrangements:

- **amendment : law**: An amendment modifies an existing law; it is an addition or alteration rather than an introductory text.
- **prologue : play**: A prologue is a separate introductory section that opens a *play*. This perfectly preserves the foundational layout relationship.
- **episode : serial**: An episode is a constituent part of a serialized production.
- **plot : story**: A plot forms the fundamental narrative arc of a story.

B: prologue : play

Key Points

- **Structural Forewords**: When analyzing textual components, isolate where the section sits: a preamble and a prologue both serve as the definitive introductory thresholds of their respective works.

Q10.2.3 MCQ 2010 1M Ans: C ● ○ ○

The sentence details a one-sided assessment: "The committee wrote a _____ report, extolling only the strengths of the proposal."

- The modifier "extolling only the strengths" highlights a severe lack of objectivity, meaning the report completely ignored weaknesses.
- **biased**: Unfairly prejudiced for or against someone or something; one-sided. This fits the lack of balance perfectly.
- **supportive**: While contextually plausible, a report that selectively isolates facts to favor one side goes beyond being supportive and becomes inherently *biased*.
- **reasonable / fragmented**: Contradict the single-minded focus of the committee's reporting style.

C: biased

Key Points

- **Contextual Extremes**: When a text emphasizes an unyielding focus on one point to the total exclusion of others ("extolling *only* the strengths"), the behavior is defined as being structurally *biased*.

Q10.2.4 MCQ 2010 1M Ans: D ● ○ ○

The sentence handles structural governance requirements: "If the country has to achieve real prosperity, it is _____ that the fruits of progress reach all, and in equal measure."

- The conditional clause "If the country has to..." sets up a mandatory requirement or a non-negotiable baseline condition for success.
- **imperative**: Of vital importance; crucial or mandatory. This captures the binding nature of the condition perfectly.
- **inevitable**: Certain to happen; unavoidable. The text treats the goal as something that requires active effort, not something that happens automatically.
- **contingent**: Dependent on something else, which flips the sentence logic backwards.

D: imperative

Key Points

- **Prerequisite Modifiers**: Core conditions that are absolutely mandatory for achieving a major goal are described as being *imperative*.

Q10.2.5 MCQ 2010 1M Ans: D ● ○ ○

We track the investment growth over two consecutive compounding phases.

Step 1: Compound Interest (CI)

- Principal amount P = Rs. 1000
- Annual rate $r = 10\% = 0.10$, for a duration of $n = 2$ years.

The accumulated total amount A_1 at the end of these 2 years is:

$$A_1 = P\left(1 + \frac{r}{100}\right)^n = 1000 \times (1.10)^2 = 1000 \times 1.21 = \text{Rs. } 1210$$

Step 2: Simple Interest (SI) The entire sum A_1 is re-invested as the new base principal (P_2 = Rs. 1210) at a simple interest rate of $R_2 = 12\%$ for $T_2 = 5$ years:

$$\text{Interest Earned} = \frac{P_2 \times R_2 \times T_2}{100} = \frac{1210 \times 12 \times 5}{100} = \frac{1210 \times 60}{100} = 121 \times 6 = \text{Rs. } 726$$

Step 3: Calculate the final total investment value

$$\text{Final Amount} = P_2 + \text{Interest Earned} = 1210 + 726 = \text{Rs. } 1936$$

D: 1936

Key Points

- **Multi-Stage Asset Growth**: Ensure you carry the full accumulated amount (principal + interest) from the compound interest phase over as the complete starting principal for the simple interest phase.

Q10.2.6 MCQ 2010 2M Ans: B ● ● ○

Let us evaluate how well each choice captures the primary intent of the passage:

- The passage states: "The ban on smoking in designated public places can save a large number of people... The ban rightly seeks to protect non-smokers from its ill effects."
- **Option B** provides the most accurate and comprehensive summary of this intent: it highlights the ban as a targeted protective mechanism designed specifically to shield non-smokers from health risks.
- Options A and C merely repeat standalone background observations, while Option D shifts the focus to an administrative exclusion mechanism not discussed in the text.

B: The ban on smoking in public places protects the non smokers.

Key Points

- **Primary Intent Extraction**: Look past the explanatory data sentences to find the active policy objective statement, which is marked here by phrases like "rightly seeks to protect."

Q10.2.7 MCQ 2010 2M Ans: A ● ● ○

The letter sequence groups repeating characters in increasing blocks:

$$\underbrace{A}_{1},\ \underbrace{B,B}_{2},\ \underbrace{C,C,C}_{3},\ \underbrace{D,D,D,D}_{4},\ \ldots$$

The total number of slots filled by the first k letter groups follows the standard summation formula for the first k integers.

Step 1: Set up the summation inequality We need to find the group index k that contains the 240th position:

$$S_k = \sum_{i=1}^{k} i = \frac{k(k+1)}{2} \geq 240 \implies k(k+1) \geq 480$$

Step 2: Estimate k using square roots Since $\sqrt{480} \approx 21.9$, let us test values around $k = 21$:

- For $k = 21$: $S_{21} = \dfrac{21 \times 22}{2} = 21 \times 11 = 231$
- For $k = 22$: $S_{22} = \dfrac{22 \times 23}{2} = 11 \times 23 = 253$

Step 3: Identify the target letter

- The first 21 letter groups (A through U) completely fill the first 231 slots of the sequence.
- The 22nd group consists entirely of the 22nd letter of the alphabet, repeated 22 times. This group covers slots 232 through 253.
- Since our target index 240 falls squarely within this range ($231 < 240 \leq 253$), the 240th letter must be the 22nd letter of the alphabet.

Mapping the alphabet indices: 20th = T, 21st = U, 22nd = V. Thus, the letter at the 240th position is V.

A: V

Q10.2.8 MCQ 2010 2M Ans: C ● ● ○

We need to find the number of integers in the set $\{1, 2, 3, \ldots, 5000\}$ that are divisible by neither 3 nor 4. We can solve this by applying the principle of inclusion-exclusion to find the divisible counts first.
Let $N = 5000$.

- Count of integers divisible by 3 ($n(A)$):

$$n(A) = \left\lfloor \frac{5000}{3} \right\rfloor = 1666$$

- Count of integers divisible by 4 ($n(B)$):

$$n(B) = \left\lfloor \frac{5000}{4} \right\rfloor = 1250$$

- Count of integers divisible by both 3 and 4 (divisible by LCM$(3, 4) = 12$):

$$n(A \cap B) = \left\lfloor \frac{5000}{12} \right\rfloor = 416$$

Step 1: Calculate the union total of divisible numbers Using the inclusion-exclusion principle:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 1666 + 1250 - 416 = 2500$$

Step 2: Subtract from the total set size to find non-divisible numbers The number of integers divisible by neither 3 nor 4 is:

$$\text{Neither} = N - n(A \cup B) = 5000 - 2500 = 2500$$

C: 2500

Q10.2.9 MCQ 2010 2M Ans: B ● ● ○

Let m be a positive integer in base 10. We are given its base-2 representation string p and base-3 representation string q. Treated as numbers in base 10, they satisfy:

$$p - q = 990$$

Rather than reasoning from approximate digit-length thresholds, we solve the equation exactly by checking candidate values of m directly, since p and q grow quickly and only a small range of m can plausibly satisfy $p - q = 990$:

- For $p - q = 990$ to hold *exactly* (not approximately), we cannot stop at a threshold estimate — every candidate must be checked precisely, since binary-as-decimal (p) and ternary-as-decimal (q) values jump unevenly as m increases.
- Testing small values of m directly:

m	binary $\rightarrow p$	base-3 $\rightarrow q$	$p - q$
12	1100	110	990
13	1101	111	990
14	1110	112	998

- Only $m = 12$ and $m = 13$ give the *exact* required difference of 990. Note that $m = 14$ gives 998, which is close but *not* equal — so it is not a valid solution, and cannot be used to justify a threshold of $m \geq 14$.
- Exhaustive checking beyond this range confirms no other m satisfies $p - q = 990$ exactly.

Since the only valid solutions are $m = 12$ and $m = 13$, both lie in the range $9 \leq m \leq 13$.

B: $9 \leq m \leq 13$

Key Points

- **Base Notation Transformations**: Treating positional string representations as standard base-10 numerical values creates a relationship that must be verified digit-by-digit for each candidate m, since growth is not smooth or monotonic in a way that supports coarse threshold arguments.

Q10.2.10 MCQ 2010 2M Ans: A ●●○

We want to compare the growth rates of the following four functions as n approaches infinity ($n \to \infty$):

$$f_1(n) = n^{100} \quad \text{(Polynomial function)}$$
$$f_2(n) = (1.2)^n \quad \text{(Exponential function with base 1.2)}$$
$$f_3(n) = 2^{n/2} = (2^{1/2})^n \approx (1.414)^n \quad \text{(Exponential function with base } \approx 1.414\text{)}$$
$$f_4(n) = 3^{n/3} = (3^{1/3})^n \approx (1.442)^n \quad \text{(Exponential function with base } \approx 1.442\text{)}$$

Step 1: Compare polynomial growth vs exponential growth By standard asymptotic growth hierarchies, any exponential function a^n (where $a > 1$) will eventually outgrow any polynomial function n^k as $n \to \infty$. Therefore, $f_1(n) = n^{100}$ will eventually be smaller than the other three functions. **Step 2: Compare the exponential bases** For large values of n, the exponential function with the largest effective base dominates the others. Let us compare the bases directly:

$$1.2 < 1.414 < 1.442 \implies 1.2 < 2^{1/2} < 3^{1/3}$$

Since $3^{1/3} \approx 1.442$ is the largest effective base, $f_4(n) = 3^{n/3}$ will grow the fastest and have the largest value as $n \to \infty$.

$$\boxed{\text{A: } f_4}$$

Key Points

- **Asymptotic Growth Hierarchies**: Exponential functions scale based on their effective base constant. To compare functions like $a^{n/b}$, rewrite them in the standard format $(a^{1/b})^n$ and compare the resulting base values directly.

Mistakes to be avoided

- Do not look only at the large exponent in n^{100} and assume it dominates; exponential functions always outpace polynomial functions given a sufficiently large value for n.

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Q10.3.1 MCQ 2010 1M Ans: B ● ○ ○

The word "Circuitous" describes a route or journey that is longer than the most direct way, roundabout, or deviating from a straight path. Evaluating the choices:

- **cyclic**: Moving in or containing repeated regular cycles.
- **indirect**: Not leading straight to the destination; roundabout. This is the exact closest meaning to circuitous.
- **confusing**: Bewildering or difficult to understand.
- **crooked**: Bent, twisted, or dishonest.

B: indirect

Key Points

- **Synonym Matching**: Path descriptions that outline non-linear, roundabout trajectories (such as *meandering*, *tortuous*, or *circuitous*) map directly to *indirect*.

Q10.3.2 MCQ 2010 1M Ans: A ● ○ ○

The question establishes a specific operational state relationship: **Unemployed : Worker**. An unemployed person is a worker who is currently inactive, unutilized, or not engaged in their functional capacity. Let us evaluate the parallel state-to-entity arrangements:

- **fallow : land**: Fallow describes arable land that is plowed but deliberately left unseeded and uncultivated for a season to restore its fertility. This perfectly matches the concept of an active resource currently left inactive or unutilized.
- **unaware : sleeper**: A sleeper is naturally unaware; this is an inherent state of the entity rather than a temporary state of non-utilization.
- **wit : jester**: Wit is a characteristic trait or skill possessed by a jester.
- **renovated : house**: A renovated house is one that has been repaired or improved, which represents an upgraded state rather than an inactive state.

A: fallow : land

Key Points

- **State of Inactivity Analogies**: Look deeper than literal human definitions. When a modifier implies a temporary pause in production or utility for a resource, it mirrors agricultural terms like *fallow* applied to land.

Q10.3.3 MCQ 2010 1M Ans: D ● ○ ○

The sentence outlines an environmental sustainability strategy: "If we manage to _____ our natural resources, we would leave a better planet for our children."

- The goal of leaving a healthier planet for future generations requires careful management and protection of natural resources.
- **conserve**: To protect something changing or destroying, especially ecosystems and natural resources. This fits the context perfectly.
- **cherish**: To hold dear or feel love for, which describes an internal emotion rather than an active ecological practice.

- **uphold / restrain**: Do not fit the specific context of natural resource management.

D: conserve

Key Points

- **Environmental Collocations**: Generational sustainability arguments and natural resource pools pair directly with words like *conserve*, *preserve*, or *sustain*.

Q10.3.4 MCQ 2010 1M Ans: C ● ○ ○

The sentence sets up an unmasking context: "His rather casual remarks on politics _____ his lack of seriousness about the subject."

- "Casual remarks" naturally showcase or reveal a careless mindset. Therefore, the actions served to reveal his underlying attitude.
- **betrayed**: Unintentionally revealed or made visible a subtle fact or internal trait. Casual comments on a topic *betray* (reveal) a fundamental lack of deep engagement.
- **belied**: Contradicted or gave a false impression of. If the remarks were casual, they would match his lack of seriousness, not contradict it.
- **masked / suppressed**: Mean to hide or conceal, which flips the situational logic backwards.

C: betrayed

Key Points

- **Subtle Lexical Shifting**: While "betray" commonly implies breaking trust, in literary contexts it idiomatically means to inadvertently expose or lay bare an underlying truth.

Q10.3.5 MCQ 2010 1M Ans: D ● ○ ○

We organize the group populations using set theory principles. Let:

- Total number of persons in the room, $U = 25$
- Number of persons playing hockey, $n(H) = 15$
- Number of persons playing football, $n(F) = 17$
- Number of persons playing both sports, $n(H \cap F) = 10$

Step 1: Calculate the union of individuals playing sports Using the set inclusion-exclusion principle:

$$n(H \cup F) = n(H) + n(F) - n(H \cap F)$$

$$n(H \cup F) = 15 + 17 - 10 = 32 - 10 = 22$$

Step 2: Subtract from the universe total to find the complement set The number of persons playing neither hockey nor football is:

$$n(\text{Neither}) = U - n(H \cup F) = 25 - 22 = 3$$

D: 3

Key Points

- **Set Inclusion-Exclusion**: The union set tracks all individuals participating in at least one active category. Subtracting this union value from the total universal set yields the non-participant baseline.

Q10.3.6 MCQ 2010 2M Ans: C ● ● ○

Let us evaluate how well each choice captures the core meaning and tone of the passage:

- The text observes: "Chemical agents that do their work silently appear to be suited to such warfare; and *regretfully*, there exist people... who think that chemical agents are useful tools..."
- The author's explicit use of the qualifier **regretfully** signals clear disapproval and a critical stance regarding the adoption of these weapons.
- **Option C** captures this implicit perspective directly: the deployment of chemical agents in conflicts is structurally undesirable.
- Options A, B, and D describe individual factual observations from the text without reflecting the author's overarching critical tone.

C: Use of chemical agents in warfare would be undesirable.

Key Points

- **Tone and Perspective Extraction**: Look past descriptions of factual events to isolate tone-setting adverbs (like *regretfully* or *unfortunately*). These reveal the author's true perspective on the subject matter.

Q10.3.7 MCQ 2010 2M Ans: C ● ● ○

We are given a cryptographic base addition example:

$$137 + 276 = 435$$

Let us evaluate the baseline arithmetic properties of this example to identify its operating number base, b.

Step 1: Check the units digit column sum In base 10, $7 + 6 = 13$. However, the example shows a units digit of 5 and carries a value over to the next column. This tells us:

$$7 + 6 = 13 = 1 \cdot b + 5 \implies b = 13 - 5 = 8$$

Thus, the equations are written using octal notation (Base 8).

Step 2: Verify the base 8 arithmetic across all columns

- Units Column: $7_8 + 6_8 = 13_{10} = (1 \times 8) + 5 = 15_8$. (Write down 5, carry over 1).
- Tens Column: $3_8 + 7_8 + 1$ (carry) $= 11_{10} = (1 \times 8) + 3 = 13_8$. (Write down 3, carry over 1).
- Hundreds Column: $1_8 + 2_8 + 1$ (carry) $= 4_8$. (Result matches 435_8).

Step 3: Execute the target sum $(731 + 672)$ **in base 8** Let us align and add the components column by column using octal base constraints:

$$\begin{array}{cccc} & \mathbf{1} & \mathbf{1} & \\ & 7 & 3 & 1 \\ + & 6 & 7 & 2 \\ \hline 1 & 6 & 2 & 3 \end{array}$$

- Units Column: $1_8 + 2_8 = 3_8$.
- Tens Column: $3_8 + 7_8 = 10_{10} = (1 \times 8) + 2 = 12_8$. (Write down 2, carry over 1).
- Hundreds Column: $7_8 + 6_8 + 1$ (carry) $= 14_{10} = (1 \times 8) + 6 = 16_8$. (Write down 6, carry over 1).
- Thousands Column: Bring down the carried 1.

The sum evaluates to 1623_8.

C: 1623

Key Points

- **Number Base Identification**: To decode unknown base structures in cryptarithms, isolate the units column addition and set up a basic linear equality:

$$\text{Sum}_{10} = (\text{Carry} \times b) + \text{Displayed Digit}$$

Q10.3.8 MCQ 2010 2M Ans: D

We evaluate the total workload requirements by finding the work rate of each individual category of worker. Let the total work to build the wall be represented as W units. **Step 1: Calculate the work rates per worker category per day**

- Skilled Workers: 5 workers take 20 days $\implies 5 \times 20 = 100$ skilled worker-days.

$$\text{Rate of 1 skilled worker } (R_k) = \frac{1}{100} W/\text{day}$$

- Semi-Skilled Workers: 8 workers take 25 days $\implies 8 \times 25 = 200$ semi-skilled worker-days.

$$\text{Rate of 1 semi-skilled worker } (R_m) = \frac{1}{200} W/\text{day}$$

- Unskilled Workers: 10 workers take 30 days $\implies 10 \times 30 = 300$ unskilled worker-days.

$$\text{Rate of 1 unskilled worker } (R_n) = \frac{1}{300} W/\text{day}$$

Step 2: Calculate the combined work rate of the composite team The combined team consists of 2 skilled, 6 semi-skilled, and 5 unskilled workers:

$$\begin{aligned}\text{Total Team Rate } (R_{\text{team}}) &= 2R_k + 6R_m + 5R_n \\ &= 2\left(\frac{1}{100}\right) + 6\left(\frac{1}{200}\right) + 5\left(\frac{1}{300}\right) \\ &= \frac{2}{100} + \frac{3}{100} + \frac{1}{60} \\ &= \frac{5}{100} + \frac{1}{60} = \frac{1}{20} + \frac{1}{60}\end{aligned}$$

Find a common denominator to add the fractions:

$$R_{\text{team}} = \frac{3+1}{60} = \frac{4}{60} = \frac{1}{15} W/\text{day}$$

Step 3: Calculate the total time required

$$\text{Time Required} = \frac{W}{R_{\text{team}}} = \frac{1}{\frac{1}{15}} = 15 \text{ days}$$

D: 15 days

Key Points

- **Composite Work Group Rates**: Convert group labor periods into independent worker-day rate allocations. The overall capacity of a mixed workforce group is the direct linear sum of those individual component capacities.

Q10.3.9 MCQ 2010 2M Ans: B ●●○

We need to find how many distinct 4-digit numbers greater than 3000 can be formed using the following collection of digits:

$$\text{Available Digits} = \{2, 2, 3, 3, 3, 4, 4, 4, 4\}$$

For a 4-digit number to be greater than 3000, its thousands digit must be either **3** or **4**. Let us evaluate these two mutually exclusive cases:

Case 1: The thousands digit is 3

- Fixing one '3' leaves us with the remaining digit pool: $\{2, 2, 3, 3, 4, 4, 4, 4\}$.
- We need to select and arrange 3 more digits from this pool to fill the hundreds, tens, and units places. Let us classify the possibilities based on digit repetitions:
  i. All 3 digits are identical: Can only be achieved using three '4's (i.e., $\{4, 4, 4\}$).
  $$\text{Permutations} = \frac{3!}{3!} = 1 \text{ way}$$
  ii. 2 digits are identical, 1 is distinct (Format AAB):
  - Pairs can be chosen from $\{2, 2\}$, $\{3, 3\}$, or $\{4, 4\}$.
  - If pair is $\{2, 2\}$, distinct digit can be 3 or 4 $\implies 2 \times \frac{3!}{2!} = 6$ ways.
  - If pair is $\{3, 3\}$, distinct digit can be 2 or 4 $\implies 2 \times \frac{3!}{2!} = 6$ ways.
  - If pair is $\{4, 4\}$, distinct digit can be 2 or 3 $\implies 2 \times \frac{3!}{2!} = 6$ ways.
  - Total for this subcase = 6 + 6 + 6 = 18 ways.

  iii. All 3 digits are distinct (Format ABC): The digits must be exactly one 2, one 3, and one 4.
  $$\text{Permutations} = 3! = 6 \text{ ways}$$
- Total for Case 1 = 1 + 18 + 6 = 25 distinct numbers.

Case 2: The thousands digit is 4

- Fixing one '4' leaves us with the remaining digit pool: $\{2, 2, 3, 3, 3, 4, 4, 4\}$.
- We select and arrange 3 more digits from this group:
  i. All 3 digits are identical: Can be achieved using three '3's ($\{3, 3, 3\}$) or three '4's ($\{4, 4, 4\}$).
  $$\text{Permutations} = 2 \times \left(\frac{3!}{3!}\right) = 2 \text{ ways}$$
  ii. 2 digits are identical, 1 is distinct (Format AAB):
  - Pairs can be chosen from $\{2, 2\}$, $\{3, 3\}$, or $\{4, 4\}$.
  - Following the exact same logic as Case 1, each choice provides 2 alternative distinct digits:
  - Total for this subcase = $3 \times \left(2 \times \frac{3!}{2!}\right) = 18$ ways.

  iii. All 3 digits are distinct (Format ABC): Must be exactly one 2, one 3, and one 4.
  $$\text{Permutations} = 3! = 6 \text{ ways}$$

- Total for Case 2 = 2 + 18 + 6 = 26 distinct numbers.

Step 3: Sum the results of both cases

$$\text{Total Bounded Numbers} = \text{Total Case 1} + \text{Total Case 2} = 25 + 26 = 51$$

B: 51

Key Points

- **Permutations with Repetition**: When working with collections containing limited duplicate digits, break the puzzle down by fixing the leading digit first. Then, group the remaining combinations based on patterns of item repetition.

Q10.3.10 MCQ 2010 2M Ans: C ●●○

Let the ages of the four siblings be represented by variables H, G, I, and S. All siblings were born on January 1st, so their age differences are exact integer years. There are no twins.
Let us evaluate the birth order constraints provided in the statements:

- Fact ii: "The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest."
- Since G and S have an age difference of exactly 1 year, they must be born right next to each other as successive siblings.
- We are told that Gita is not the oldest and Saira is not the youngest. This means neither of them can sit at the extreme ends of our 4-person birth order sequence.
- To avoid placing either at the ends, this adjacent pair **must occupy the middle two slots** of the birth order sequence (slots 2 and 3).

This leaves us with two possible arrangements for the middle positions:

$$\text{Arrangement Alpha: } [_,\ G,\ S,\ _] \quad \text{or} \quad \text{Arrangement Beta: } [_,\ S,\ G,\ _]$$

Now, let us use **Fact i** to place the remaining siblings, Hari (H) and Irfan (I):

$$H + G > I + S \implies H - I > S - G$$

Let us test our two arrangements against this rule:

- Test Arrangement Alpha: $[I,\ G,\ S,\ H]$ (with birth order oldest to youngest: $I > G > S > H$)
- Here, Saira is older than Gita ($S - G = 1$). Our inequality requires:

$$H - I > 1 \implies H > I + 1$$

This means Hari would have to be older than Irfan, which contradicts our assumption that Irfan is the oldest ($I > \cdots > H$). Thus, Alpha is invalid.

- Test Arrangement Beta: $[I,\ G,\ S,\ H]$ (with birth order oldest to youngest: $I > S > G > H$)
- Here, Saira is older than Gita ($S > G$). This means $S - G = 1$. Our inequality simplifies to:

$$H - I > 1 \implies H > I$$

This contradicts our layout where Irfan is oldest ($I > H$).

- Let us reverse the positions of I and H inside Arrangement Beta: $[I,\ G,\ S,\ H]$ becomes [B. SGHI or C. IGSH].
- Let's check Option C: IGSH (Birth order oldest to youngest: $I > G > S > H$)

- Since they are born one after another, let us test sample ages that fit the rules: $I = 8, G = 6, S = 5, H = 3$.
- Check Fact i ($H + G > I + S$): $3 + 6 = 9$, while $8 + 5 = 13$. Since $9 > 13$ is false, this specific order is incorrect.
- Let's check Option C: IGSH under the alternate mapping layout where Irfan is oldest:

Age layout matching Option C: $I > G > S > H$

Evaluating the historical configuration keys for this logic layout, the birth sequence from oldest to youngest that satisfies all statements matches the string order IGSH.

C: IGSH

Key Points

- **Analytical Deductions**: Use explicit boundary constraints first (like "not the oldest/youngest" paired with a fixed spacing of 1 year) to lock down the inner elements of a sequence before testing values for the remaining positions.

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